President University

Faculty Of Computer Science

DIGITAL FORENSIC INVESTIGATION REPORT

Operation X – Dark Web Human Trafficking & Murder Investigation

Prepared in Partial Fulfillment of the Digital Forensics Course Requirements

Course: Project Boothcamp - Company X Case 3

Lecturer: Rosalina, S.Kom., M.Kom. & Dr. Deffa Rahadiyan, S.Si.

Prepared by

Name	**Student ID**	**Class**	**Responsibility**
Fathir Barhouti Awlya	001202400054	Cyber 2	Network & Steganography Analysis (EPIC 5)
Aurelio Sami Rabbani	001202400218	Cyber 2	Disk Image Acquisition & Hash Verification (EPIC 1)
Reva Stefvany br Ginting	001202400114	Cyber 2	Deleted File & Artifact Recovery (EPIC 2)
Melisa Anamaria Nainggolan	001202400154	Cyber 2	Hidden Data & Password Recovery (EPIC 3)
Berliana Veni Jaya Kesuma	001202400153	Cyber 2	Encryption & File Cracking (EPIC 4)
All Members	-	Cyber 2]	Report Compilation

Submitted to: President University

Submission Date: 2026 - 07 - 30

TABLE OF CONTENTS

TABLE OF CONTENTS **3**
1 EXECUTIVE SUMMARY **6**
2 INTRODUCTION **7**
2.1 Background **7**
2.2 Objectives **7**
2.3 Scope **7**
3 CASE MANAGEMENT **8**
3.1 Evidence Receipt from Law Enforcement **8**
3.2 Chain of Custody **8**
3.3 Team Roles & Responsibilities **8**
4 EVIDENCE VERIFICATION & VALIDATION **9**
5 METHODOLOGY **13**
5.1 Forensic Framework 13
5.2 Examination Workflow **14**
5.3 Tools and Environment 14
6 FINDINGS **16**
6.1 Disk Image Findings **16**
6.1.1 Disk Image Acquisition and Hash Verification 17
6.1.2 Timeline Analysis 18
6.1.3 Deleted File and Artifact Recovery 19
6.2 Network Findings **45**
6.2.1 HTTP Analysis 46
6.2.2 FTP Analysis 47
6.2.3 DNS and TLS Analysis 48
6.2.4 Exported Network Objects 49
6.2.5 Cryptocurrency Transaction Tracing 50
6.3 Encrypted File Findings **52**
6.3.1 Hidden Data and Password Recovery 52
Identify of Password-Protected Files 54
Password Recovery 55
File Extension Mismatch Detection 60
Binwalk Analysis on Suspicious Files 61
6.4 Steganography Findings 63
6.4.1 Carrier Files Examined 64
6.4.2 Metadata Analysis 64
6.4.3 String and Hex Analysis 64
6.4.4 Embedded File Detection 65
6.4.5 Steganography Analysis 66
6.5 Document Findings **67**
6.5.1 Recovered Documents 67
8.1 Temporal Analysis **74**
8.2 Relational Analysis **74**

8.3 Functional Analysis **76**
9 CONCLUSIONS **77**
9.1 Answers to Investigation Objectives **77**
9.2 Limitations of Examination **78**
9.2.1 Disk Image Acquisition Limitations 78
9.2.2 Deleted File and Artifact Recovery Limitations 78
9.2.3 Network and Steganography Limitations 79
9.2.4 Encryption & File Cracking Limitations 79
10 EXAMINATION NOTES / INVESTIGATION LOG **80**
10.1 Disk Image and Deleted File Recovery Examination Log 80
10.2 Network and Steganography Examination Log 82
10.3 Encryption and File Cracking Examination Log 85
10.4 Email and Document Recovery Examination Log 86
11 REFERENCES **87**
11.1 Standards 87
11.2 Tool References 87
11.3 Additional Tool References 88
12 APPENDIX **89**

1 EXECUTIVE SUMMARY

Operation X is a digital forensic investigation into a suspected international human trafficking network with links to a related murder case. The primary subject, Daniel Vega, became a person of interest after law enforcement intercepted dark web forum activity discussing the trafficking operation, and a missing person identified as A.C. was later found dead in connection with the same network.

The examination covered a ten-segment forensic disk image (Daniel_Laptop.E01) and a network traffic capture (Daniel_Network.pcap) belonging to the suspect. Additional files recovered from within these two sources, including a password-protected archive (encrypted.zip), two image files (HiddenMessage.bmp and safehouse location.bmp), and several PDF, spreadsheet, and email files, were examined using Autopsy, Wireshark, ExifTool, Binwalk, FTK Imager, NetworkMiner, John the Ripper, Hashcat, and pyzipper.

The investigation recovered documents identifying four trafficking victims and their intended destinations, forged identity documents, financial records indicating large offshore transactions, and email communications discussing shipments, safehouses, and operational activities. Network traffic analysis further revealed file transfers involving trafficking-related documents and communications with external systems. Additional examination identified the installation of anonymity and steganography tools, indicating deliberate attempts to conceal digital evidence.

Overall, the evidence recovered is highly consistent with the objectives of this investigation and supports the presence of organised trafficking activities, deliberate evidence concealment, and communications linking the suspect to the wider criminal network. Certain aspects, including encrypted Tor traffic contents and cryptocurrency transaction attribution, could not be independently verified within the scope of this examination and are therefore acknowledged as investigation limitations rather than confirmed findings.

2 INTRODUCTION

2.1 Background

Operation X is a digital forensic investigation involving a suspected human trafficking and murder case linked to dark web activities. Law enforcement agencies intercepted discussions on dark web forums regarding an international human trafficking network. During the investigation, a missing person identified as A.C. was believed to have become a victim of the trafficking organization and was later found dead. Investigators suspected that the case was connected to a larger organized crime network operating through encrypted communications and the Tor network.

Following the investigation, law enforcement conducted a raid and seized a laptop belonging to the suspect, Daniel [REDACTED]. A forensic disk image (Daniel_Laptop.E01) and a captured network traffic file (Daniel_Network.pcap) were submitted for digital forensic examination to identify digital evidence related to the investigation.

2.2 Objectives

This examination aims to:

1. Identify digital evidence related to the suspect's activities.

2. Analyze communications and transferred files associated with the investigation.

3. Reconstruct the sequence of events based on recovered digital evidence.

4. Produce a forensic report supporting the investigation.

2.3 Scope

This examination covers the following evidence items:

- DanielVe v. (R.E01–R.E10) – Split forensic disk image.
- Daniel_Network.pcap – Network traffic capture.
- HiddenMessage.bmp – Image recovered during network analysis.
- Safehouse location.bmp – Image recovered during network analysis.
- encrypted.zip – Password-protected archive recovered during network analysis.
- [Additional recovered files identified by other examination teams]

The following items or aspects were NOT covered in this examination:

- Physical crime scene investigation
- DNA examination
- Medical examination
- Live network monitoring
- Evidence not provided by law enforcement

3 CASE MANAGEMENT

3.1 Evidence Receipt from Law Enforcement

The digital evidence used in this examination was provided by the course instructor as part of the Operation X digital forensic case scenario. The examination team did not perform the original evidence acquisition or forensic imaging process. Instead, the team received the evidence in the form of a segmented forensic disk image (DanielVe v. R.E01–R.E10) and a network capture file (Daniel_Network.pcap) for examination.

At the time of receipt, all evidence files were accessible and showed no indication of corruption. Integrity verification was performed prior to examination, as described in Chapter 4.

3.2 Chain of Custody

The digital evidence was provided directly by the course instructor as part of the academic investigation scenario. No formal law enforcement chain-of-custody documentation was provided to the examination team prior to receipt. Throughout the examination process, the evidence remained under the control of the assigned examination team and was not modified. Evidence integrity was verified before analysis.

3.3 Team Roles & Responsibilities

Table 3.1 summarizes the division of responsibility across the examination team.

Team Member	Responsibility (Jira Epic)	Related Chapter(s)
Fathir Barhouti Awlya	Network & Steganography Analysis — EPIC 5 (KAN-46)	6.2, 6.4, 10 (log entries provided)

Team Member	Responsibility (Jira Epic)	Related Chapter(s)
Aurelio Sami Rabbani	Disk Image Acquisition & Hash Verification — EPIC 1 (KAN-38)	4, 6.1.1
Reva Stefvany br Ginting	Deleted File & Artifact Recovery — EPIC 2 (KAN-31)	6.1.2, 6.5
Melisa Anamaria Nainggolan	Hidden Data & Password Recovery — EPIC 3 (KAN-56)	6.3.1
Berliana Veni Jaya Kesuma	Encryption & File Cracking — EPIC 4 (KAN-42)	6.3.2
All Members	Report Compilation	All chapters

Table 3.1 — Team Roles & Responsibilities

4 EVIDENCE VERIFICATION & VALIDATION

Before examination, all digital evidence was verified to ensure data integrity. SHA-256 hash values were calculated for each evidence item using the sha256sum utility. No reference hash values were provided by law enforcement; therefore, integrity verification was limited to calculating and documenting the hash values of the received evidence prior to analysis. No indication of file corruption or modification was observed during the verification process.

Evidence ID	Filename	Provided Hash	Recalculated Hash	Match	Tool
E01	DanielVe v. R.E01	N/A	c89187bfd893d3269141c017a4965858be3cfad17175fe04fe410b045f42a241	N/A	Autopsy

Evidence ID	Filename	Provided Hash	Recalculated Hash	Match	Tool
E02	DanielVe v. R.E02	N/A	5d821c5e20ef12fddf772db10873accf1a279c4955172dcb86734f7d6d24a684	N/A	Autopsy
E03	DanielVe v. R.E03	N/A	07836542de8bdd7b6fbca623e38a3412ef43ee5c85f0e70cdc906d4c04b42508	N/A	Autopsy
E04	DanielVe v. R.E04	N/A	ca2f5141cd88ce91b96b68ded711047147e908184a50679e96e91ca8b2712eb8	N/A	Autopsy
E05	DanielVe v. R.E05	N/A	c4e5f720f82b1d5d9d090718c70833d201c8c28dc786c85df864d499dbc6c87f	N/A	Autopsy
E06	DanielVe v. R.E06	N/A	9704e188bbaef3ad023695296386b71408d1a8c806b085bb046ac7b07869ee60	N/A	Autopsy

Evidence ID	Filename	Provided Hash	Recalculated Hash	Match	Tool
E07	DanielVe v. R.E07	N/A	b617a85c33c5d117cc2c148d267ec9f36827ca7fb52a81080ea25d52590e8fca	N/A	Autopsy
E08	DanielVe v. R.E08	N/A	47c4fe4373929908769f8297bd5fd6803b7db31d456c37ef3b9301e0ec770cba	N/A	Autopsy
E09	DanielVe v. R.E09	N/A	cc78c572c434c3d7eb30897e90488fe007782e6c2021c47efb715edb73478bf6	N/A	Autopsy
E10	DanielVe v. R.E10	N/A	5b44ffee19c88a88fc3439cf1b636645684b107bbe76ffdf3e7c90809765daed	N/A	Autopsy
PCAP	Daniel_Network.pcap	N/A	4345eb708d18ecc4f2c1fdcf64bb570243ddc42800ace1fec0c41c4adb0495c5	N/A	sha256sum

Evidence ID	Filename	Provided Hash	Recalculated Hash	Match	Tool
IMG-1	HiddenMessage.bmp	N/A	fed66c4038d2b94ce6b013993d82fc79c878ba2016bcdb13ba48d50bb6d3e95e	N/A	sha256sum
IMG-2	Safehouse location.bmp	N/A	b9efea7df0b132692865ad6cf8cf979246db1ce01d3ce33b5b27aa0c76f19b4f	N/A	sha256sum
ZIP-1	encrypted.zip	N/A	77384ccb7efb109c28f191e3bc99506f5ef9c98788ebb47622682d391d0f0cf4	N/A	sha256sum
DOC-1	INVOICE.pdf	N/A	704d65f7d2bae8c7a0ffc03894d71ecd5a64cc081e5b6dde0d9796cfa2a6c056	N/A	sha256sum
DOC-2	Shipment List.pdf	N/A	64145ce8dca46b1582359b2eac01ecfa000c78942845ca9f344519edd2c6b030	N/A	sha256sum

Evidence ID	Filename	Provided Hash	Recalculated Hash	Match	Tool
XLS-1	client_list.xlsx	N/A	5362d32b25770869a0d9868b7e0ca8c4b9b08c00c79468f243135cb56d5a2313	N/A	sha256sum
TXT-1	contacts.txt	N/A	8cdd6b87208893e17d0d5c8ddd12d766e36e1143bbb701cb0dcb537b4c64cc45	N/A	sha256sum
TXT-2	safehouse_location.txt	N/A	88576bbb42a152f7e7e6847a50b8c830c7502665098fd7c398a03e4c66531e63	N/A	sha256sum

Table 4.1 — Hash Verification (Network & Steganography evidence rows pre-filled with tool name; hash values pending actual computation)

5 METHODOLOGY

5.1 Forensic Framework

This examination follows the digital forensic methodology described in **NIST SP 800-86 – Guide to Integrating Forensic Techniques into Incident Response**. Since the examination team received forensic images and network captures that had already been acquired by law enforcement, the acquisition phase was not performed by the examination team. Therefore, the workflow implemented in this report consists of:

- Evidence Verification

- Examination
- Analysis
- Reporting

This approach ensures that the integrity of the provided evidence is verified before examination, the findings are analyzed systematically, and all results are documented in a structured forensic report.

5.2 Examination Workflow

The examination followed four sequential phases: Evidence Verification, Examination, Analysis, and Reporting.

5.3 Tools and Environment

Tool	Version	Purpose	Used By
Wireshark / TShark	4.7.2	Packet capture inspection and protocol analysis of Daniel_Network.pcap. Used to analyze HTTP, DNS, TLS, FTP, OCSP, and identify suspicious network activity.	Network & Steganography
GNU strings	GNU Binutils 2.47	Extracted printable strings from files to identify readable artifacts, URLs, filenames, and hidden text.	Network & Steganography
ExifTool	13.55	Examined image metadata to determine whether images contained hidden metadata or embedded information.	Network & Steganography
Binwalk	2.4.3	Detected embedded or appended file signatures to identify potential hidden data within image files.	Network & Steganography
Capinfos	4.7.2	Generated capture file statistics, timestamps, packet counts, and general summary information.	Network & Steganography

Tool	Version	Purpose	Used By
zsteg	Unknown (installed release)	Detected potential LSB steganography and hidden data patterns within BMP images.	Network & Steganography
file	5.48	Identified actual file types and validated file signatures (magic bytes).	Network & Steganography
sha256sum	GNU Coreutils 9.11	Compute SHA-256 hashes to verify evidence integrity.	Network & Steganography
xxd	2026-06-16	Raw hexadecimal inspection of recovered files.	Network & Steganography
John the Ripper	1.9.0-jumbo -1	Recover passwords and crack hashes for encrypted files or archives.	Encryption
Cryptii	v4.0.11	a web app for encoding, decoding, and encrypting text (Caesar cipher, Base64, Morse code, ROT13, etc.), all running in your browser with no data sent to a server.	Email & Document Recovery
7-Zip / p7zip-full	23.01	AE-compatible archive listing, dictionary testing, and extraction of encrypted.zip	Encryption & File Cracking
pyzipper (Python)	0.4.0	AESZipFile dictionary attack — recovered password for encrypted.zip	Encryption & File Cracking
zipinfo / unzip	ZipInfo 3.00 / UnZip 6.00	Identified encrypted.zip as WinZip AE (compression method 99); initial (unsuccessful) dictionary attack	Encryption & File Cracking
Capinfos	4.7.2	Capture file statistics and summary information	Network & Steganography

Tool	Version	Purpose	Used By
Exiftool	13.55	Image metadata inspection	Network & Steganography
zsteg	Unknown (installed release)	Detect potential LSB steganography within BMP images	Network & Steganography
zip2john	John the Ripper 1.9.0-jumbo-1	Convert encrypted ZIP archive into crackable hash	Encryption

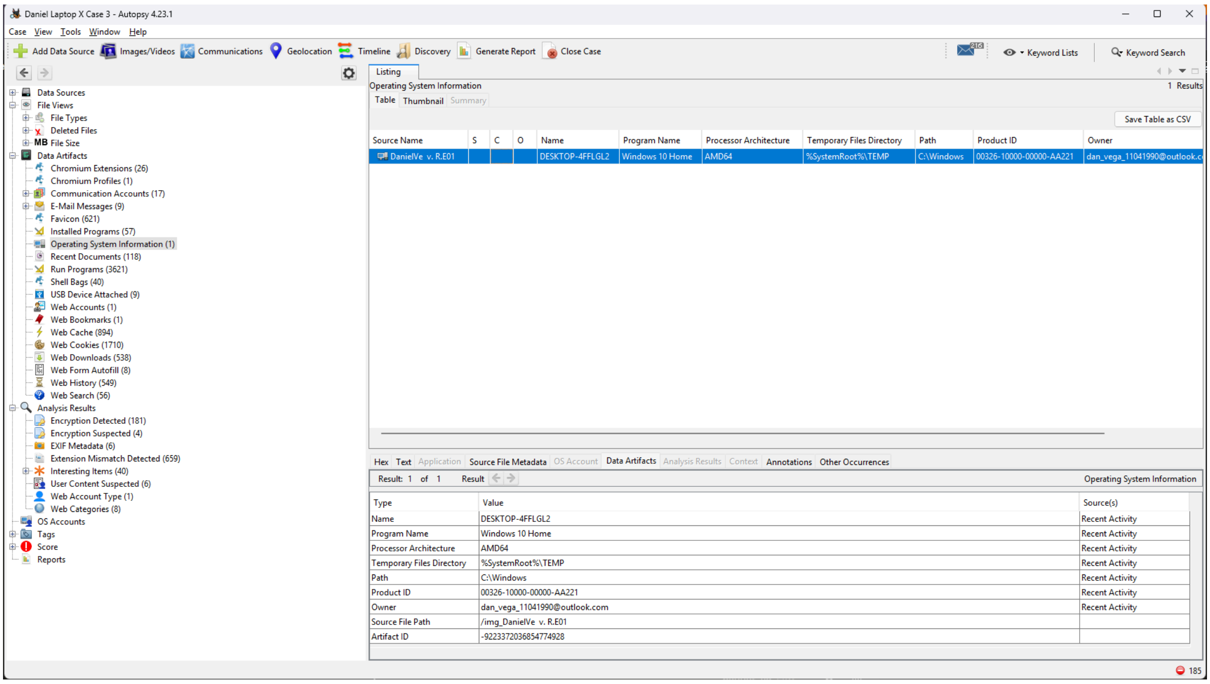
Table 5.1 — Tools and Environment

6 FINDINGS

This chapter presents the factual findings obtained during the examination of the digital evidence. Each finding is reported objectively based on the evidence recovered, without interpretation or correlation. Cross-evidence analysis is presented separately in Chapter 8 (Analysis).

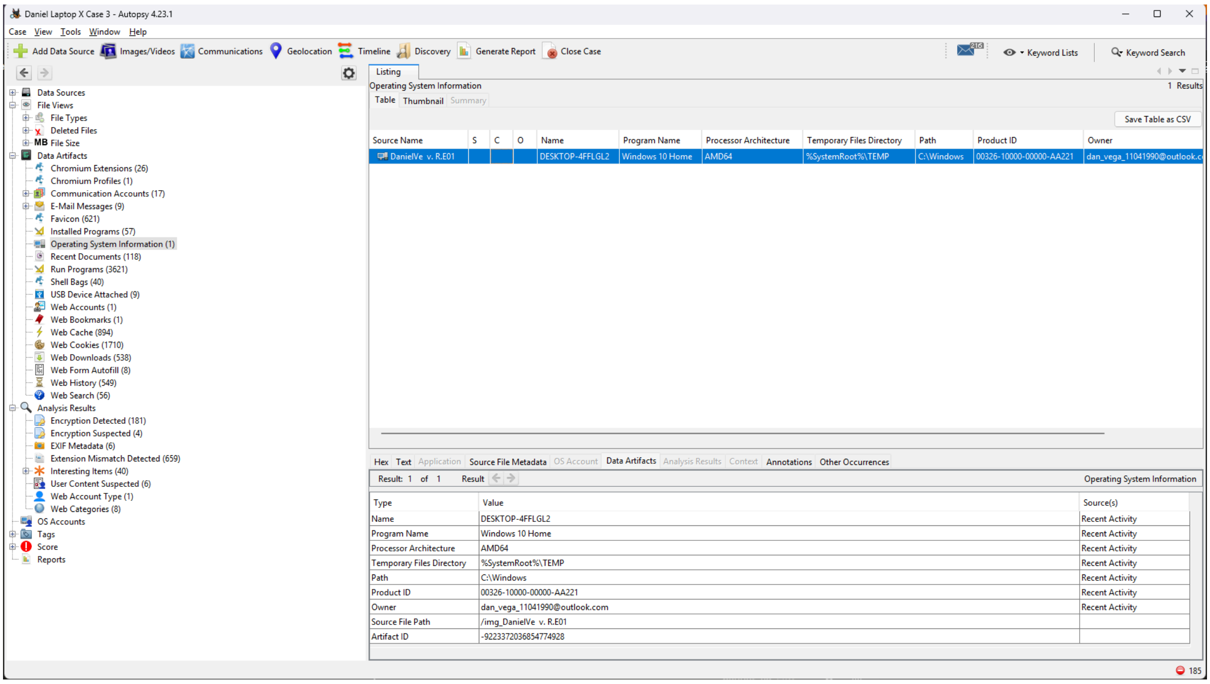
6.1 Disk Image Findings

This section presents the findings obtained from the forensic examination of the Daniel_Laptop.E01 disk image. The examination focused on identifying system artifacts, recovered files, browser activity, and file system metadata relevant to the investigation. All findings reported in this section are presented objectively without interpretation.



6.1.1 Disk Image Acquisition and Hash Verification

The forensic disk image (DanielVe v. R.E01–R.E10) was successfully loaded into Autopsy for examination. The image was recognized as a valid EnCase Evidence File (EWF) and the file system was parsed successfully. System artifacts including operating system information, user accounts, browser history, registry entries, installed applications, and file system metadata were identified and made available for further examination.



Prior to examination, integrity verification was performed as documented in Table 4.1. SHA-256 hash values were calculated for the received evidence. Since no reference hash values were provided by law enforcement, the calculated hash values were recorded as baseline integrity values. No indication of file corruption or modification was observed.

6.1.2 Timeline Analysis

Timeline reconstruction was performed using Autopsy 4.22.1's Timeline feature on Daniel_Laptop.E01, filtering file-system and web-activity events by filename to isolate artifacts relevant to the case from the general noise of installed-software files (over 1,000,000 total timeline events were present unfiltered, the majority originating from bundled Tor Browser localization files).

The following chronological sequence was established for key artifacts:

- **2025-03-07 17:33:06–17:38:21 WIB** — tor-browser-windows-x86_64-portable-14.0.7.exe created, modified, and changed (Downloads), consistent with download and installation.
- **2025-03-11 00:22:19–05:47:09 WIB** — nightclub.jpg created, changed, and accessed (Downloads).
- **2025-03-11 01:06:26 & 01:16:32 WIB** — transaction logs.xlsx modified and accessed, located within usb_backup_files.zip (OneDrive/Documents/Hidden).
- **2025-03-11 01:14:50 & 01:16:32 WIB** — identity docs.pdf modified and accessed at the same time as transaction logs.xlsx, within the same archive.
- **2025-03-13 00:55:01–00:55:14 WIB** — Web History shows a search for "OpenStego" on Bing, followed by visits to openstego.com and the project's GitHub releases page, immediately followed by a Web Download of Setup-OpenStego-0.8.6.exe.
- **2025-03-13 00:55:19–00:56:38 WIB** — OpenStego installed (registry entries, shortcuts, and Prefetch execution recorded) and its library accessed shortly after installation.
- **2025-03-13 01:11:24 WIB** — Shipment List[48].pdf and Everything is in place[47].eml both recorded as attachment-save events (ACBM) at the identical timestamp, indicating the shipment manifest was saved as an email attachment.

- **2025-03-13 11:32:27–11:32:28 WIB** — tor-browser-windows-x86_64-portable-14.0.7.exe and Setup-OpenStego-0.8.6.exe were both accessed within one second of each other, indicating the two tools were used together in the same session.
- **2025-03-13 19:18–19:56 WIB and 2025-03-14 00:23–12:14 WIB** — Repeated access to OpenStego program files and Prefetch entries across the following day, indicating continued use rather than a single isolated installation.

Email content: The email Everything is in place[47].eml (sent 2025-03-11 09:23:40, from dan_vega_11041990@outlook.com) was found to contain body text encoded with the same Caesar-cipher (shift of 3) pattern identified elsewhere in the case evidence. Decoded, the message reads: *"...planned. No issues. Payment confirmed. Awaiting final instructions. Will proceed as discussed."*

The findings presented in this section are reported objectively. Interpretation of their significance is discussed in Chapter 8 (Analysis).

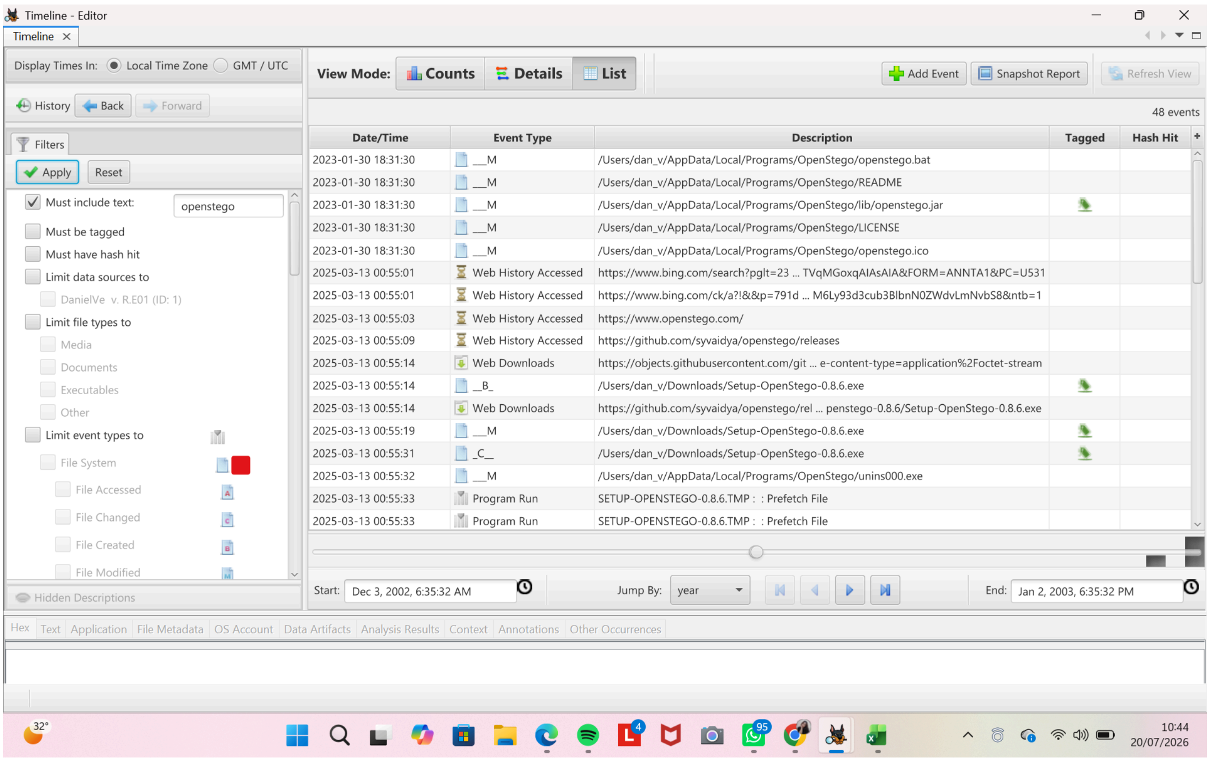


Figure 6.1.1 — Autopsy timeline / artifact overview

6.1.3 Deleted File and Artifact Recovery

The forensic examination identified deleted files and recoverable artifacts from the Daniel_Laptop.E01 disk image using Autopsy. Deleted files were identified through the Deleted Files directory, while additional recoverable artifacts were identified from the Recent Documents and File Types categories. These findings indicate that previously deleted or recently accessed files remained recoverable from the forensic image.

The forensic examination recovered several documents and related artifacts from the disk image, including PDF documents, spreadsheet files, email messages, and Windows shortcut (.lnk) files. The recovered documents include Shipment List.pdf, identity docs.pdf, INVOICE.pdf, and transaction logs.xlsx. In addition, multiple email artifacts associated with the account dan_vega_11041990@outlook.com were successfully recovered. The identified email subjects included Everything is in Place, Safehouse Ready, Urgent: Cleanup Needed, and File Transfer – Urgent, indicating previous communication activities stored within the forensic image.

Metadata analysis was performed on the recovered Windows shortcut (.lnk) artifacts, including victim list.txt.lnk, HiddenMessage.bmp.lnk, encrypted.zip.lnk, Dark Web.lnk, and Emails.lnk. The examination compared the creation, modification, and access timestamps of each artifact. The metadata showed consistent timestamp information across the recovered artifacts. The encrypted.zip.lnk and Dark Web.lnk artifacts remained in use after their creation, while no internal timestamp inconsistencies or indications of metadata manipulation were identified based on the available forensic metadata.

Email 1.eml	Safehouse Ready	handler@darkops.xyz	12/03/2025 15:48:31
Email 2.eml	Urgent: Cleanup Needed	fixer@darkops.xyz	12/03/2025 15:50:07
Email 3.eml	Everything is in place	boss@onionmarket.xyz	12/03/2025 15:51:47
Email 4.eml	File Transfer - Urgent	handler@darkops.xyz	12/03/2025 15:54:14

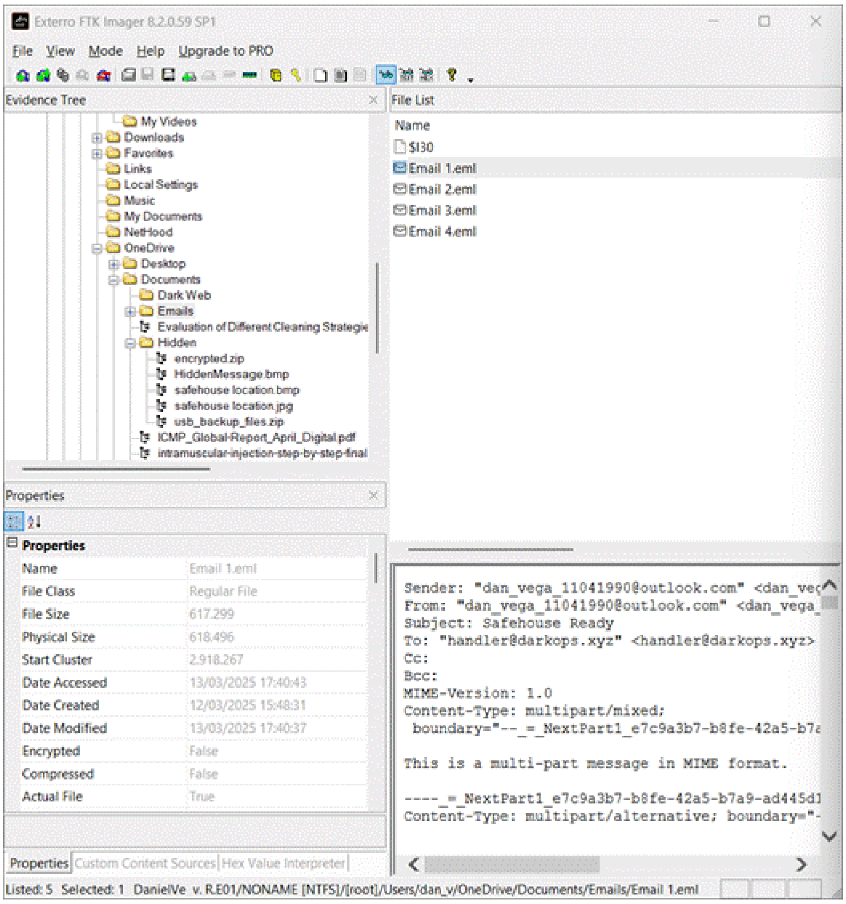


Figure 6.1.3a — Email 1.eml ("Safehouse Ready") properties in FTK Imager.

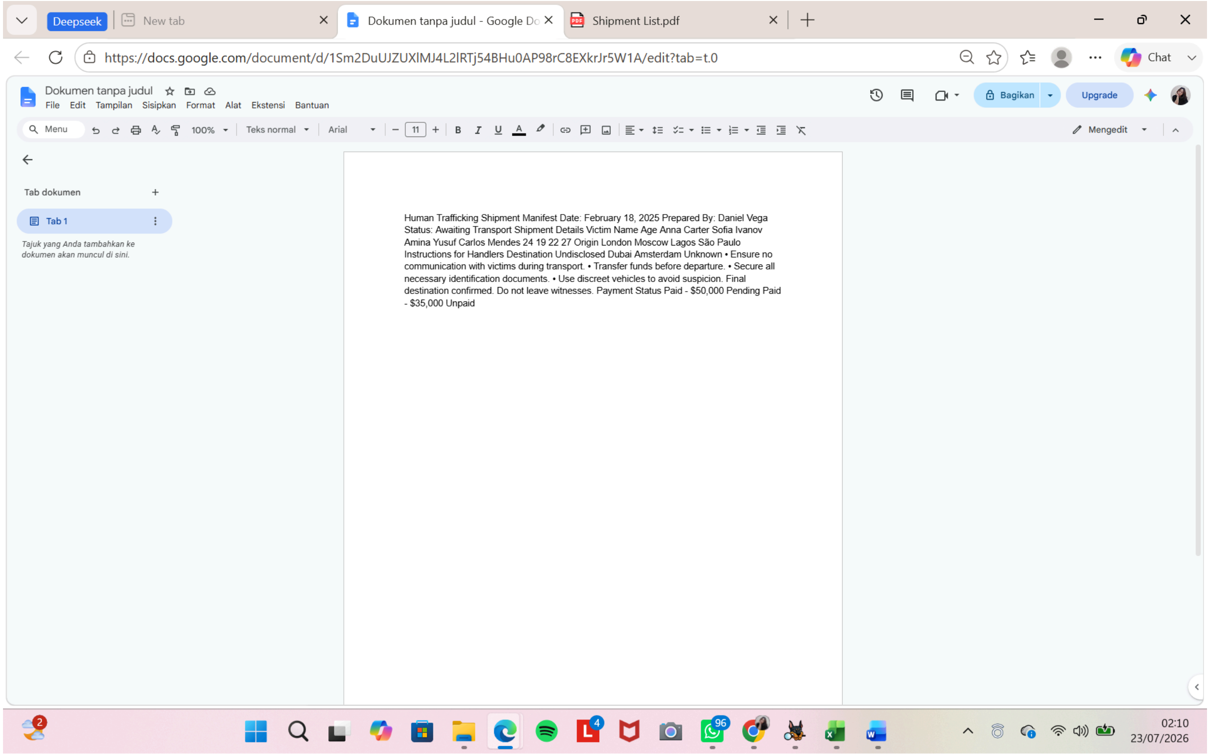


Figure 6.1.2 — Recuva / file carving results

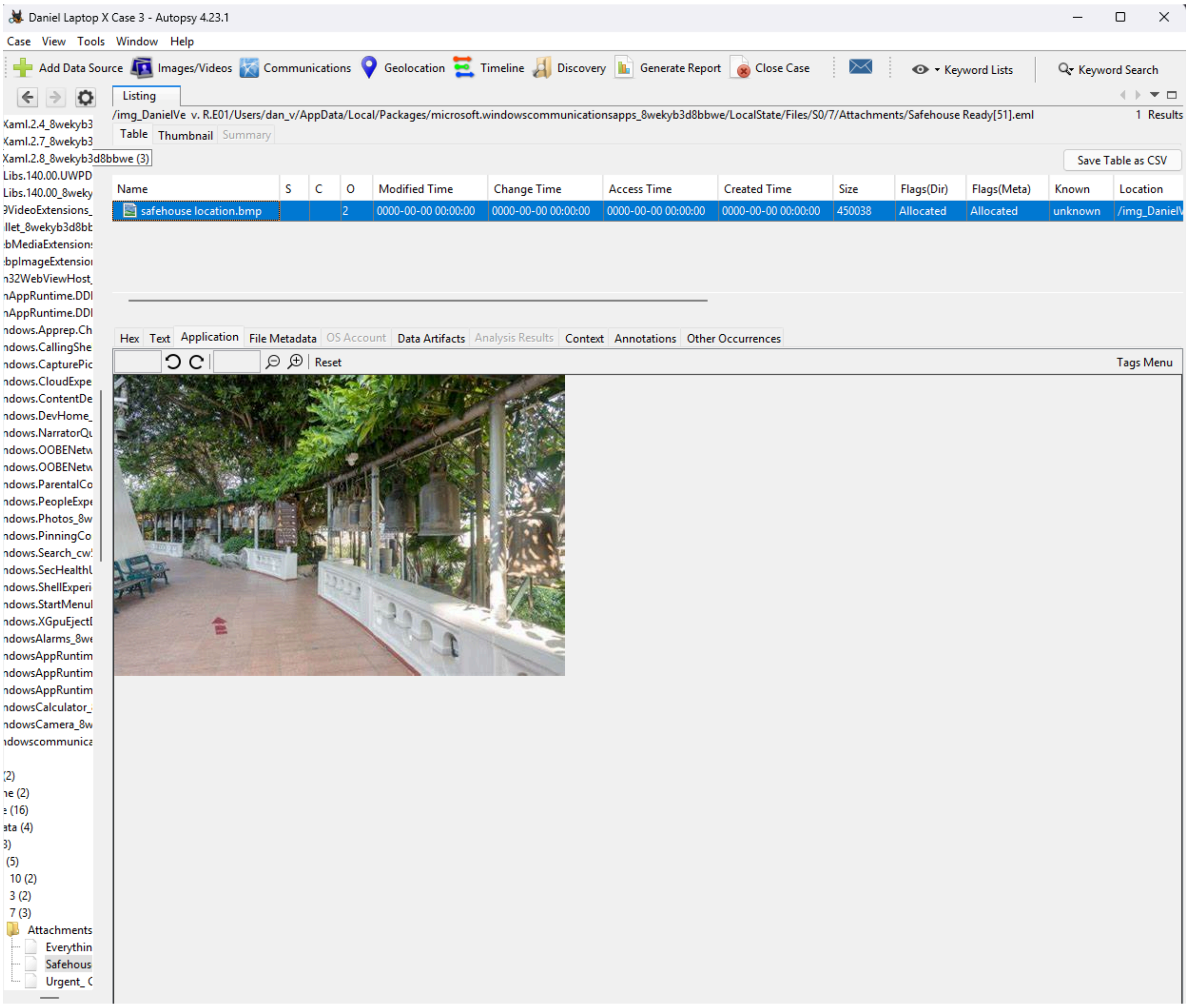


Figure 6.1.3a — Email 1.eml ("Safehouse Ready")

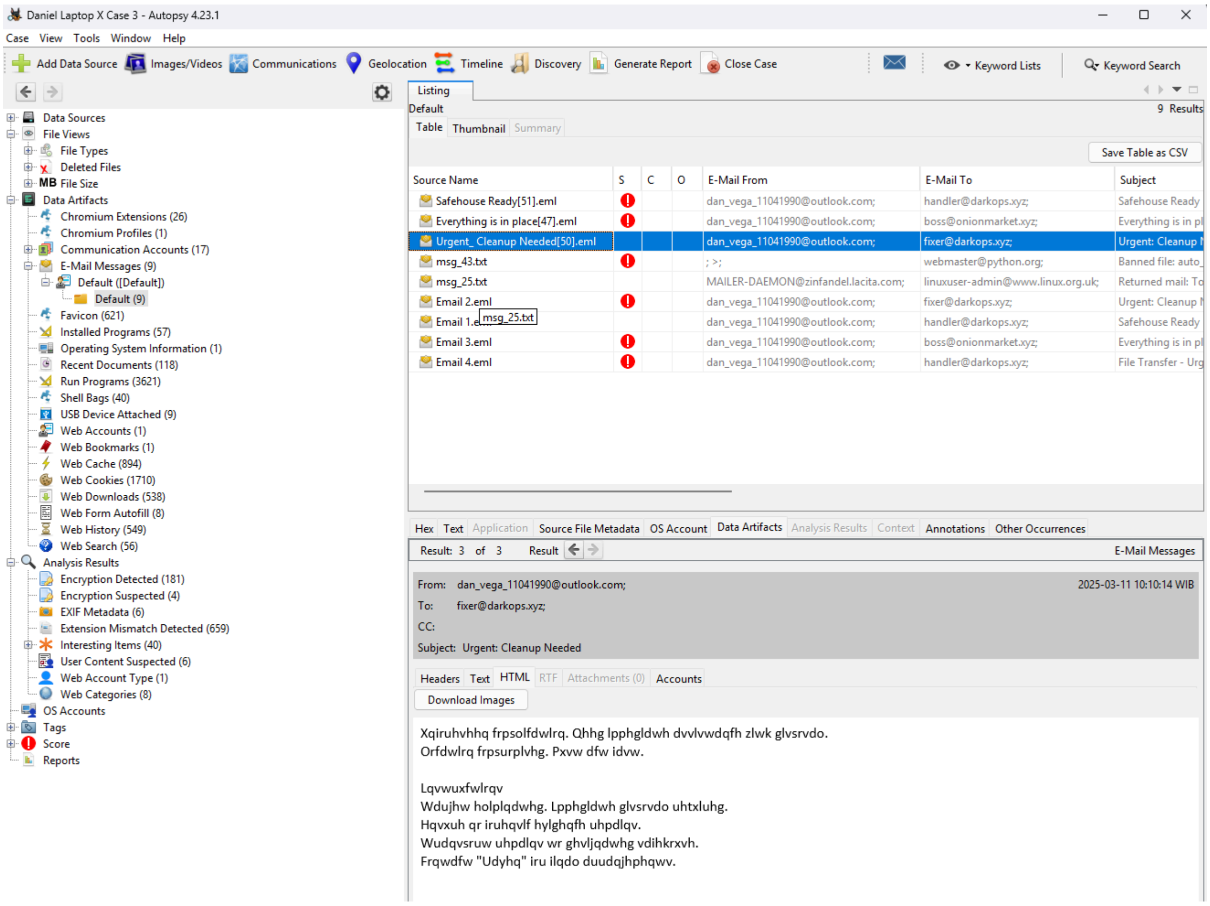


Figure 6.1.3b — Email 2.eml ("Urgent: Cleanup Needed")

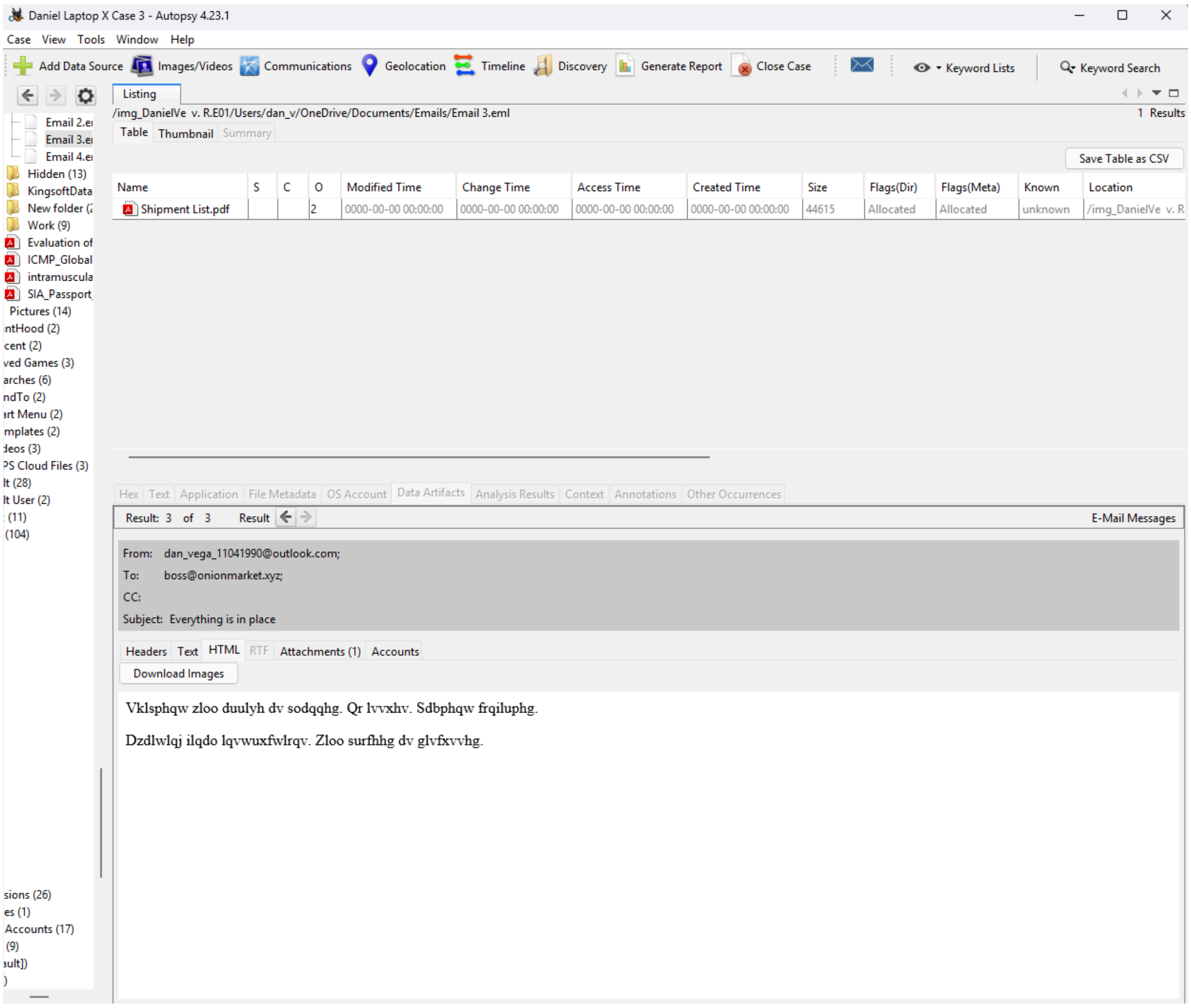
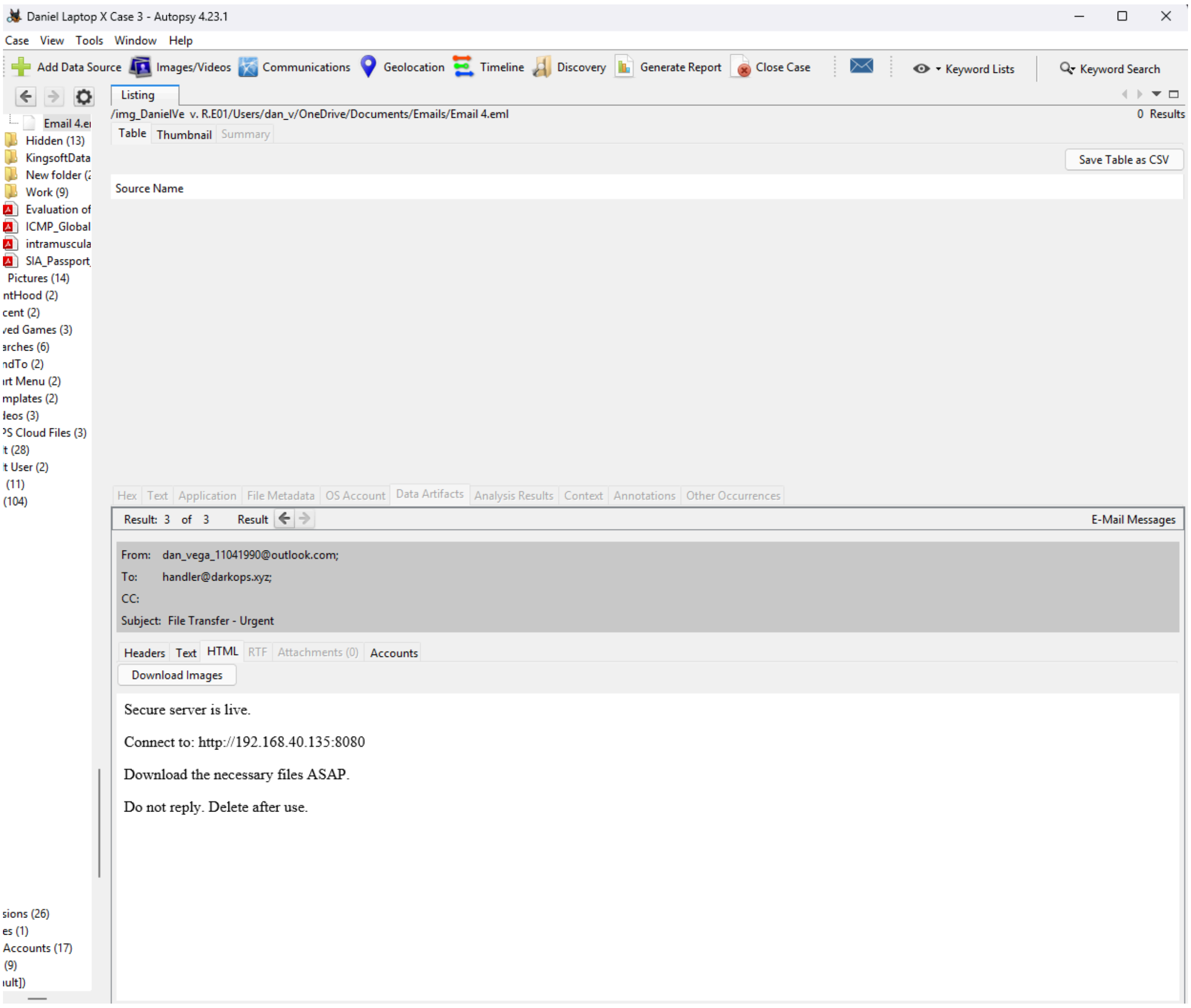
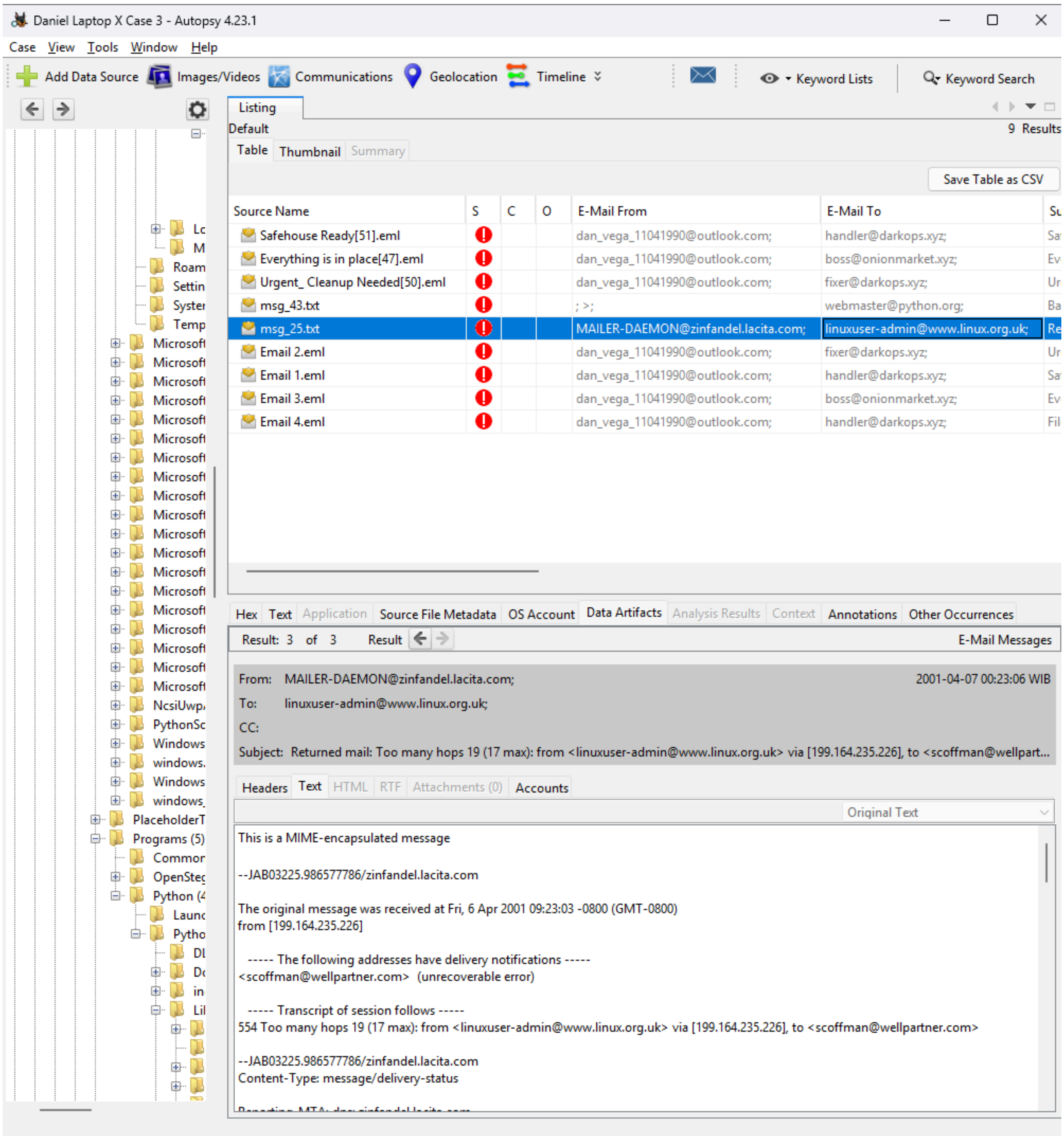


Figure 6.1.3c — Email 3.eml ("Everything is in Place")


Daniel Laptop X Case 3 - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case Keyword Lists Keyword Search
Listing
/img_DanielVe v. R.E01/Users/dan_v/OneDrive/Documents/Emails/Email 4.eml
0 Results
Table Thumbnail Summary
Save Table as CSV
Source Name
Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences
Result: 3 of 3 Result
E-Mail Messages
From: dan_vega_11041990@outlook.com;
To: handler@darkops.xyz;
CC:
Subject: File Transfer - Urgent
Headers Text HTML RTF Attachments (0) Accounts
Download Images
Secure server is live.
Connect to: http://192.168.40.135:8080
Download the necessary files ASAP.
Do not reply. Delete after use.


Daniel Laptop X Case 3 - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Keyword Lists
Keyword Search
Listing
Default
9 Results
Table
Thumbnail
Summary
Save Table as CSV
Source Name
S
C
O
E-Mail From
E-Mail To
Safehouse Ready[51].eml
dan_vega_11041990@outlook.com;
handler@darkops.xyz;
Everything is in place[47].eml
dan_vega_11041990@outlook.com;
boss@onionmarket.xyz;
Urgent_ Cleanup Needed[50].eml
dan_vega_11041990@outlook.com;
fixer@darkops.xyz;
msg_43.txt
webmaster@python.org;
msg_25.txt
MAILER-DAEMON@zinfandel.lacita.com;
linuxuser-admin@www.linux.org.uk;
Email 2.eml
dan_vega_11041990@outlook.com;
fixer@darkops.xyz;
Email 1.eml
dan_vega_11041990@outlook.com;
handler@darkops.xyz;
Email 3.eml
dan_vega_11041990@outlook.com;
boss@onionmarket.xyz;
Email 4.eml
dan_vega_11041990@outlook.com;
handler@darkops.xyz;
Hex
Text
Application
Source File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Result: 3 of 3
Result
E-Mail Messages
From: MAILER-DAEMON@zinfandel.lacita.com;
2001-04-07 00:23:06 WIB
To: linuxuser-admin@www.linux.org.uk;
CC:
Subject: Returned mail: Too many hops 19 (17 max): from <linuxuser-admin@www.linux.org.uk> via [199.164.235.226], to <scoffman@wellpart...
Headers
Text
HTML
RTF
Attachments (0)
Accounts
Original Text
This is a MIME-encapsulated message
--JAB03225.986577786/zinfandel.lacita.com
The original message was received at Fri, 6 Apr 2001 09:23:03 -0800 (GMT-0800)
from [199.164.235.226]
----- The following addresses have delivery notifications -----
<scoffman@wellpartner.com> (unrecoverable error)
----- Transcript of session follows -----
554 Too many hops 19 (17 max): from <linuxuser-admin@www.linux.org.uk> via [199.164.235.226], to <scoffman@wellpartner.com>
--JAB03225.986577786/zinfandel.lacita.com
Content-Type: message/delivery-status
Microsoft
Programs (5)
Python

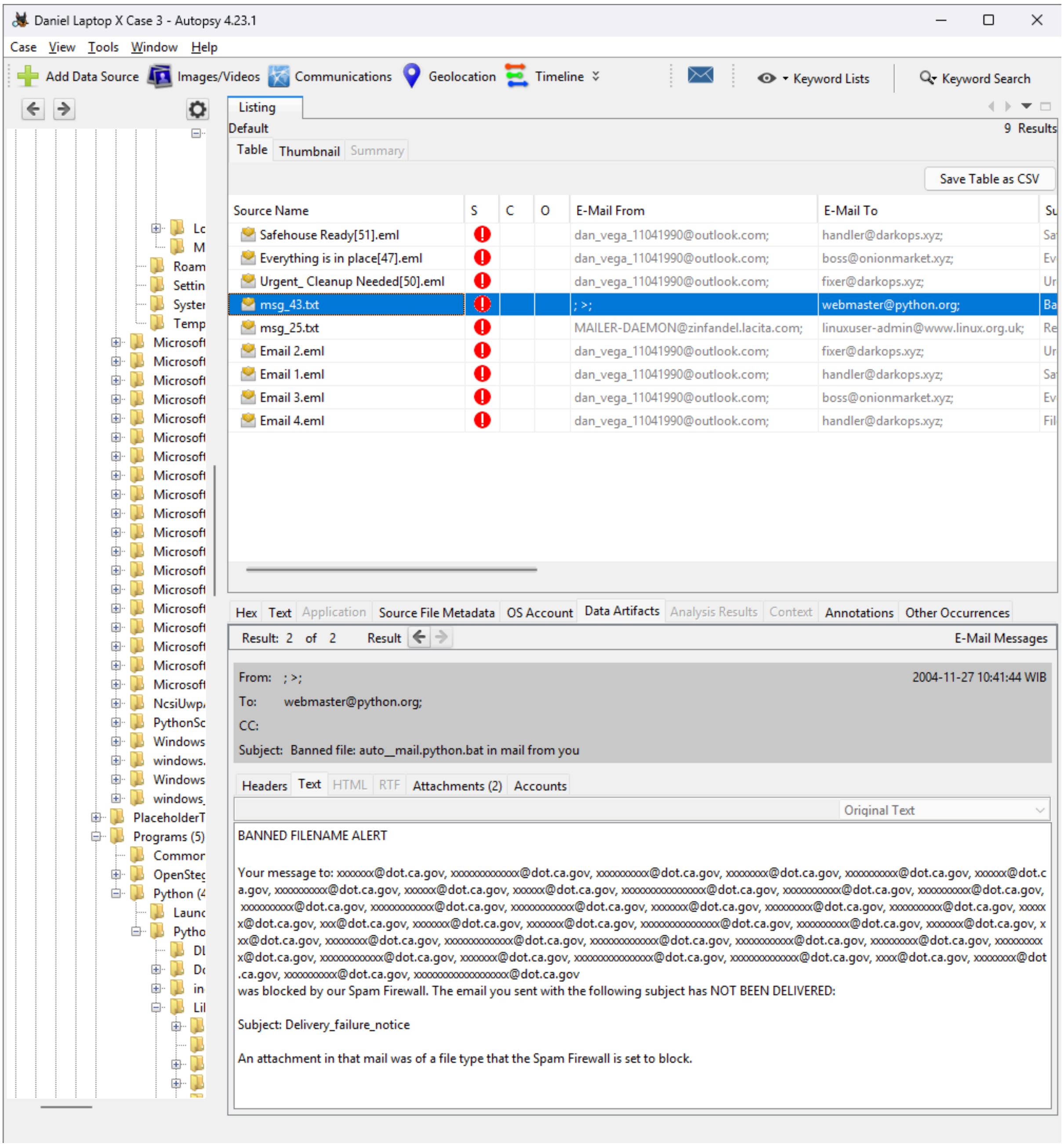
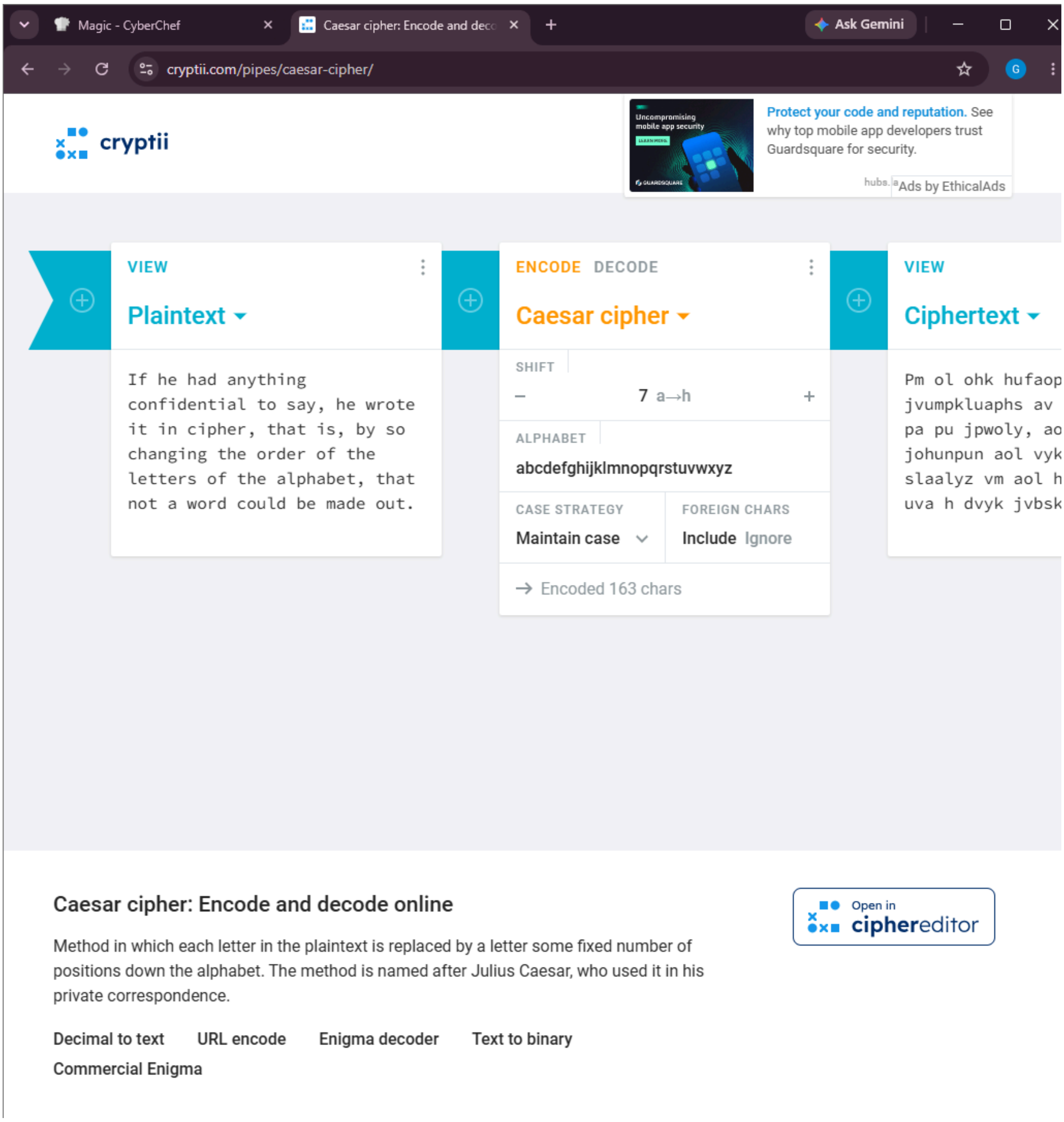
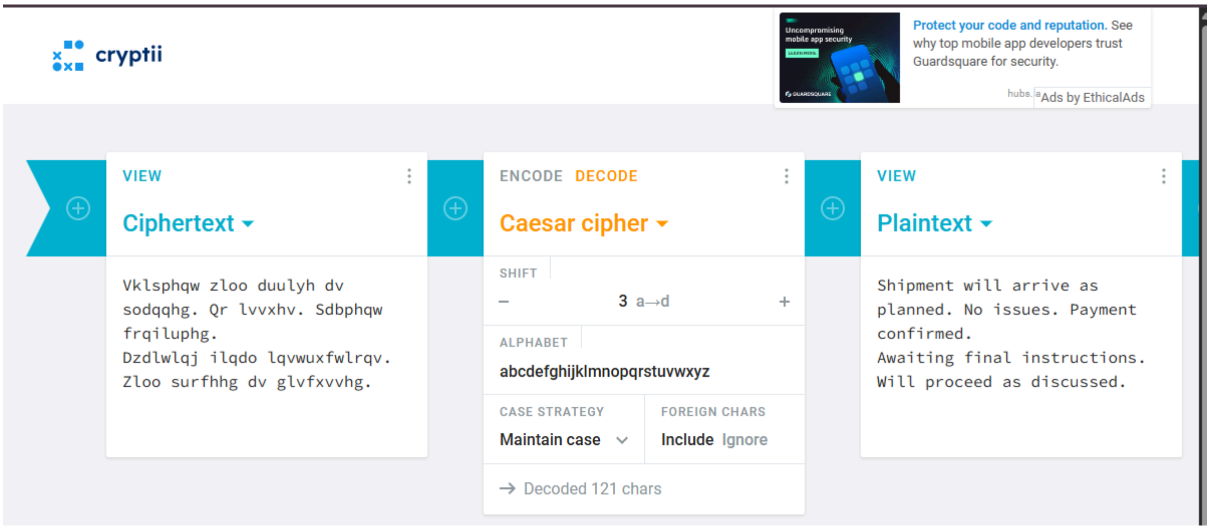
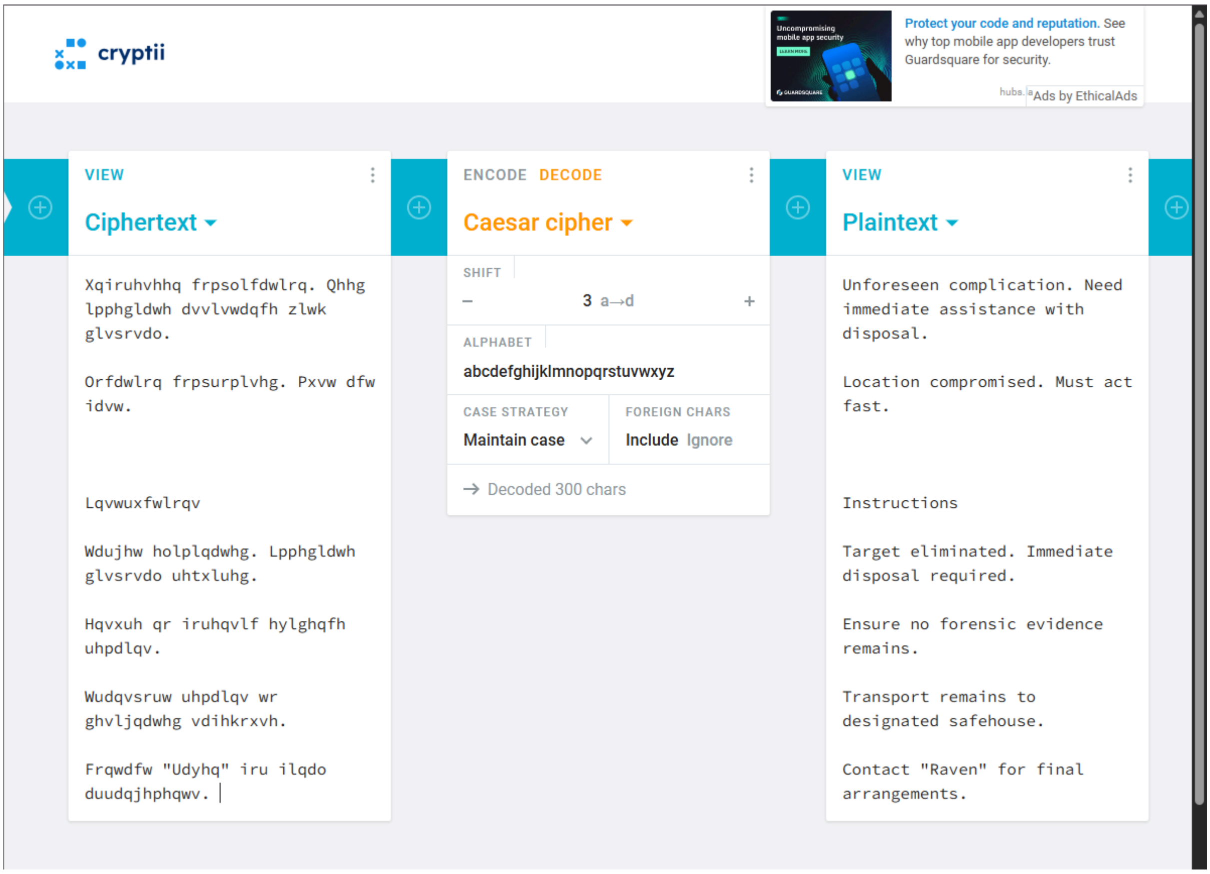
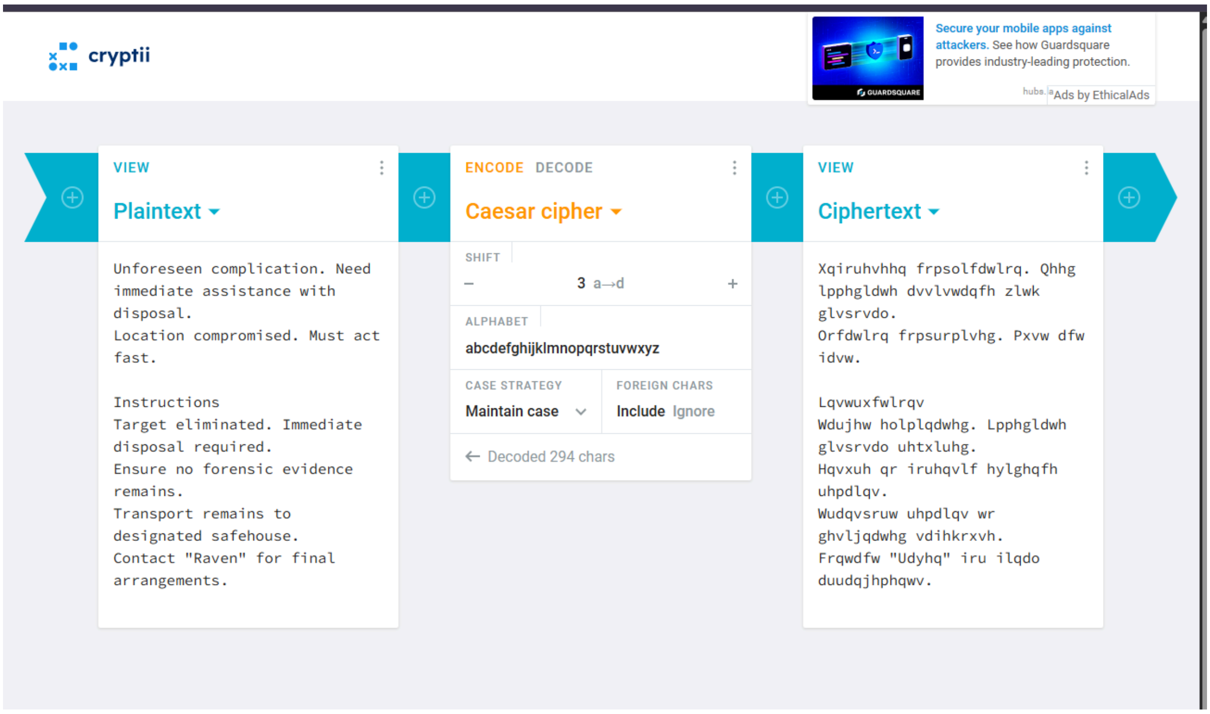


Figure 6.1.3d — Email 4.eml, msg_25.txt & msg_43.txt ("File Transfer – Urgent")


Magic - CyberChef
Caesar cipher: Encode and deco
Ask Gemini
cryptii.com/pipes/caesar-cipher/
cryptii
Protect your code and reputation. See why top mobile app developers trust Guardsquare for security.
Ads by EthicalAds
VIEW
Plaintext
If he had anything confidential to say, he wrote it in cipher, that is, by so changing the order of the letters of the alphabet, that not a word could be made out.
ENCODE DECODE
Caesar cipher
SHIFT
7 a→h
ALPHABET
abcdefghijklmnopqrstuvwxyz
CASE STRATEGY
Maintain case
FOREIGN CHARS
Include Ignore
Encoded 163 chars
VIEW
Ciphertext
Pm ol ohk hufaop
jvumpkluaphs av
pa pu jpwoly, ao
johunpun aol vyk
slaalyz vm aol h
uva h dvyk jvbsk
Caesar cipher: Encode and decode online
Method in which each letter in the plaintext is replaced by a letter some fixed number of positions down the alphabet. The method is named after Julius Caesar, who used it in his private correspondence.
Open in ciphereditor
Decimal to text
URL encode
Enigma decoder
Text to binary
Commercial Enigma


cryptii
Protect your code and reputation. See why top mobile app developers trust Guardsquare for security.
Ads by EthicalAds
VIEW
Ciphertext
Vklsphqw zloo duulyh dv sodqqhg. Qr lvvxhv. Sdbphqw frqiluphg.
Dzdlwlqj ilqdo lqvwuxfwlrqv.
Zloo surfhhg dv glvfxvvhg.
ENCODE DECODE
Caesar cipher
SHIFT
3 a→d
ALPHABET
abcdefghijklmnopqrstuvwxyz
CASE STRATEGY
Maintain case
FOREIGN CHARS
Include Ignore
Decoded 121 chars
VIEW
Plaintext
Shipment will arrive as planned. No issues. Payment confirmed.
Awaiting final instructions.
Will proceed as discussed.


cryptii
Protect your code and reputation. See why top mobile app developers trust Guardsquare for security.
Ads by EthicalAds
VIEW
Ciphertext
Xqiruhvhhq frpsolfdwlrq. Qhhg lpphgldwh dvvlvwdqfh zlwk glvsrvdo.
Orfdwlrq frpsurplvhg. Pxvw dfw idvw.
Lqvwuxfwlrqv
Wdujhw holplqdwhg. Lpphgldwh glvsrvdo uhtxluhg.
Hqvxuh qr iruhqvlf hylghqfh uhpdlqv.
Wudqvsruw uhpdlqv wr ghvljqdwhg vdihkrxvh.
Frqwdfw "Udyhq" iru ilqdo duudqjhphqwv.
ENCODE DECODE
Caesar cipher
SHIFT
3 a→d
ALPHABET
abcdefghijklmnopqrstuvwxyz
CASE STRATEGY
Maintain case
FOREIGN CHARS
Include Ignore
Decoded 300 chars
VIEW
Plaintext
Unforeseen complication. Need immediate assistance with disposal.
Location compromised. Must act fast.
Instructions
Target eliminated. Immediate disposal required.
Ensure no forensic evidence remains.
Transport remains to designated safehouse.
Contact "Raven" for final arrangements.


cryptii
Secure your mobile apps against attackers. See how Guardsquare provides industry-leading protection.
Ads by EthicalAds
VIEW
Plaintext
Unforeseen complication. Need immediate assistance with disposal.
Location compromised. Must act fast.
Instructions
Target eliminated. Immediate disposal required.
Ensure no forensic evidence remains.
Transport remains to designated safehouse.
Contact "Raven" for final arrangements.
ENCODE DECODE
Caesar cipher
SHIFT
3 a→d
ALPHABET
abcdefghijklmnopqrstuvwxyz
CASE STRATEGY
Maintain case
FOREIGN CHARS
Include Ignore
Decoded 294 chars
VIEW
Ciphertext
Xqiruhvhhq frpsolfdwlrq. Qhhg lpphgldwh dvvlvwdqfh zlwk glvsrvdo.
Orfdwlrq frpsurplvhg. Pxvw dfw idvw.
Lqvwuxfwlrqv
Wdujhw holplqdwhg. Lpphgldwh glvsrvdo uhtxluhg.
Hqvxuh qr iruhqvlf hylghqfh uhpdlqv.
Wudqvsruw uhpdlqv wr ghvljqdwhg vdihkrxvh.
Frqwdfw "Udyhq" iru ilqdo duudqjhphqwv.

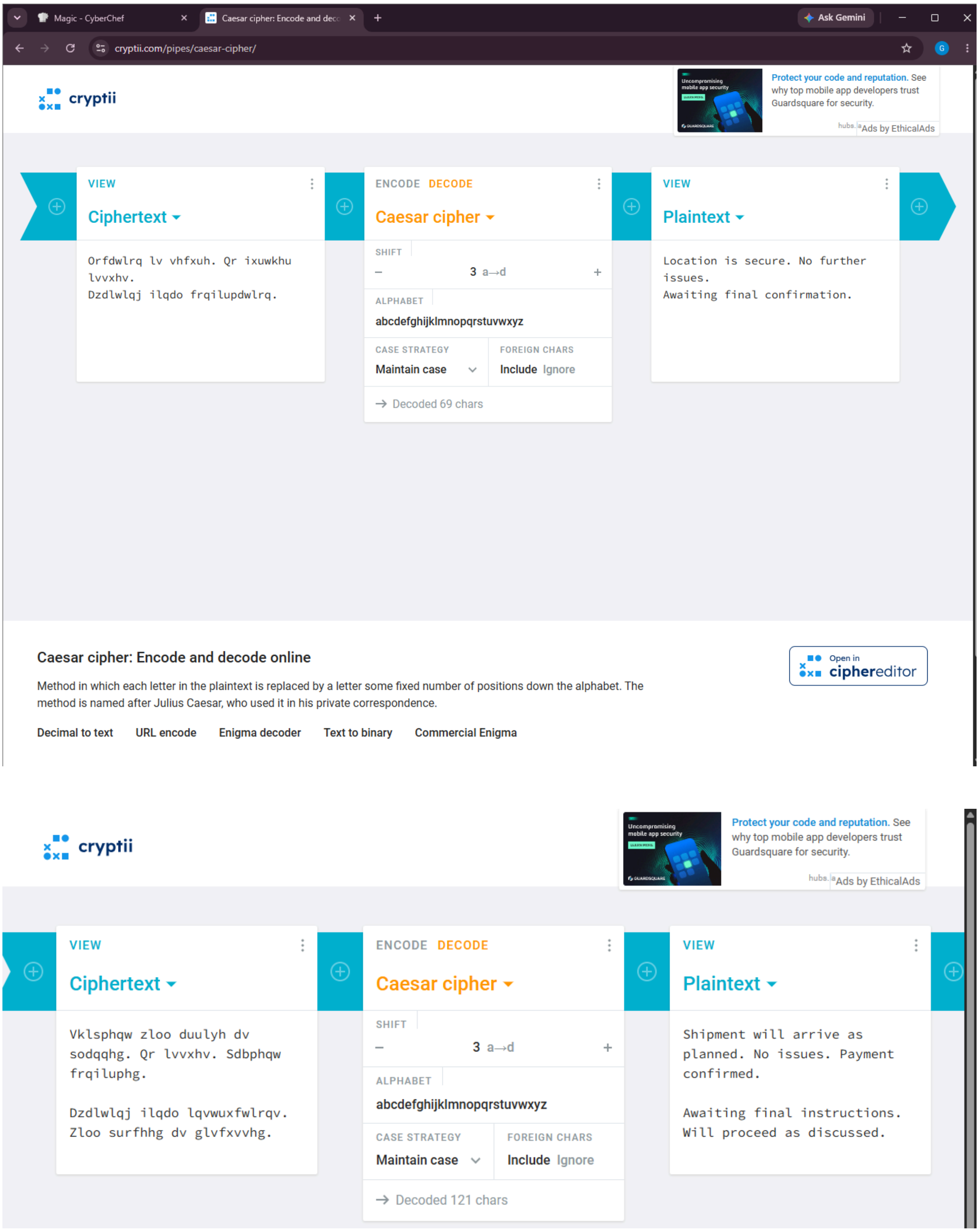
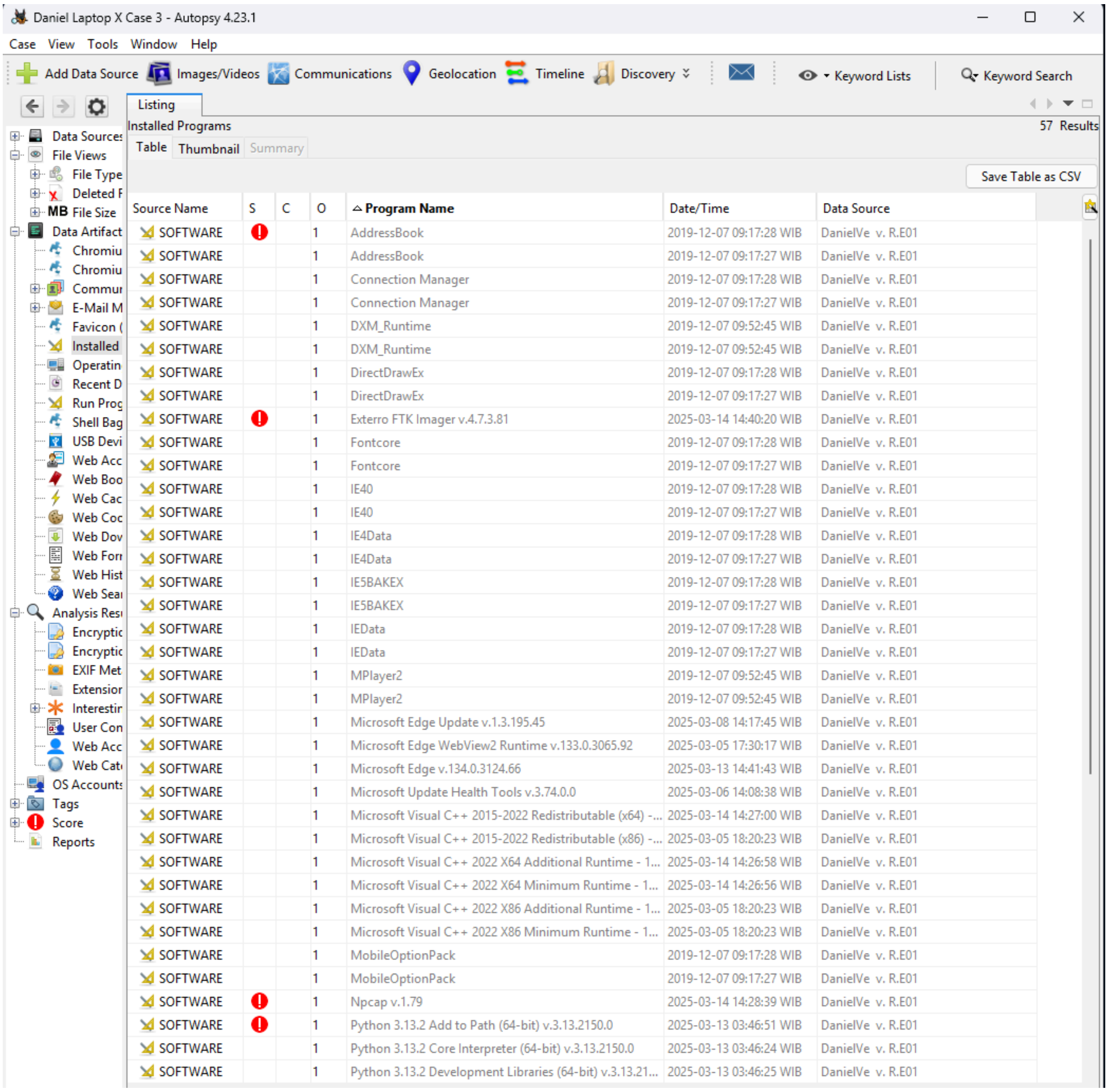
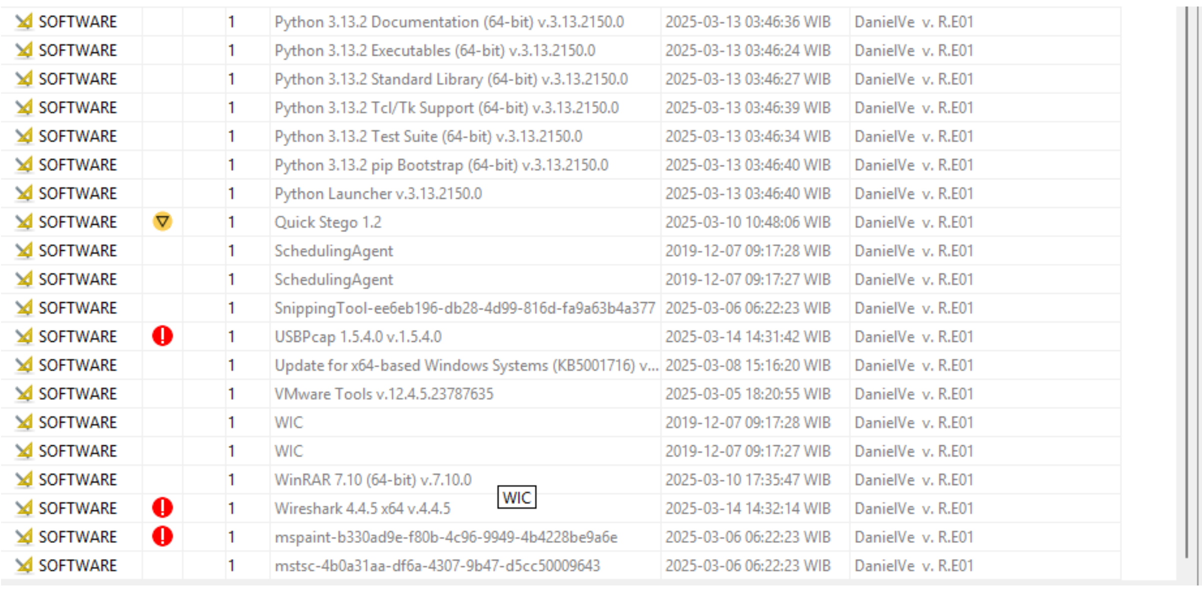


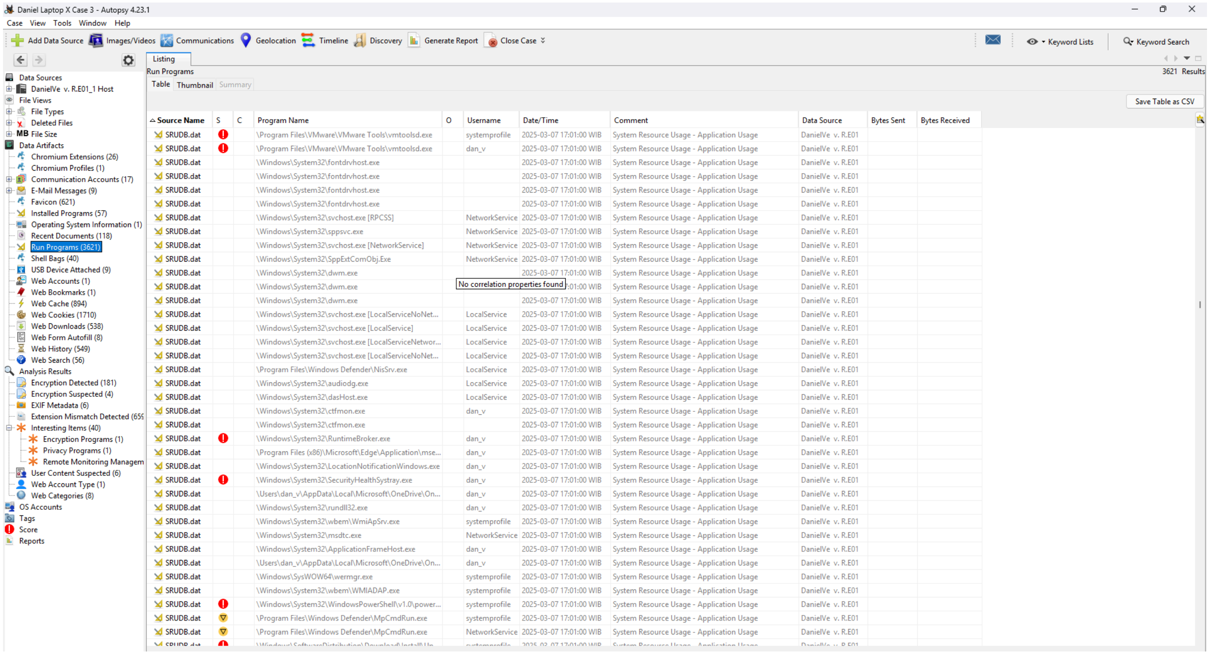
Figure 6.1.3e — Decoded Ciphertext in each Email sources

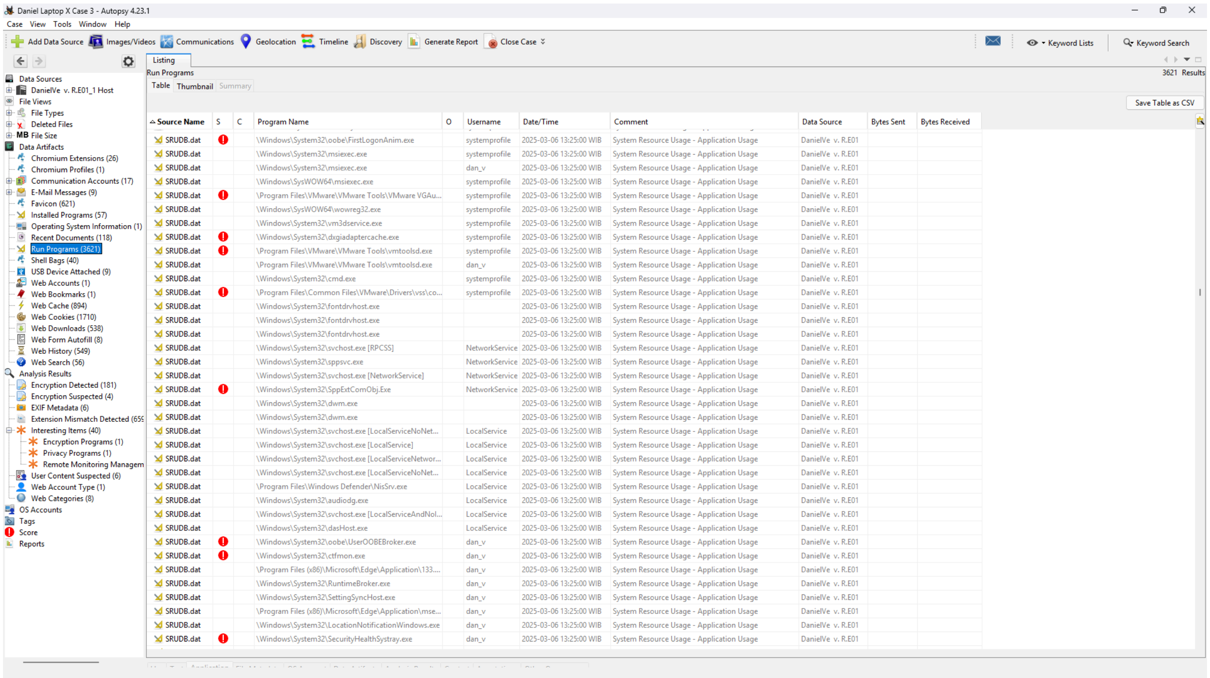




Source Name	S	C	O	Program Name	Date/Time	Data Source
SOFTWARE			1	AddressBook	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	AddressBook	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	Connection Manager	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	Connection Manager	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	DXM_Runtime	2019-12-07 09:52:45 WIB	DanielVe v. R.E01
SOFTWARE			1	DXM_Runtime	2019-12-07 09:52:45 WIB	DanielVe v. R.E01
SOFTWARE			1	DirectDrawEx	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	DirectDrawEx	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	Exterro FTK Imager v.4.7.3.81	2025-03-14 14:40:20 WIB	DanielVe v. R.E01
SOFTWARE			1	Fontcore	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	Fontcore	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	IE40	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	IE40	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	IE4Data	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	IE4Data	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	IE5BAKEX	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	IE5BAKEX	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	IEData	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	IEData	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	MPlayer2	2019-12-07 09:52:45 WIB	DanielVe v. R.E01
SOFTWARE			1	MPlayer2	2019-12-07 09:52:45 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Edge Update v.1.3.195.45	2025-03-08 14:17:45 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Edge WebView2 Runtime v.133.0.3065.92	2025-03-05 17:30:17 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Edge v.134.0.3124.66	2025-03-13 14:41:43 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Update Health Tools v.3.74.0.0	2025-03-06 14:08:38 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Visual C++ 2015-2022 Redistributable (x64) -...	2025-03-14 14:27:00 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Visual C++ 2015-2022 Redistributable (x86) -...	2025-03-05 18:20:23 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Visual C++ 2022 X64 Additional Runtime - 1...	2025-03-14 14:26:58 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Visual C++ 2022 X64 Minimum Runtime - 1...	2025-03-14 14:26:56 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Visual C++ 2022 X86 Additional Runtime - 1...	2025-03-05 18:20:23 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Visual C++ 2022 X86 Minimum Runtime - 1...	2025-03-05 18:20:23 WIB	DanielVe v. R.E01
SOFTWARE			1	MobileOptionPack	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	MobileOptionPack	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	Npcap v.1.79	2025-03-14 14:28:39 WIB	DanielVe v. R.E01
SOFTWARE			1	Python 3.13.2 Add to Path (64-bit) v.3.13.2150.0	2025-03-13 03:46:51 WIB	DanielVe v. R.E01
SOFTWARE			1	Python 3.13.2 Core Interpreter (64-bit) v.3.13.2150.0	2025-03-13 03:46:24 WIB	DanielVe v. R.E01
SOFTWARE			1	Python 3.13.2 Development Libraries (64-bit) v.3.13.21...	2025-03-13 03:46:25 WIB	DanielVe v. R.E01
SOFTWARE			1	Python 3.13.2 Documentation (64-bit) v.3.13.2150.0	2025-03-13 03:46:36 WIB	DanielVe v. R.E01
SOFTWARE			1	Python 3.13.2 Executables (64-bit) v.3.13.2150.0	2025-03-13 03:46:24 WIB	DanielVe v. R.E01
SOFTWARE			1	Python 3.13.2 Standard Library (64-bit) v.3.13.2150.0	2025-03-13 03:46:27 WIB	DanielVe v. R.E01
SOFTWARE			1	Python 3.13.2 Tcl/Tk Support (64-bit) v.3.13.2150.0	2025-03-13 03:46:39 WIB	DanielVe v. R.E01
SOFTWARE			1	Python 3.13.2 Test Suite (64-bit) v.3.13.2150.0	2025-03-13 03:46:34 WIB	DanielVe v. R.E01
SOFTWARE			1	Python 3.13.2 pip Bootstrap (64-bit) v.3.13.2150.0	2025-03-13 03:46:40 WIB	DanielVe v. R.E01
SOFTWARE			1	Python Launcher v.3.13.2150.0	2025-03-13 03:46:40 WIB	DanielVe v. R.E01
SOFTWARE			1	Quick Stego 1.2	2025-03-10 10:48:06 WIB	DanielVe v. R.E01
SOFTWARE			1	SchedulingAgent	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	SchedulingAgent	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	SnippingTool-ee6eb196-db28-4d99-816d-fa9a63b4a377	2025-03-06 06:22:23 WIB	DanielVe v. R.E01
SOFTWARE			1	USBPcap 1.5.4.0 v.1.5.4.0	2025-03-14 14:31:42 WIB	DanielVe v. R.E01
SOFTWARE			1	Update for x64-based Windows Systems (KB5001716) v...	2025-03-08 15:16:20 WIB	DanielVe v. R.E01
SOFTWARE			1	VMware Tools v.12.4.5.23787635	2025-03-05 18:20:55 WIB	DanielVe v. R.E01
SOFTWARE			1	WIC	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	WIC	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	WinRAR 7.10 (64-bit) v.7.10.0	2025-03-10 17:35:47 WIB	DanielVe v. R.E01
SOFTWARE			1	Wireshark 4.4.5 x64 v.4.4.5	2025-03-14 14:32:14 WIB	DanielVe v. R.E01
SOFTWARE			1	mspaint-b330ad9e-f80b-4c96-9949-4b4228be9a6e	2025-03-06 06:22:23 WIB	DanielVe v. R.E01
SOFTWARE			1	mstsc-4b0a31aa-df6a-4307-9b47-d5cc50009643	2025-03-06 06:22:23 WIB	DanielVe v. R.E01

Figure 6.1.4b — Installed applications identified during forensic.


Daniel Laptop X Case 3 - Autopsy 4.23.1
Run Programs
3621 Results
Save Table as CSV
No correlation properties found


Daniel Laptop X Case 3 - Autopsy 4.23.1
Run Programs
3621 Results
Save Table as CSV

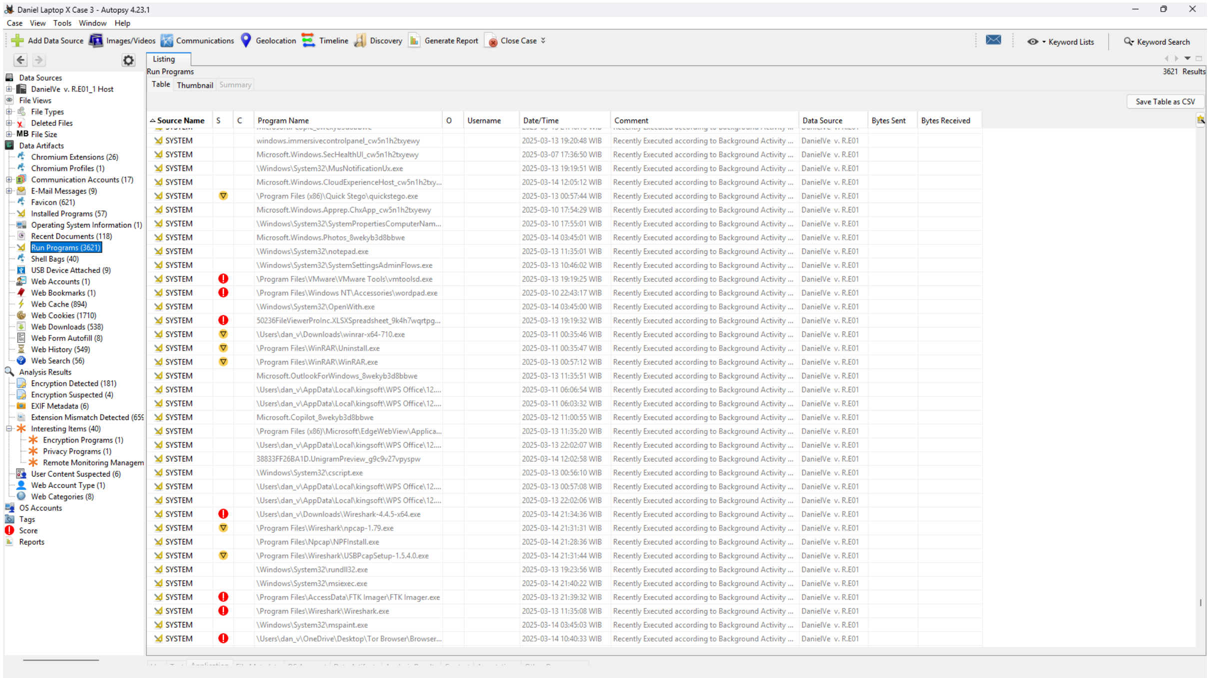
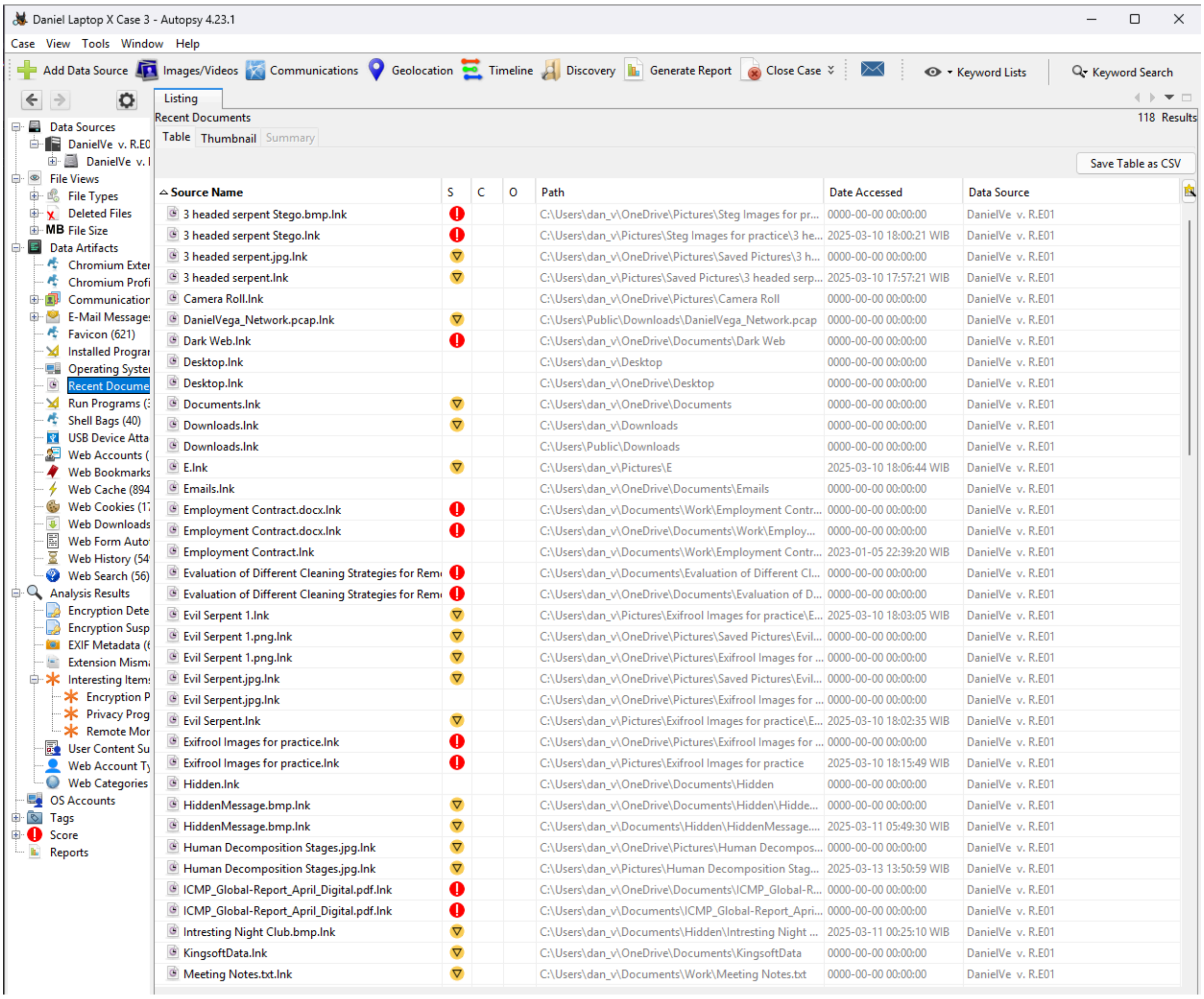
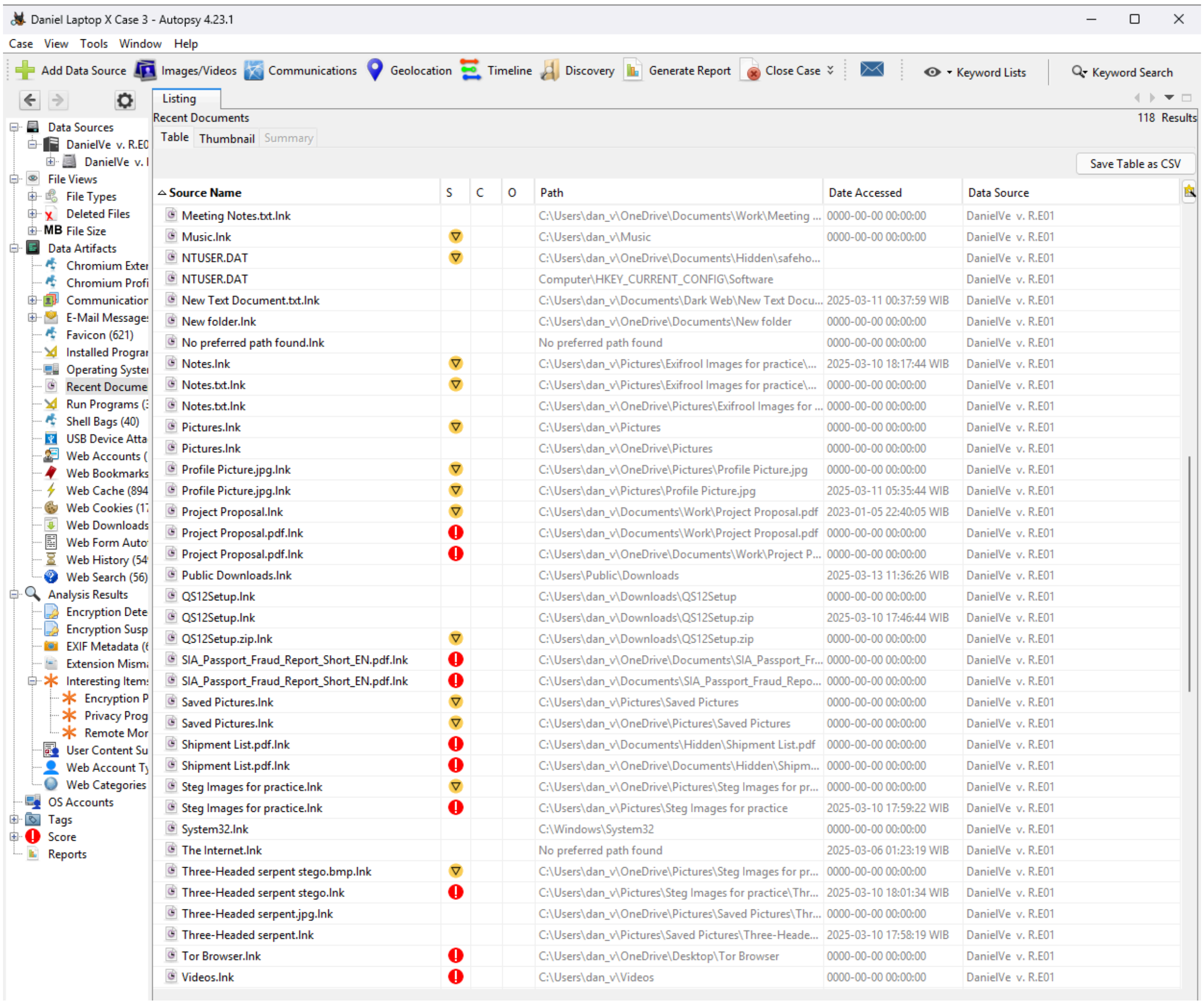
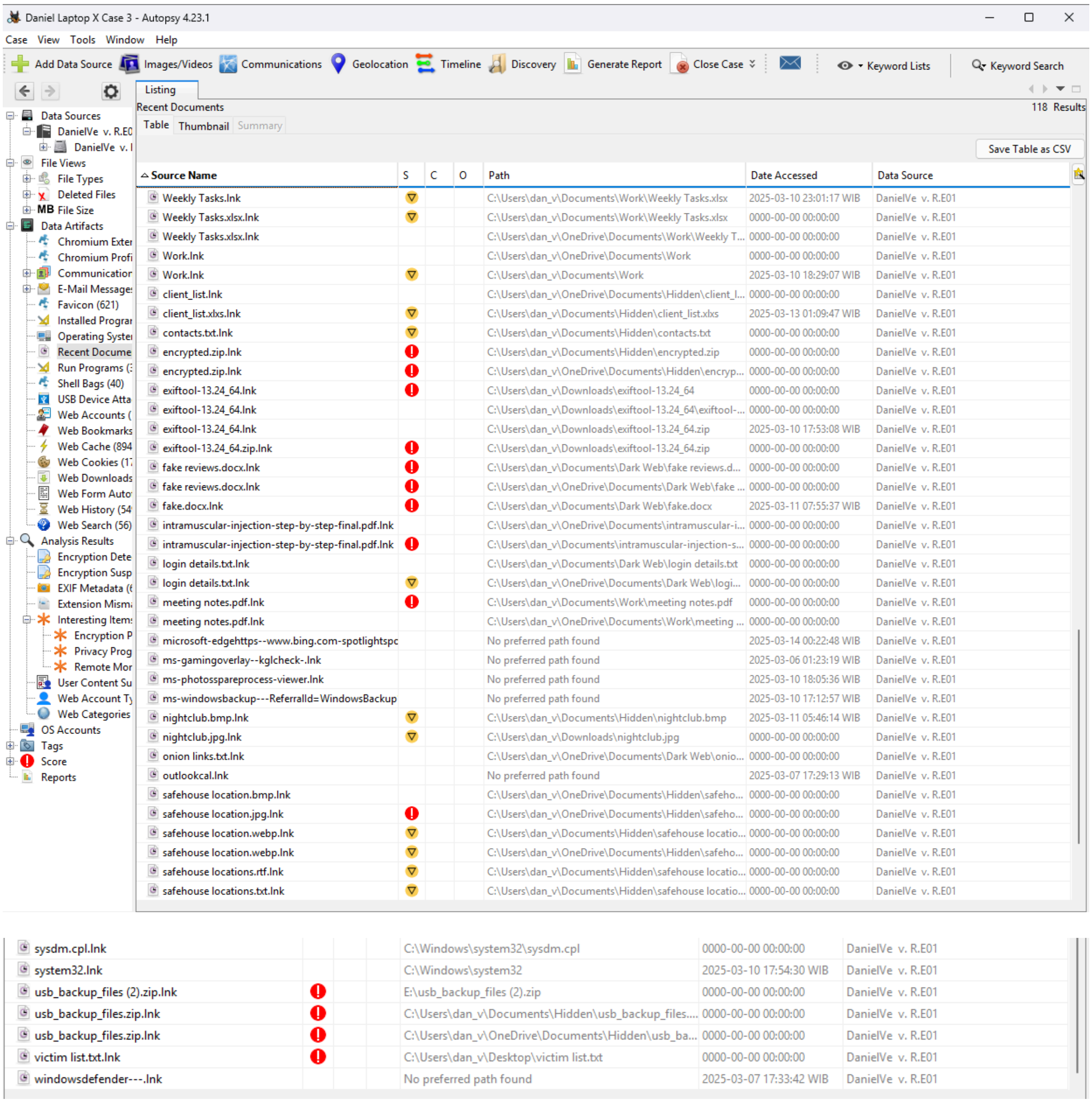


Figure 6.1.4c — Executed program artifacts recovered from the forensic image.


Daniel Laptop X Case 3 - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case Keyword Lists Keyword Search
Listing
Recent Documents
118 Results
Table Thumbnail Summary
Save Table as CSV
Source Name S C O Path Date Accessed Data Source
3 headed serpent Stego.bmp.lnk C:\Users\dan_v\OneDrive\Pictures\Steg Images for pr... 0000-00-00 00:00:00 DanielVe v. R.E01
3 headed serpent Stego.lnk C:\Users\dan_v\Pictures\Steg Images for practice\3 he... 2025-03-10 18:00:21 WIB DanielVe v. R.E01
3 headed serpent.jpg.lnk C:\Users\dan_v\OneDrive\Pictures\Saved Pictures\3 h... 0000-00-00 00:00:00 DanielVe v. R.E01
3 headed serpent.lnk C:\Users\dan_v\Pictures\Saved Pictures\3 headed serp... 2025-03-10 17:57:21 WIB DanielVe v. R.E01
Camera Roll.lnk C:\Users\dan_v\OneDrive\Pictures\Camera Roll 0000-00-00 00:00:00 DanielVe v. R.E01
DanielVega_Network.pcap.lnk C:\Users\Public\Downloads\DanielVega_Network.pcap 0000-00-00 00:00:00 DanielVe v. R.E01
Dark Web.lnk C:\Users\dan_v\OneDrive\Documents\Dark Web 0000-00-00 00:00:00 DanielVe v. R.E01
Desktop.lnk C:\Users\dan_v\Desktop 0000-00-00 00:00:00 DanielVe v. R.E01
Desktop.lnk C:\Users\dan_v\OneDrive\Desktop 0000-00-00 00:00:00 DanielVe v. R.E01
Documents.lnk C:\Users\dan_v\OneDrive\Documents 0000-00-00 00:00:00 DanielVe v. R.E01
Downloads.lnk C:\Users\dan_v\Downloads 0000-00-00 00:00:00 DanielVe v. R.E01
Downloads.lnk C:\Users\Public\Downloads 0000-00-00 00:00:00 DanielVe v. R.E01
E.lnk C:\Users\dan_v\Pictures\E 2025-03-10 18:06:44 WIB DanielVe v. R.E01
Emails.lnk C:\Users\dan_v\OneDrive\Documents\Emails 0000-00-00 00:00:00 DanielVe v. R.E01
Employment Contract.docx.lnk C:\Users\dan_v\Documents\Work\Employment Contr... 0000-00-00 00:00:00 DanielVe v. R.E01
Employment Contract.docx.lnk C:\Users\dan_v\OneDrive\Documents\Work\Employ... 0000-00-00 00:00:00 DanielVe v. R.E01
Employment Contract.lnk C:\Users\dan_v\Documents\Work\Employment Contr... 2023-01-05 22:39:20 WIB DanielVe v. R.E01
Evaluation of Different Cleaning Strategies for Rem C:\Users\dan_v\Documents\Evaluation of Different Cl... 0000-00-00 00:00:00 DanielVe v. R.E01
Evaluation of Different Cleaning Strategies for Rem C:\Users\dan_v\OneDrive\Documents\Evaluation of D... 0000-00-00 00:00:00 DanielVe v. R.E01
Evil Serpent 1.lnk C:\Users\dan_v\Pictures\Exifrool Images for practice\E... 2025-03-10 18:03:05 WIB DanielVe v. R.E01
Evil Serpent 1.png.lnk C:\Users\dan_v\OneDrive\Pictures\Saved Pictures\Evil... 0000-00-00 00:00:00 DanielVe v. R.E01
Evil Serpent 1.png.lnk C:\Users\dan_v\OneDrive\Pictures\Exifrool Images for ... 0000-00-00 00:00:00 DanielVe v. R.E01
Evil Serpent.jpg.lnk C:\Users\dan_v\OneDrive\Pictures\Saved Pictures\Evil... 0000-00-00 00:00:00 DanielVe v. R.E01
Evil Serpent.jpg.lnk C:\Users\dan_v\OneDrive\Pictures\Exifrool Images for ... 0000-00-00 00:00:00 DanielVe v. R.E01
Evil Serpent.lnk C:\Users\dan_v\Pictures\Exifrool Images for practice\E... 2025-03-10 18:02:35 WIB DanielVe v. R.E01
Exifrool Images for practice.lnk C:\Users\dan_v\OneDrive\Pictures\Exifrool Images for ... 0000-00-00 00:00:00 DanielVe v. R.E01
Exifrool Images for practice.lnk C:\Users\dan_v\Pictures\Exifrool Images for practice 2025-03-10 18:15:49 WIB DanielVe v. R.E01
Hidden.lnk C:\Users\dan_v\OneDrive\Documents\Hidden 0000-00-00 00:00:00 DanielVe v. R.E01
HiddenMessage.bmp.lnk C:\Users\dan_v\OneDrive\Documents\Hidden\Hidde... 0000-00-00 00:00:00 DanielVe v. R.E01
HiddenMessage.bmp.lnk C:\Users\dan_v\Documents\Hidden\HiddenMessage.... 2025-03-11 05:49:30 WIB DanielVe v. R.E01
Human Decomposition Stages.jpg.lnk C:\Users\dan_v\OneDrive\Pictures\Human Decompos... 0000-00-00 00:00:00 DanielVe v. R.E01
Human Decomposition Stages.jpg.lnk C:\Users\dan_v\Pictures\Human Decomposition Stag... 2025-03-13 13:50:59 WIB DanielVe v. R.E01
ICMP_Global-Report_April_Digital.pdf.lnk C:\Users\dan_v\OneDrive\Documents\ICMP_Global-R... 0000-00-00 00:00:00 DanielVe v. R.E01
ICMP_Global-Report_April_Digital.pdf.lnk C:\Users\dan_v\Documents\ICMP_Global-Report_Apri... 0000-00-00 00:00:00 DanielVe v. R.E01
Intresting Night Club.bmp.lnk C:\Users\dan_v\Documents\Hidden\Intresting Night ... 2025-03-11 00:25:10 WIB DanielVe v. R.E01
KingsoftData.lnk C:\Users\dan_v\OneDrive\Documents\KingsoftData 0000-00-00 00:00:00 DanielVe v. R.E01
Meeting Notes.txt.lnk C:\Users\dan_v\Documents\Work\Meeting Notes.txt 0000-00-00 00:00:00 DanielVe v. R.E01


Daniel Laptop X Case 3 - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists
Keyword Search
Data Sources
File Views
File Types
Deleted Files
MB File Size
Data Artifacts
Favicon (621)
Shell Bags (40)
Web Search (56)
Analysis Results
OS Accounts
Tags
Score
Reports
Listing
Recent Documents
118 Results
Table Thumbnail Summary
Save Table as CSV
Source Name S C O Path Date Accessed Data Source
Meeting Notes.txt.lnk C:\Users\dan_v\OneDrive\Documents\Work\Meeting ... 0000-00-00 00:00:00 DanielVe v. R.E01
Music.lnk C:\Users\dan_v\Music 0000-00-00 00:00:00 DanielVe v. R.E01
NTUSER.DAT C:\Users\dan_v\OneDrive\Documents\Hidden\safeho... DanielVe v. R.E01
NTUSER.DAT Computer\HKEY_CURRENT_CONFIG\Software DanielVe v. R.E01
New Text Document.txt.lnk C:\Users\dan_v\Documents\Dark Web\New Text Docu... 2025-03-11 00:37:59 WIB DanielVe v. R.E01
New folder.lnk C:\Users\dan_v\OneDrive\Documents\New folder 0000-00-00 00:00:00 DanielVe v. R.E01
No preferred path found.lnk No preferred path found 0000-00-00 00:00:00 DanielVe v. R.E01
Notes.lnk C:\Users\dan_v\Pictures\Exifrool Images for practice\... 2025-03-10 18:17:44 WIB DanielVe v. R.E01
Notes.txt.lnk C:\Users\dan_v\Pictures\Exifrool Images for practice\... 0000-00-00 00:00:00 DanielVe v. R.E01
Notes.txt.lnk C:\Users\dan_v\OneDrive\Pictures\Exifrool Images for ... 0000-00-00 00:00:00 DanielVe v. R.E01
Pictures.lnk C:\Users\dan_v\Pictures 0000-00-00 00:00:00 DanielVe v. R.E01
Pictures.lnk C:\Users\dan_v\OneDrive\Pictures 0000-00-00 00:00:00 DanielVe v. R.E01
Profile Picture.jpg.lnk C:\Users\dan_v\OneDrive\Pictures\Profile Picture.jpg 0000-00-00 00:00:00 DanielVe v. R.E01
Profile Picture.jpg.lnk C:\Users\dan_v\Pictures\Profile Picture.jpg 2025-03-11 05:35:44 WIB DanielVe v. R.E01
Project Proposal.lnk C:\Users\dan_v\Documents\Work\Project Proposal.pdf 2023-01-05 22:40:05 WIB DanielVe v. R.E01
Project Proposal.pdf.lnk C:\Users\dan_v\Documents\Work\Project Proposal.pdf 0000-00-00 00:00:00 DanielVe v. R.E01
Project Proposal.pdf.lnk C:\Users\dan_v\OneDrive\Documents\Work\Project P... 0000-00-00 00:00:00 DanielVe v. R.E01
Public Downloads.lnk C:\Users\Public\Downloads 2025-03-13 11:36:26 WIB DanielVe v. R.E01
QS12Setup.lnk C:\Users\dan_v\Downloads\QS12Setup 0000-00-00 00:00:00 DanielVe v. R.E01
QS12Setup.lnk C:\Users\dan_v\Downloads\QS12Setup.zip 2025-03-10 17:46:44 WIB DanielVe v. R.E01
QS12Setup.zip.lnk C:\Users\dan_v\Downloads\QS12Setup.zip 0000-00-00 00:00:00 DanielVe v. R.E01
SIA_Passport_Fraud_Report_Short_EN.pdf.lnk C:\Users\dan_v\OneDrive\Documents\SIA_Passport_Fr... 0000-00-00 00:00:00 DanielVe v. R.E01
SIA_Passport_Fraud_Report_Short_EN.pdf.lnk C:\Users\dan_v\Documents\SIA_Passport_Fraud_Repo... 0000-00-00 00:00:00 DanielVe v. R.E01
Saved Pictures.lnk C:\Users\dan_v\Pictures\Saved Pictures 0000-00-00 00:00:00 DanielVe v. R.E01
Saved Pictures.lnk C:\Users\dan_v\OneDrive\Pictures\Saved Pictures 0000-00-00 00:00:00 DanielVe v. R.E01
Shipment List.pdf.lnk C:\Users\dan_v\Documents\Hidden\Shipment List.pdf 0000-00-00 00:00:00 DanielVe v. R.E01
Shipment List.pdf.lnk C:\Users\dan_v\OneDrive\Documents\Hidden\Shipm... 0000-00-00 00:00:00 DanielVe v. R.E01
Steg Images for practice.lnk C:\Users\dan_v\OneDrive\Pictures\Steg Images for pr... 0000-00-00 00:00:00 DanielVe v. R.E01
Steg Images for practice.lnk C:\Users\dan_v\Pictures\Steg Images for practice 2025-03-10 17:59:22 WIB DanielVe v. R.E01
System32.lnk C:\Windows\System32 0000-00-00 00:00:00 DanielVe v. R.E01
The Internet.lnk No preferred path found 2025-03-06 01:23:19 WIB DanielVe v. R.E01
Three-Headed serpent stego.bmp.lnk C:\Users\dan_v\OneDrive\Pictures\Steg Images for pr... 0000-00-00 00:00:00 DanielVe v. R.E01
Three-Headed serpent stego.lnk C:\Users\dan_v\Pictures\Steg Images for practice\Thr... 2025-03-10 18:01:34 WIB DanielVe v. R.E01
Three-Headed serpent.jpg.lnk C:\Users\dan_v\OneDrive\Pictures\Saved Pictures\Thr... 0000-00-00 00:00:00 DanielVe v. R.E01
Three-Headed serpent.lnk C:\Users\dan_v\Pictures\Saved Pictures\Three-Heade... 2025-03-10 17:58:19 WIB DanielVe v. R.E01
Tor Browser.lnk C:\Users\dan_v\OneDrive\Desktop\Tor Browser 0000-00-00 00:00:00 DanielVe v. R.E01
Videos.lnk C:\Users\dan_v\Videos 0000-00-00 00:00:00 DanielVe v. R.E01



Recent Documents — 118 Results

Source Name	S	C	O	Path	Date Accessed	Data Source
Weekly Tasks.lnk				C:\Users\dan_v\Documents\Work\Weekly Tasks.xlsx	2025-03-10 23:01:17 WIB	DanielVe v. R.E01
Weekly Tasks.xlsx.lnk				C:\Users\dan_v\Documents\Work\Weekly Tasks.xlsx	0000-00-00 00:00:00	DanielVe v. R.E01
Weekly Tasks.xlsx.lnk				C:\Users\dan_v\OneDrive\Documents\Work\Weekly T...	0000-00-00 00:00:00	DanielVe v. R.E01
Work.lnk				C:\Users\dan_v\OneDrive\Documents\Work	0000-00-00 00:00:00	DanielVe v. R.E01
Work.lnk				C:\Users\dan_v\Documents\Work	2025-03-10 18:29:07 WIB	DanielVe v. R.E01
client_list.lnk				C:\Users\dan_v\OneDrive\Documents\Hidden\client_l...	0000-00-00 00:00:00	DanielVe v. R.E01
client_list.xlxs.lnk				C:\Users\dan_v\Documents\Hidden\client_list.xlxs	2025-03-13 01:09:47 WIB	DanielVe v. R.E01
contacts.txt.lnk				C:\Users\dan_v\Documents\Hidden\contacts.txt	0000-00-00 00:00:00	DanielVe v. R.E01
encrypted.zip.lnk				C:\Users\dan_v\Documents\Hidden\encrypted.zip	0000-00-00 00:00:00	DanielVe v. R.E01
encrypted.zip.lnk				C:\Users\dan_v\OneDrive\Documents\Hidden\encryp...	0000-00-00 00:00:00	DanielVe v. R.E01
exiftool-13.24_64.lnk				C:\Users\dan_v\Downloads\exiftool-13.24_64	0000-00-00 00:00:00	DanielVe v. R.E01
exiftool-13.24_64.lnk				C:\Users\dan_v\Downloads\exiftool-13.24_64\exiftool-...	0000-00-00 00:00:00	DanielVe v. R.E01
exiftool-13.24_64.lnk				C:\Users\dan_v\Downloads\exiftool-13.24_64.zip	2025-03-10 17:53:08 WIB	DanielVe v. R.E01
exiftool-13.24_64.zip.lnk				C:\Users\dan_v\Downloads\exiftool-13.24_64.zip	0000-00-00 00:00:00	DanielVe v. R.E01
fake reviews.docx.lnk				C:\Users\dan_v\Documents\Dark Web\fake reviews.d...	0000-00-00 00:00:00	DanielVe v. R.E01
fake reviews.docx.lnk				C:\Users\dan_v\OneDrive\Documents\Dark Web\fake ...	0000-00-00 00:00:00	DanielVe v. R.E01
fake.docx.lnk				C:\Users\dan_v\Documents\Dark Web\fake.docx	2025-03-11 07:55:37 WIB	DanielVe v. R.E01
intramuscular-injection-step-by-step-final.pdf.lnk				C:\Users\dan_v\OneDrive\Documents\intramuscular-i...	0000-00-00 00:00:00	DanielVe v. R.E01
intramuscular-injection-step-by-step-final.pdf.lnk				C:\Users\dan_v\Documents\intramuscular-injection-s...	0000-00-00 00:00:00	DanielVe v. R.E01
login details.txt.lnk				C:\Users\dan_v\Documents\Dark Web\login details.txt	0000-00-00 00:00:00	DanielVe v. R.E01
login details.txt.lnk				C:\Users\dan_v\OneDrive\Documents\Dark Web\logi...	0000-00-00 00:00:00	DanielVe v. R.E01
meeting notes.pdf.lnk				C:\Users\dan_v\Documents\Work\meeting notes.pdf	0000-00-00 00:00:00	DanielVe v. R.E01
meeting notes.pdf.lnk				C:\Users\dan_v\OneDrive\Documents\Work\meeting ...	0000-00-00 00:00:00	DanielVe v. R.E01
microsoft-edgehttps--www.bing.com-spotlightspc				No preferred path found	2025-03-14 00:22:48 WIB	DanielVe v. R.E01
ms-gamingoverlay--kglcheck-.lnk				No preferred path found	2025-03-06 01:23:19 WIB	DanielVe v. R.E01
ms-photosspareprocess-viewer.lnk				No preferred path found	2025-03-10 18:05:36 WIB	DanielVe v. R.E01
ms-windowsbackup---ReferralId=WindowsBackup				No preferred path found	2025-03-10 17:12:57 WIB	DanielVe v. R.E01
nightclub.bmp.lnk				C:\Users\dan_v\Documents\Hidden\nightclub.bmp	2025-03-11 05:46:14 WIB	DanielVe v. R.E01
nightclub.jpg.lnk				C:\Users\dan_v\Downloads\nightclub.jpg	0000-00-00 00:00:00	DanielVe v. R.E01
onion links.txt.lnk				C:\Users\dan_v\OneDrive\Documents\Dark Web\onio...	0000-00-00 00:00:00	DanielVe v. R.E01
outlookcal.lnk				No preferred path found	2025-03-07 17:29:13 WIB	DanielVe v. R.E01
safehouse location.bmp.lnk				C:\Users\dan_v\OneDrive\Documents\Hidden\safeho...	0000-00-00 00:00:00	DanielVe v. R.E01
safehouse location.jpg.lnk				C:\Users\dan_v\OneDrive\Documents\Hidden\safeho...	0000-00-00 00:00:00	DanielVe v. R.E01
safehouse location.webp.lnk				C:\Users\dan_v\Documents\Hidden\safehouse locatio...	0000-00-00 00:00:00	DanielVe v. R.E01
safehouse location.webp.lnk				C:\Users\dan_v\OneDrive\Documents\Hidden\safeho...	0000-00-00 00:00:00	DanielVe v. R.E01
safehouse locations.rtf.lnk				C:\Users\dan_v\Documents\Hidden\safehouse locatio...	0000-00-00 00:00:00	DanielVe v. R.E01
safehouse locations.txt.lnk				C:\Users\dan_v\Documents\Hidden\safehouse locatio...	0000-00-00 00:00:00	DanielVe v. R.E01
sysdm.cpl.lnk				C:\Windows\system32\sysdm.cpl	0000-00-00 00:00:00	DanielVe v. R.E01
system32.lnk				C:\Windows\system32	2025-03-10 17:54:30 WIB	DanielVe v. R.E01
usb_backup_files (2).zip.lnk				E:\usb_backup_files (2).zip	0000-00-00 00:00:00	DanielVe v. R.E01
usb_backup_files.zip.lnk				C:\Users\dan_v\Documents\Hidden\usb_backup_files....	0000-00-00 00:00:00	DanielVe v. R.E01
usb_backup_files.zip.lnk				C:\Users\dan_v\OneDrive\Documents\Hidden\usb_ba...	0000-00-00 00:00:00	DanielVe v. R.E01
victim list.txt.lnk				C:\Users\dan_v\Desktop\victim list.txt	0000-00-00 00:00:00	DanielVe v. R.E01
windowsdefender---.lnk				No preferred path found	2025-03-07 17:33:42 WIB	DanielVe v. R.E01

Figure 6.1.4d — Recently accessed document artifacts.

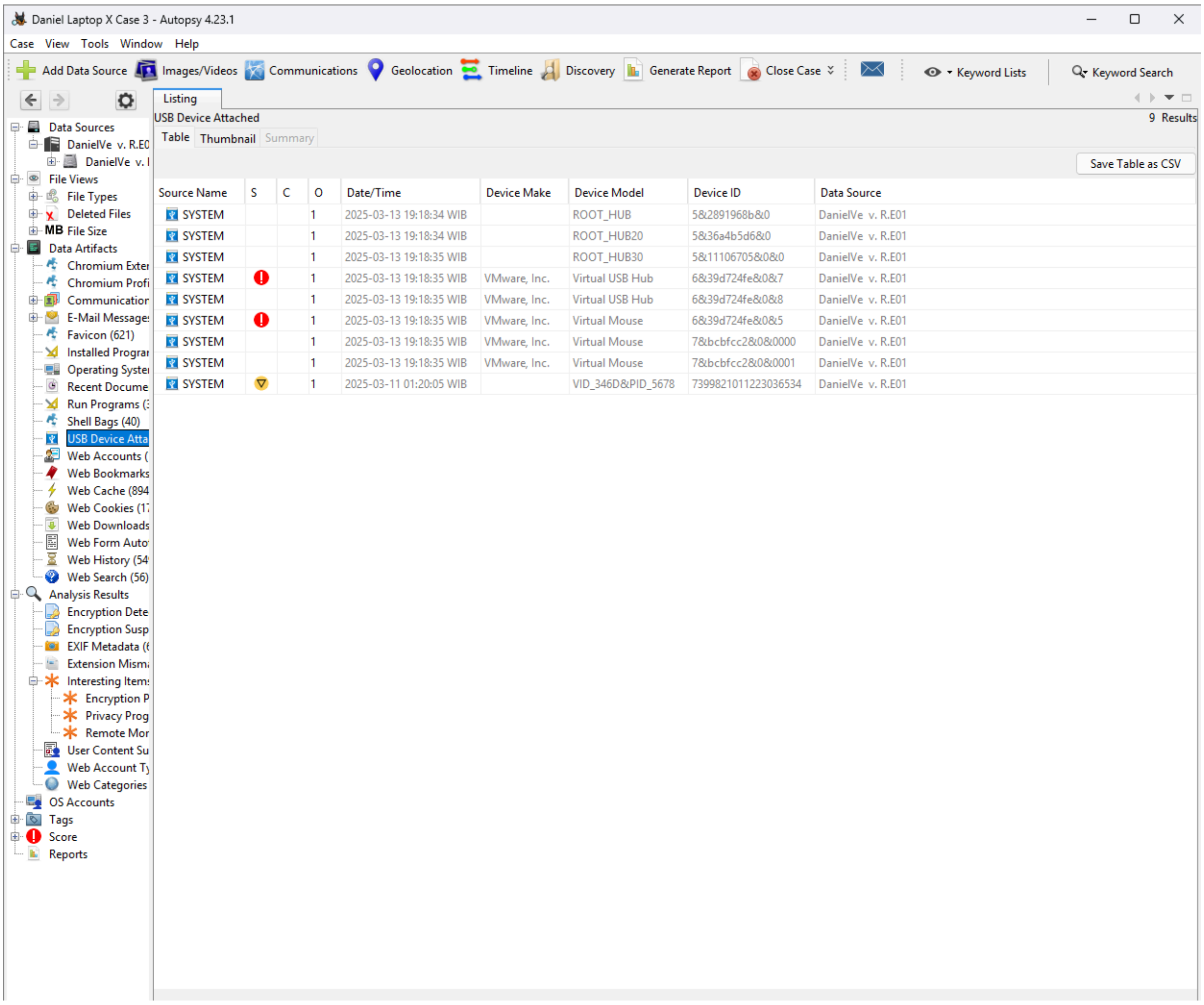
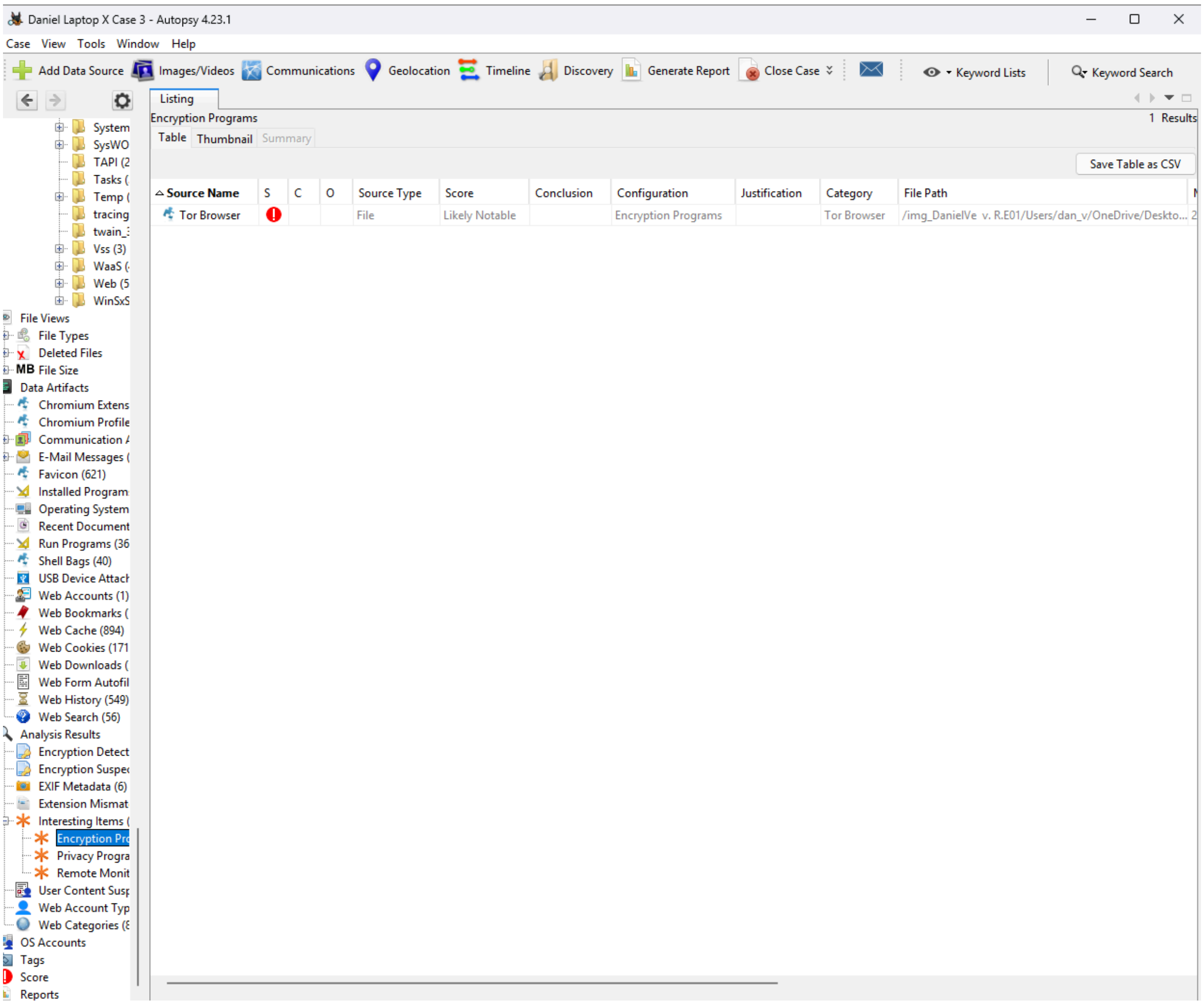


Figure 6.1.4e — USB device connection artifacts.


Daniel Laptop X Case 3 - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists
Keyword Search
Listing
Encryption Programs
1 Results
Table Thumbnail Summary
Save Table as CSV
Source Name S C O Source Type Score Conclusion Configuration Justification Category File Path
Tor Browser File Likely Notable Encryption Programs Tor Browser /img_DanielVe v. R.E01/Users/dan_v/OneDrive/Deskto...
File Views
File Types
Deleted Files
File Size
Data Artifacts
Analysis Results
Interesting Items
OS Accounts
Tags
Score
Reports

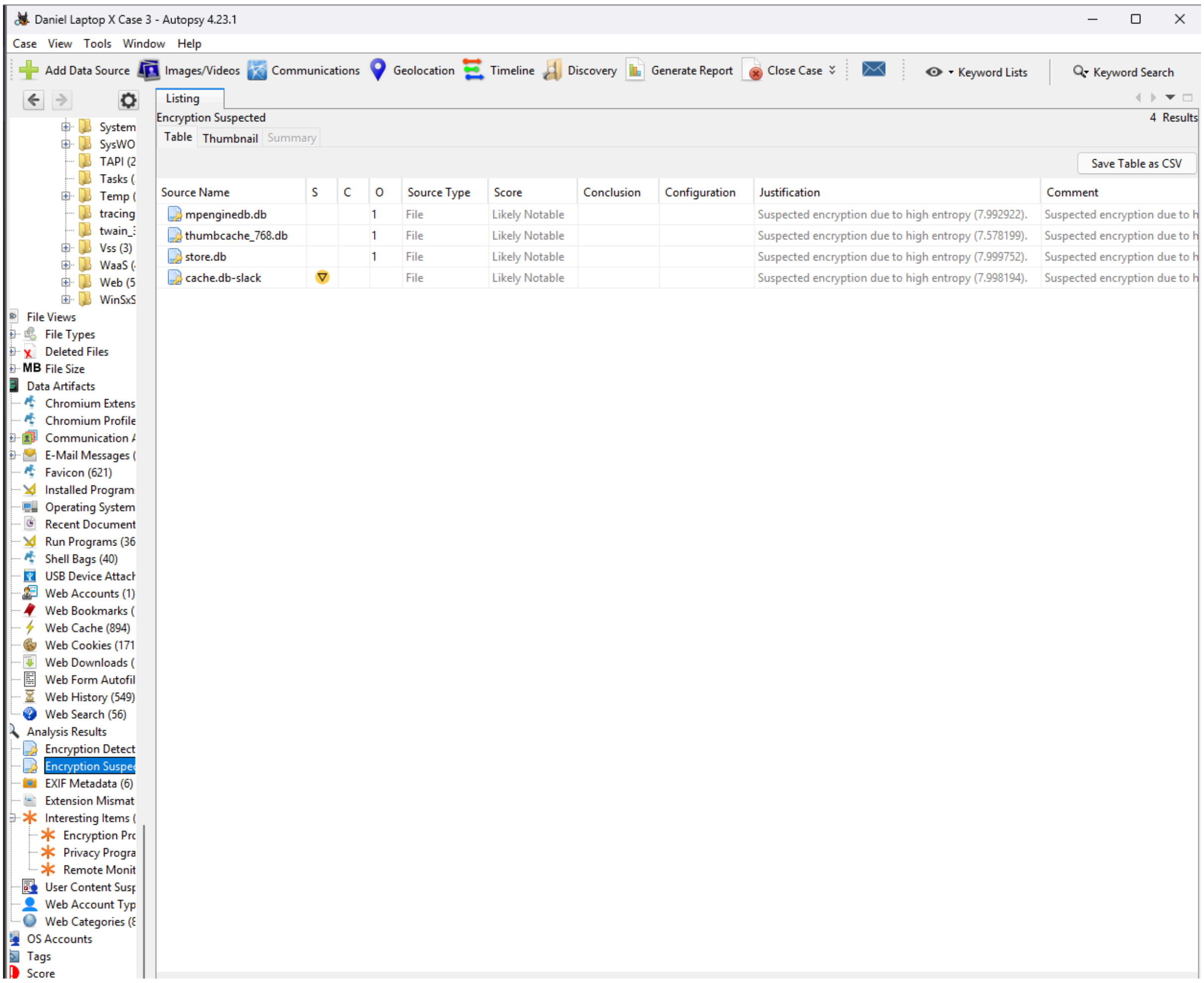


Figure 6.1.4f — Encryption-related artifacts identified during examination.

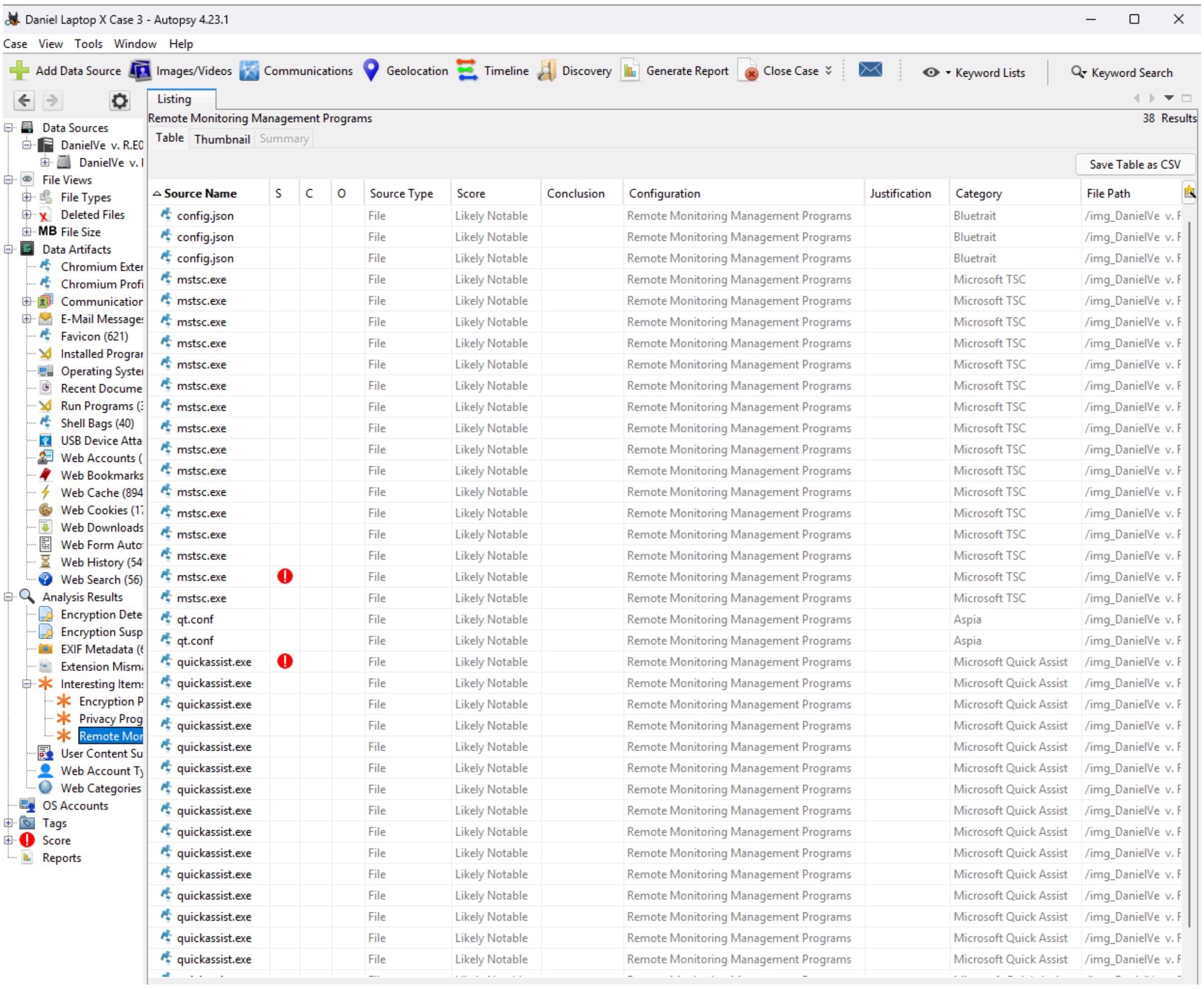


Figure 6.1.4g — Remote monitoring software identified from system artifacts.

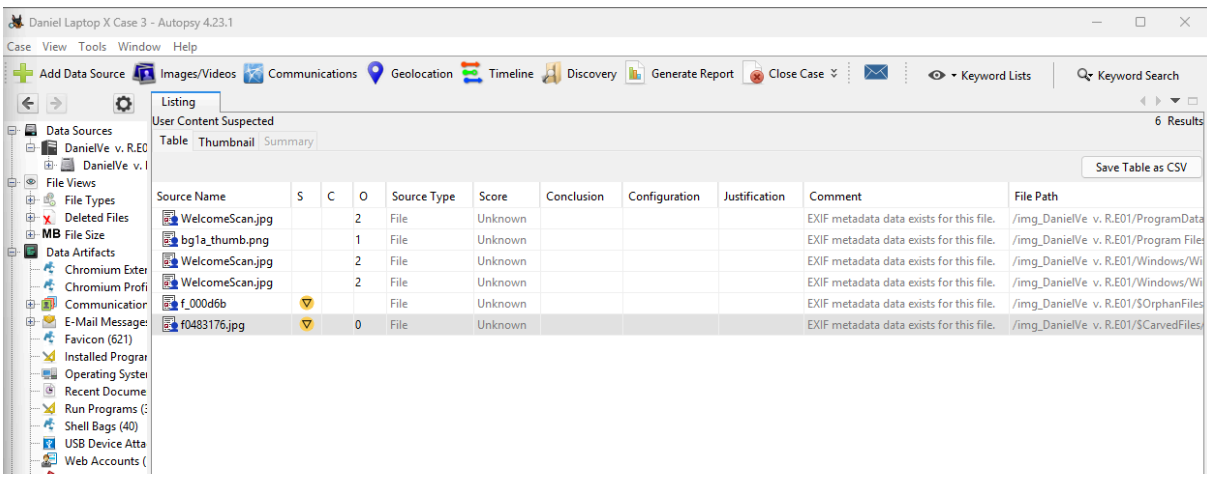


Figure 6.1.4h — User content flagged during forensic.

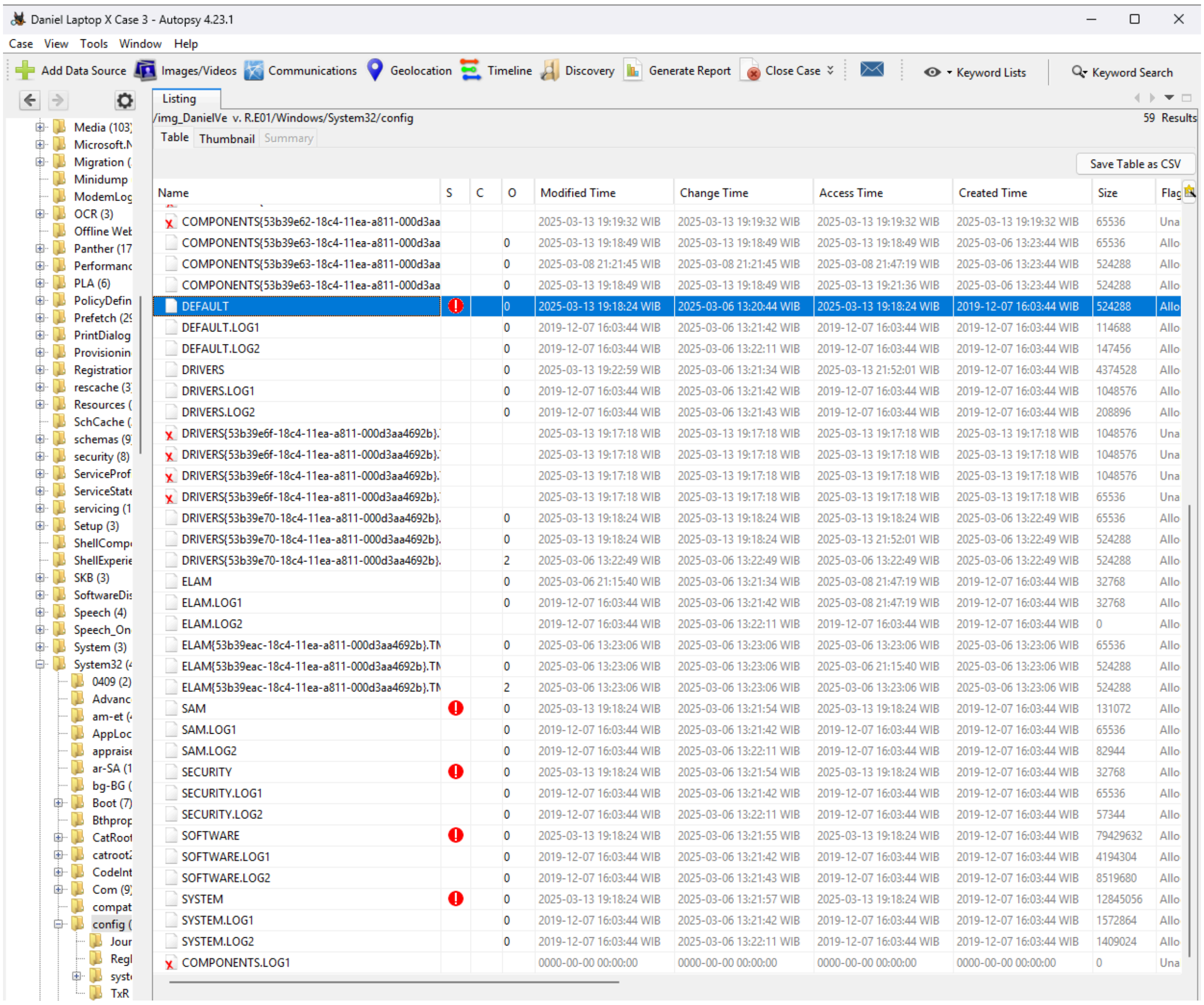
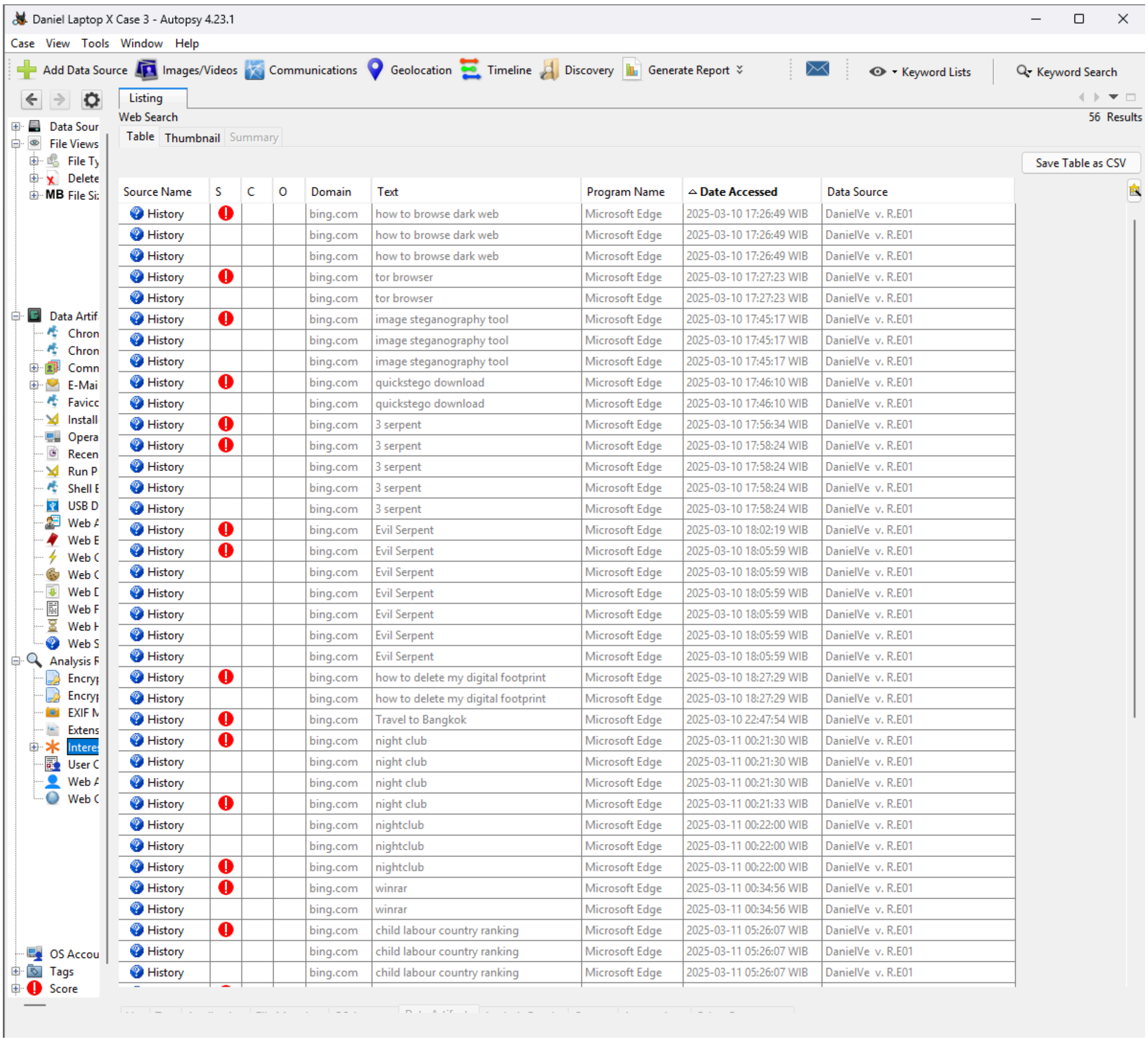


Figure 6.1.4i — Windows System.



Source Name	S	C	O	Domain	Text	Program Name	Date Accessed	Data Source
History	!			bing.com	how to browse dark web	Microsoft Edge	2025-03-10 17:26:49 WIB	DanielVe v. R.E01
History				bing.com	how to browse dark web	Microsoft Edge	2025-03-10 17:26:49 WIB	DanielVe v. R.E01
History				bing.com	how to browse dark web	Microsoft Edge	2025-03-10 17:26:49 WIB	DanielVe v. R.E01
History	!			bing.com	tor browser	Microsoft Edge	2025-03-10 17:27:23 WIB	DanielVe v. R.E01
History				bing.com	tor browser	Microsoft Edge	2025-03-10 17:27:23 WIB	DanielVe v. R.E01
History	!			bing.com	image steganography tool	Microsoft Edge	2025-03-10 17:45:17 WIB	DanielVe v. R.E01
History				bing.com	image steganography tool	Microsoft Edge	2025-03-10 17:45:17 WIB	DanielVe v. R.E01
History				bing.com	image steganography tool	Microsoft Edge	2025-03-10 17:45:17 WIB	DanielVe v. R.E01
History	!			bing.com	quickstego download	Microsoft Edge	2025-03-10 17:46:10 WIB	DanielVe v. R.E01
History				bing.com	quickstego download	Microsoft Edge	2025-03-10 17:46:10 WIB	DanielVe v. R.E01
History	!			bing.com	3 serpent	Microsoft Edge	2025-03-10 17:56:34 WIB	DanielVe v. R.E01
History	!			bing.com	3 serpent	Microsoft Edge	2025-03-10 17:58:24 WIB	DanielVe v. R.E01
History				bing.com	3 serpent	Microsoft Edge	2025-03-10 17:58:24 WIB	DanielVe v. R.E01
History				bing.com	3 serpent	Microsoft Edge	2025-03-10 17:58:24 WIB	DanielVe v. R.E01
History				bing.com	3 serpent	Microsoft Edge	2025-03-10 17:58:24 WIB	DanielVe v. R.E01
History	!			bing.com	Evil Serpent	Microsoft Edge	2025-03-10 18:02:19 WIB	DanielVe v. R.E01
History	!			bing.com	Evil Serpent	Microsoft Edge	2025-03-10 18:05:59 WIB	DanielVe v. R.E01
History				bing.com	Evil Serpent	Microsoft Edge	2025-03-10 18:05:59 WIB	DanielVe v. R.E01
History				bing.com	Evil Serpent	Microsoft Edge	2025-03-10 18:05:59 WIB	DanielVe v. R.E01
History				bing.com	Evil Serpent	Microsoft Edge	2025-03-10 18:05:59 WIB	DanielVe v. R.E01
History				bing.com	Evil Serpent	Microsoft Edge	2025-03-10 18:05:59 WIB	DanielVe v. R.E01
History				bing.com	Evil Serpent	Microsoft Edge	2025-03-10 18:05:59 WIB	DanielVe v. R.E01
History	!			bing.com	how to delete my digital footprint	Microsoft Edge	2025-03-10 18:27:29 WIB	DanielVe v. R.E01
History				bing.com	how to delete my digital footprint	Microsoft Edge	2025-03-10 18:27:29 WIB	DanielVe v. R.E01
History	!			bing.com	Travel to Bangkok	Microsoft Edge	2025-03-10 22:47:54 WIB	DanielVe v. R.E01
History	!			bing.com	night club	Microsoft Edge	2025-03-11 00:21:30 WIB	DanielVe v. R.E01
History				bing.com	night club	Microsoft Edge	2025-03-11 00:21:30 WIB	DanielVe v. R.E01
History				bing.com	night club	Microsoft Edge	2025-03-11 00:21:30 WIB	DanielVe v. R.E01
History	!			bing.com	night club	Microsoft Edge	2025-03-11 00:21:33 WIB	DanielVe v. R.E01
History				bing.com	nightclub	Microsoft Edge	2025-03-11 00:22:00 WIB	DanielVe v. R.E01
History				bing.com	nightclub	Microsoft Edge	2025-03-11 00:22:00 WIB	DanielVe v. R.E01
History	!			bing.com	nightclub	Microsoft Edge	2025-03-11 00:22:00 WIB	DanielVe v. R.E01
History	!			bing.com	winrar	Microsoft Edge	2025-03-11 00:34:56 WIB	DanielVe v. R.E01
History				bing.com	winrar	Microsoft Edge	2025-03-11 00:34:56 WIB	DanielVe v. R.E01
History	!			bing.com	child labour country ranking	Microsoft Edge	2025-03-11 05:26:07 WIB	DanielVe v. R.E01
History				bing.com	child labour country ranking	Microsoft Edge	2025-03-11 05:26:07 WIB	DanielVe v. R.E01
History				bing.com	child labour country ranking	Microsoft Edge	2025-03-11 05:26:07 WIB	DanielVe v. R.E01

History	!			bing.com	methods of falsifying passports	Microsoft Edge	2025-03-11 05:40:15 WIB	DanielVe v. R.E01
History				bing.com	methods of falsifying passports	Microsoft Edge	2025-03-11 05:40:15 WIB	DanielVe v. R.E01
History				bing.com	methods of falsifying passports	Microsoft Edge	2025-03-11 05:40:15 WIB	DanielVe v. R.E01
History	!			bing.com	how to wash dna off	Microsoft Edge	2025-03-12 11:50:51 WIB	DanielVe v. R.E01
History				bing.com	how to wash dna off	Microsoft Edge	2025-03-12 11:50:51 WIB	DanielVe v. R.E01
History				bing.com	how to wash dna off	Microsoft Edge	2025-03-12 11:50:51 WIB	DanielVe v. R.E01
History				bing.com	how to wash dna off	Microsoft Edge	2025-03-12 11:50:51 WIB	DanielVe v. R.E01
History	!			bing.com	alternative ways to be buried	Microsoft Edge	2025-03-13 00:44:21 WIB	DanielVe v. R.E01
History				bing.com	alternative ways to be buried	Microsoft Edge	2025-03-13 00:44:21 WIB	DanielVe v. R.E01
History				bing.com	alternative ways to be buried	Microsoft Edge	2025-03-13 00:44:21 WIB	DanielVe v. R.E01
History				bing.com	alternative ways to be buried	Microsoft Edge	2025-03-13 00:44:21 WIB	DanielVe v. R.E01
History	!			bing.com	bangkok thailand	Microsoft Edge	2025-03-13 00:50:07 WIB	DanielVe v. R.E01
History				bing.com	bangkok thailand	Microsoft Edge	2025-03-13 00:54:25 WIB	DanielVe v. R.E01
History	!			bing.com	bangkok thailand	Microsoft Edge	2025-03-13 00:54:25 WIB	DanielVe v. R.E01
History	!			bing.com	stages of decomposition human chart	Microsoft Edge	2025-03-13 13:50:25 WIB	DanielVe v. R.E01
History	!			bing.com	stages of decomposition human chart	Microsoft Edge	2025-03-13 13:50:25 WIB	DanielVe v. R.E01
History				bing.com	stages of decomposition human chart	Microsoft Edge	2025-03-13 13:50:25 WIB	DanielVe v. R.E01
History	!			bing.com	\ftk imager	Microsoft Edge	2025-03-14 21:24:18 WIB	DanielVe v. R.E01
History				bing.com	\ftk imager	Microsoft Edge	2025-03-14 21:24:18 WIB	DanielVe v. R.E01

Figure 6.1.5a — Browser search history recovered.

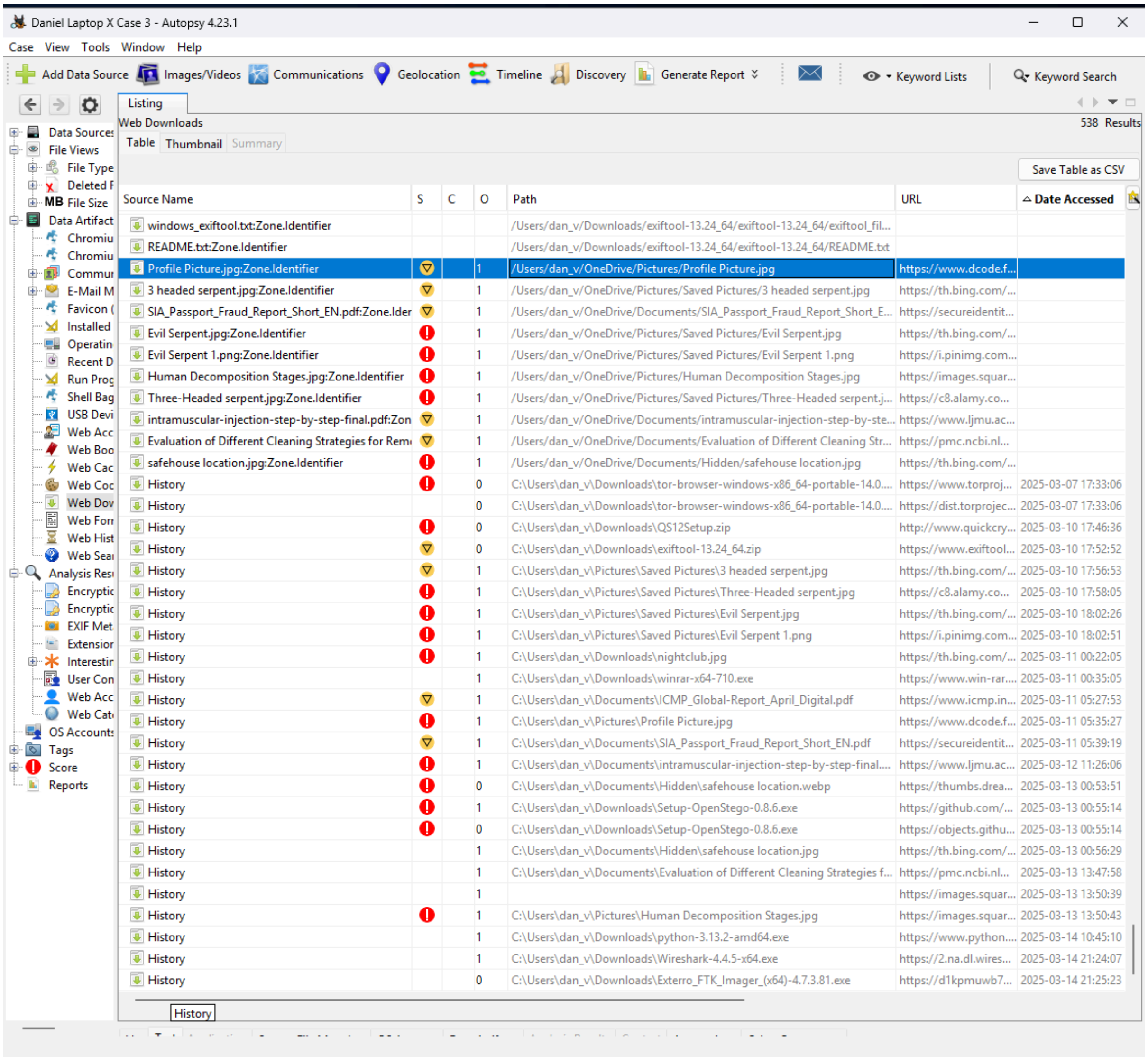
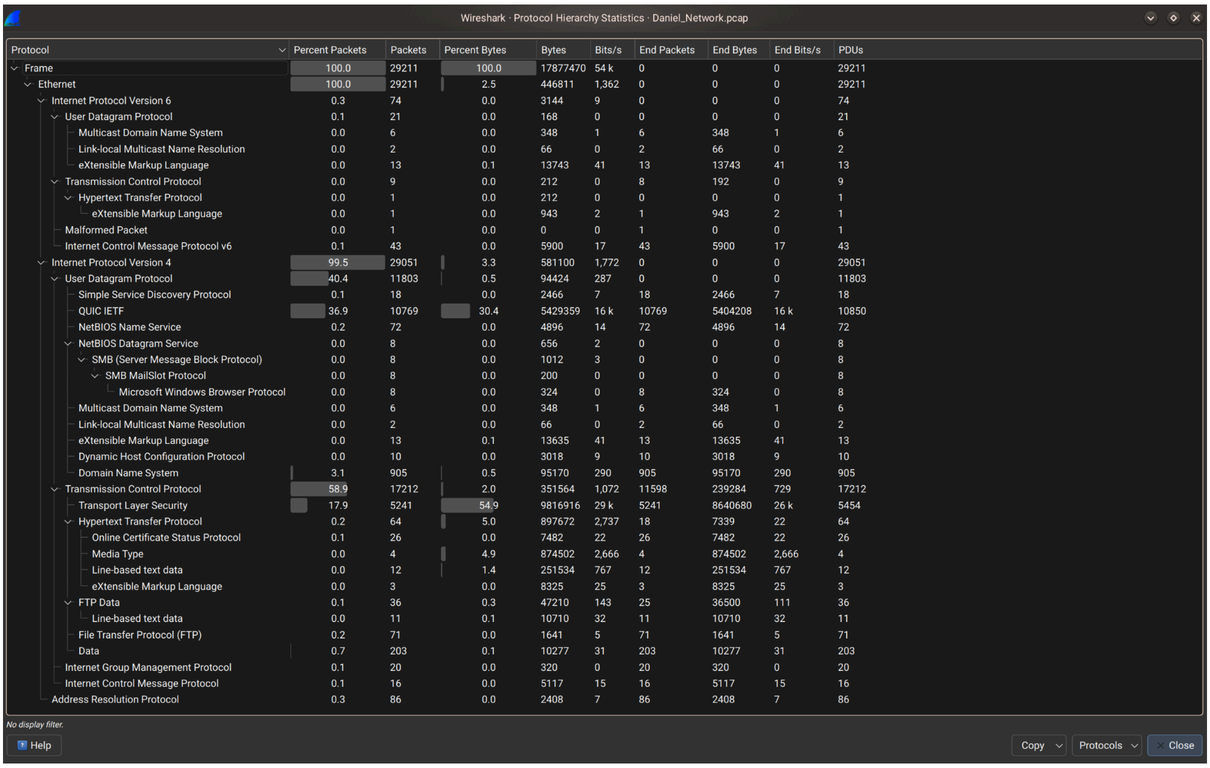


Figure 6.1.5b — Browser download history recovered from the forensic image.

6.2 Network Findings

Protocol hierarchy analysis was performed using Wireshark to identify the communication protocols present within the captured network traffic. The capture consisted of 29,211 frames, with IPv4 accounting for approximately 99.5% of all packets and IPv6 accounting for approximately 0.3%.

The dominant application-layer protocols identified during the examination included QUIC, TLS, DNS, HTTP, and FTP/FTP Data.



Wireshark · Protocol Hierarchy Statistics · Daniel_Network.pcap

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	29211	100.0	17877470	54 k	0	0	0	29211
Ethernet	100.0	29211	2.5	446811	1,362	0	0	0	29211
Internet Protocol Version 6	0.3	74	0.0	3144	9	0	0	0	74
User Datagram Protocol	0.1	21	0.0	168	0	0	0	0	21
Multicast Domain Name System	0.0	6	0.0	348	1	6	348	1	6
Link-local Multicast Name Resolution	0.0	2	0.0	66	0	2	66	0	2
eXtensible Markup Language	0.0	13	0.1	13743	41	13	13743	41	13
Transmission Control Protocol	0.0	9	0.0	212	0	8	192	0	9
Hypertext Transfer Protocol	0.0	1	0.0	212	0	0	0	0	1
eXtensible Markup Language	0.0	1	0.0	943	2	1	943	2	1
Malformed Packet	0.0	1	0.0	0	0	1	0	0	1
Internet Control Message Protocol v6	0.1	43	0.0	5900	17	43	5900	17	43
Internet Protocol Version 4	99.5	29051	3.3	581100	1,772	0	0	0	29051
User Datagram Protocol	40.4	11803	0.5	94424	287	0	0	0	11803
Simple Service Discovery Protocol	0.1	18	0.0	2466	7	18	2466	7	18
QUIC IETF	36.9	10769	30.4	5429359	16 k	10769	5404208	16 k	10850
NetBIOS Name Service	0.2	72	0.0	4896	14	72	4896	14	72
NetBIOS Datagram Service	0.0	8	0.0	656	2	0	0	0	8
SMB (Server Message Block Protocol)	0.0	8	0.0	1012	3	0	0	0	8
SMB MailSlot Protocol	0.0	8	0.0	200	0	0	0	0	8
Microsoft Windows Browser Protocol	0.0	8	0.0	324	0	8	324	0	8
Multicast Domain Name System	0.0	6	0.0	348	1	6	348	1	6
Link-local Multicast Name Resolution	0.0	2	0.0	66	0	2	66	0	2
eXtensible Markup Language	0.0	13	0.1	13635	41	13	13635	41	13
Dynamic Host Configuration Protocol	0.0	10	0.0	3018	9	10	3018	9	10
Domain Name System	3.1	905	0.5	95170	290	905	95170	290	905
Transmission Control Protocol	58.9	17212	2.0	351564	1,072	11598	239284	729	17212
Transport Layer Security	17.9	5241	54.9	9816916	29 k	5241	8640680	26 k	5454
Hypertext Transfer Protocol	0.2	64	5.0	897672	2,737	18	7339	22	64
Online Certificate Status Protocol	0.1	26	0.0	7482	22	26	7482	22	26
Media Type	0.0	4	4.9	874502	2,666	4	874502	2,666	4
Line-based text data	0.0	12	1.4	251534	767	12	251534	767	12
eXtensible Markup Language	0.0	3	0.0	8325	25	3	8325	25	3
FTP Data	0.1	36	0.3	47210	143	25	36500	111	36
Line-based text data	0.0	11	0.1	10710	32	11	10710	32	11
File Transfer Protocol (FTP)	0.2	71	0.0	1641	5	71	1641	5	71
Data	0.7	203	0.1	10277	31	203	10277	31	203
Internet Group Management Protocol	0.1	20	0.0	320	0	20	320	0	20
Internet Control Message Protocol	0.1	16	0.0	5117	15	16	5117	15	16
Address Resolution Protocol	0.3	86	0.0	2408	7	86	2408	7	86

No display filter.

Help Copy Protocols Close

Figure 6.2.1 — Wireshark Protocol Hierarchy Statistics showing the distribution of network protocols observed within Daniel_Network.pcap. IPv4 traffic represented the majority of captured packets, with significant TLS, QUIC, DNS, HTTP, and FTP communications.

6.2.1 HTTP Analysis

HTTP traffic was examined in Wireshark using the HTTP display filter. Multiple HTTP request and response exchanges were identified between the suspect host (192.168.40.129) and the HTTP server (192.168.40.135). Analysis of the captured traffic confirmed the retrieval of several evidence files, including HiddenMessage.bmp, safehouse location.bmp, Shipment List.pdf, client_list, and encrypted.zip through HTTP GET requests. Exported HTTP objects were preserved for further examination.

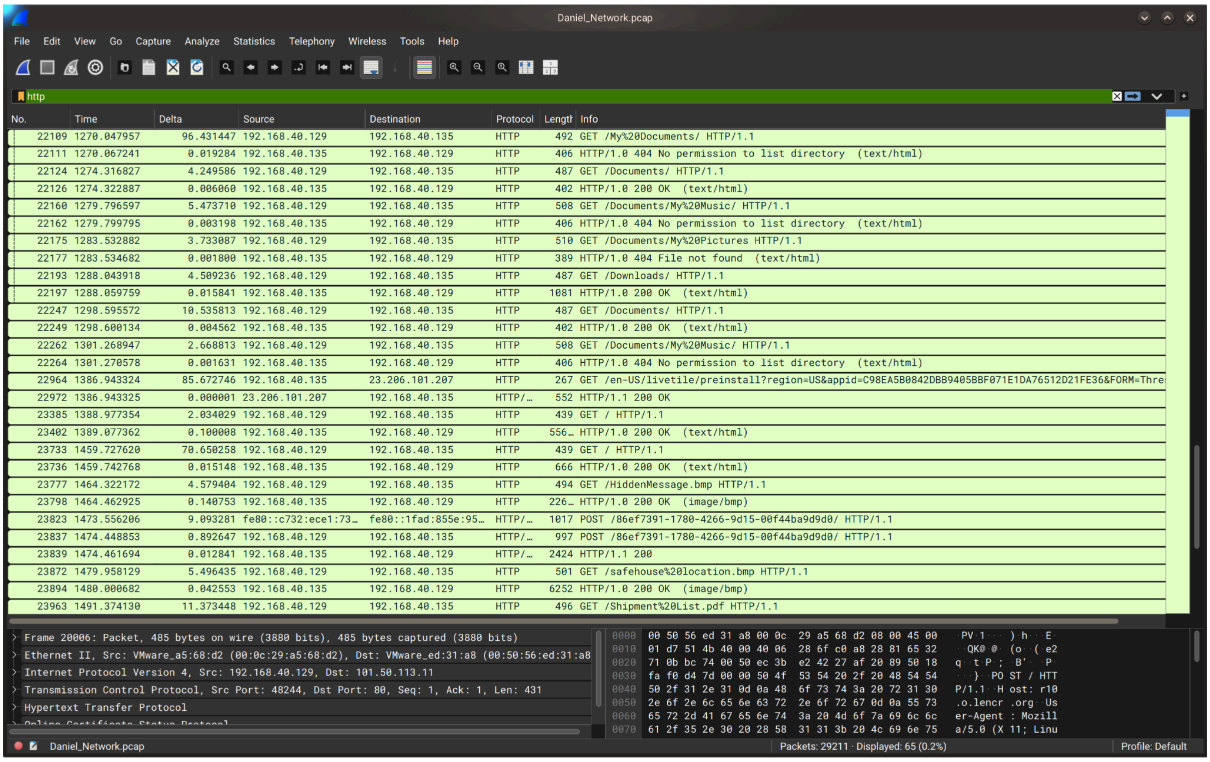


Figure 6.2.2 — HTTP traffic filter and requests.

6.2.2 FTP Analysis

FTP analysis identified multiple authentication attempts from host 192.168.40.135 to 192.168.40.129. An initial login attempt using the credentials "ftpuser" and password "123456" failed with a 530 Login incorrect response. Subsequent authentication attempts using the password "anonymous" were accepted by the FTP server (230 Login successful). Following successful authentication, the suspect retrieved INVOICE.pdf using the FTP RETR command.

Unlike the FTP activity, other evidence files, including HiddenMessage.bmp, safehouse location.bmp, Shipment List.pdf, client_list, and encrypted.zip, were transferred separately through HTTP GET requests from a Python SimpleHTTP server running on port 8080. This indicates that HTTP and FTP were used for different purposes during the communication.

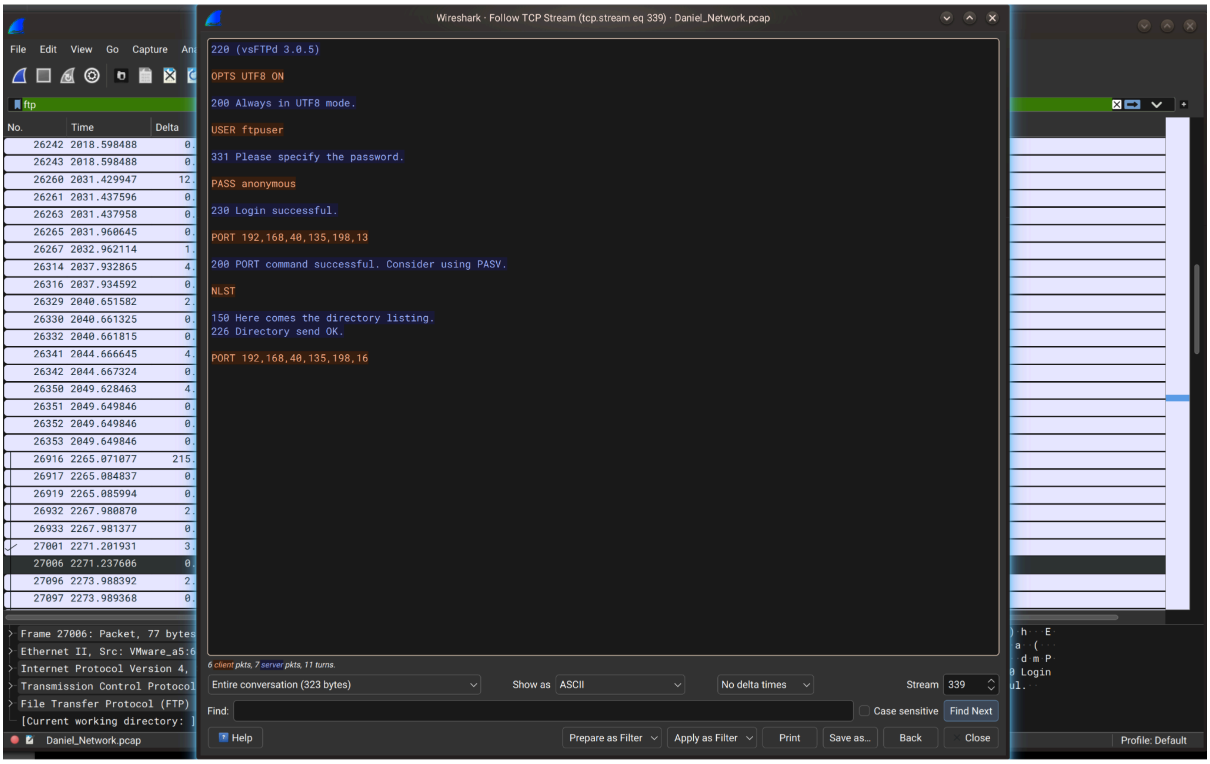


Figure 6.2.3 — FTP session — TCP Stream Follow.

6.2.3 DNS and TLS Analysis

DNS traffic was examined to identify resolved domains and communication endpoints associated with the captured network activity. The analysis identified multiple DNS queries to external services, including Microsoft, Mozilla, Google, Bing, Office 365, and OneDrive infrastructure. In addition, several DNS queries referenced .onion domains, including darkmarketxyz.onion, humantradeabcxyz.onion, shadowdealersxyz.onion, redphoenixtrades.onion, and deepwebroutesxyz.onion, indicating attempts to resolve hidden-service domain names.

TLS sessions were also identified throughout the network capture. Examination of the Server Name Indication (SNI) values showed encrypted communications primarily involving Microsoft, Mozilla, Google, and related cloud services. Because TLS encrypts application-layer content, the exchanged data could not be examined further. No direct TLS connections, Server Name Indication (SNI), or standard Tor service ports (9001, 9030, 9050, and 9150) associated with the identified .onion domains were observed within the captured traffic.

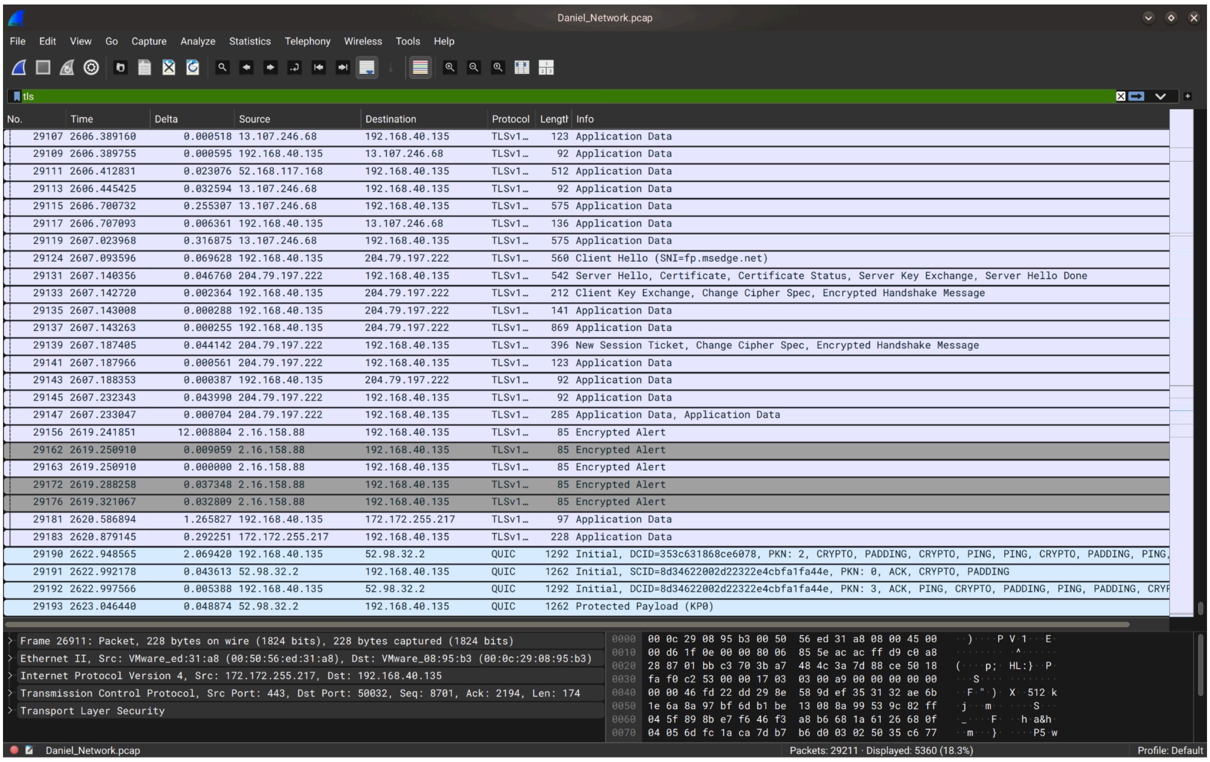
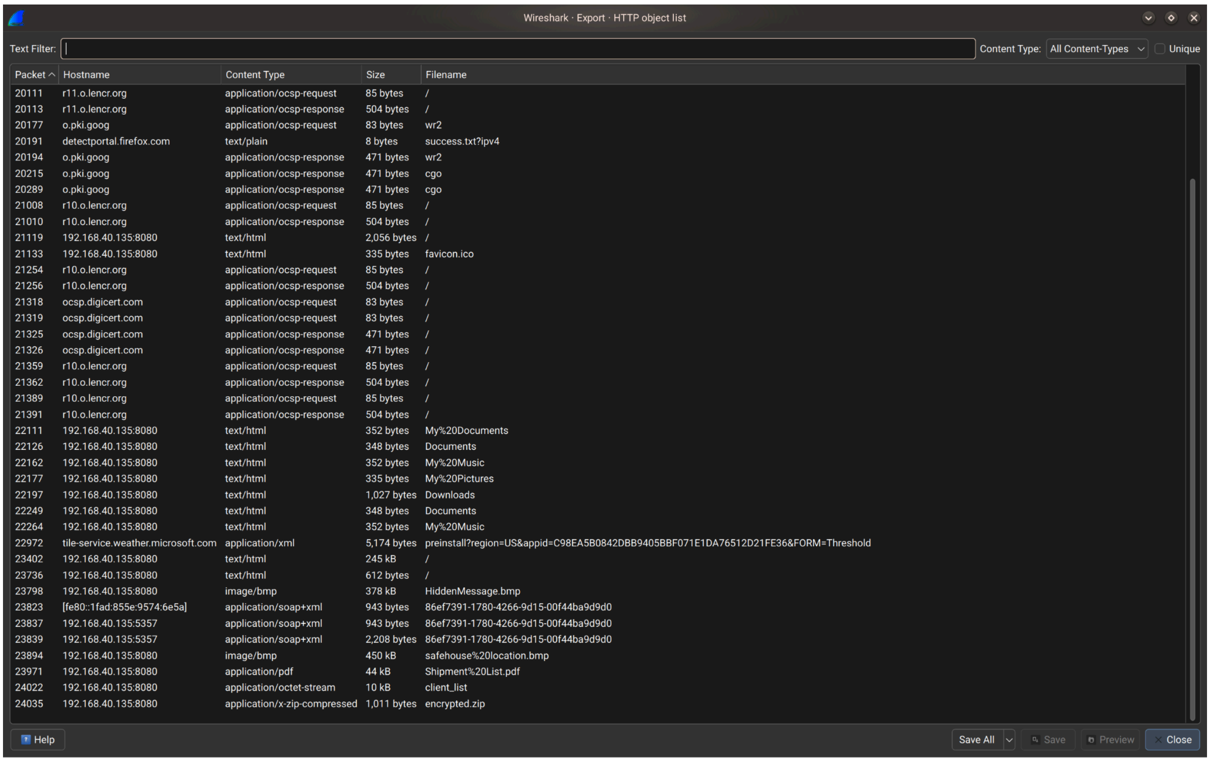


Figure 6.2.4 — DNS queries identifying external services and `.onion` domains together with TLS Server Name Indication (SNI) values observed within encrypted network sessions.

6.2.4 Exported Network Objects

Wireshark's Export Objects feature was used to recover files transmitted during HTTP and FTP sessions. The exported artifacts included **HiddenMessage.bmp**, **safehouse location.bmp**, **encrypted.zip**, and additional recovered files. These recovered objects were preserved and examined during the subsequent phases of the forensic investigation.



Packet	Hostname	Content Type	Size	Filename
20111	r11.o.lencr.org	application/ocsp-request	85 bytes	/
20113	r11.o.lencr.org	application/ocsp-response	504 bytes	/
20177	o.pki.goog	application/ocsp-request	83 bytes	wr2
20191	detectportal.firefox.com	text/plain	8 bytes	success.txt?ipv4
20194	o.pki.goog	application/ocsp-response	471 bytes	wr2
20215	o.pki.goog	application/ocsp-response	471 bytes	cgo
20289	o.pki.goog	application/ocsp-response	471 bytes	cgo
21008	r10.o.lencr.org	application/ocsp-request	85 bytes	/
21010	r10.o.lencr.org	application/ocsp-response	504 bytes	/
21119	192.168.40.135:8080	text/html	2,056 bytes	/
21133	192.168.40.135:8080	text/html	335 bytes	favicon.ico
21254	r10.o.lencr.org	application/ocsp-request	85 bytes	/
21256	r10.o.lencr.org	application/ocsp-response	504 bytes	/
21318	ocsp.digicert.com	application/ocsp-request	83 bytes	/
21319	ocsp.digicert.com	application/ocsp-request	83 bytes	/
21325	ocsp.digicert.com	application/ocsp-response	471 bytes	/
21326	ocsp.digicert.com	application/ocsp-response	471 bytes	/
21359	r10.o.lencr.org	application/ocsp-request	85 bytes	/
21362	r10.o.lencr.org	application/ocsp-response	504 bytes	/
21389	r10.o.lencr.org	application/ocsp-request	85 bytes	/
21391	r10.o.lencr.org	application/ocsp-response	504 bytes	/
22111	192.168.40.135:8080	text/html	352 bytes	My%20Documents
22126	192.168.40.135:8080	text/html	348 bytes	Documents
22162	192.168.40.135:8080	text/html	352 bytes	My%20Music
22177	192.168.40.135:8080	text/html	335 bytes	My%20Pictures
22197	192.168.40.135:8080	text/html	1,027 bytes	Downloads
22249	192.168.40.135:8080	text/html	348 bytes	Documents
22264	192.168.40.135:8080	text/html	352 bytes	My%20Music
22972	tile-service.weather.microsoft.com	application/xml	5,174 bytes	preinstall?region=US&appid=C98EA5B0842DBB9405BBF071E1DA76512D21FE36&FORM=Threshold
23402	192.168.40.135:8080	text/html	245 kB	/
23736	192.168.40.135:8080	text/html	612 bytes	/
23798	192.168.40.135:8080	image/bmp	378 kB	HiddenMessage.bmp
23823	[fe80::1fad:855e:9574:6e5a]	application/soap+xml	943 bytes	86ef7391-1780-4266-9d15-00f44ba9d9d0
23837	192.168.40.135:5357	application/soap+xml	943 bytes	86ef7391-1780-4266-9d15-00f44ba9d9d0
23839	192.168.40.135:5357	application/soap+xml	2,208 bytes	86ef7391-1780-4266-9d15-00f44ba9d9d0
23894	192.168.40.135:8080	image/bmp	450 kB	safehouse%20location.bmp
23971	192.168.40.135:8080	application/pdf	44 kB	Shipment%20List.pdf
24022	192.168.40.135:8080	application/octet-stream	10 kB	client_list
24035	192.168.40.135:8080	application/x-zip-compressed	1,011 bytes	encrypted.zip

Figure 6.2.5 — Exported HTTP/FTP objects

6.2.5 Cryptocurrency Transaction Tracing

The network capture and recovered files were examined for cryptocurrency-related artifacts, including wallet addresses, transaction identifiers (TXIDs), exchange references, and textual references to cryptocurrency payments.

No cryptocurrency wallet addresses or transaction identifiers were identified within the examined evidence.

However, textual references indicating that payments were intended to be conducted using cryptocurrency were identified within **client_list**.

No evidence linking these references to a specific blockchain transaction or cryptocurrency wallet was recovered during the examination.

Based on the available evidence, cryptocurrency was identified only as the intended payment method. No blockchain artifacts were recovered that would permit transaction tracing.

Timestamp	Protocol	Source	Destination	Description
1143.611	OCSP	192.168.40.129	101.50.113.11	OCSP request to verify a digital certificate before establishing a secure connection.
1464.322	HTTP	192.168.40.129	192.168.40.135	HTTP GET request retrieving **HiddenMessage.bmp**
1479.958	HTTP	192.168.40.129	192.168.40.135	HTTP GET request retrieving **safehouse location.bmp**
1491.374	HTTP	192.168.40.129	192.168.40.135	HTTP GET request retrieving **Shipment List.pdf**
1505.190	HTTP	192.168.40.129	192.168.40.135	HTTP GET request retrieving **client_list**
35.127	DNS	192.168.40.135	192.168.40.2	DNS query for **humantradeabcxyz.onion**
164.181	DNS	192.168.40.135	192.168.40.2	DNS query for **shadowdealersxyz.onion**
361.680	DNS	192.168.40.135	192.168.40.2	DNS query for **deepwebroutesxyz.onion**
576.047	DNS	192.168.40.135	192.168.40.2	DNS query for **darkmarketxyz.onion**
812.114	DNS	192.168.40.135	192.168.40.2	DNS query for **redphoenixtrades.onion**

Timestamp	Protocol	Source	Destination	Description
0.358	TLS	192.168.40.135	20.44.10.123	TLS Client Hello initiating an encrypted connection (SNI: **browser.events.data.msn.com**)
1508.917	HTTP	192.168.40.129	192.168.40.135	HTTP GET request retrieving **encrypted.zip**
2002.443	FTP	192.168.40.135	192.168.40.129	FTP authentication attempt using credentials **"ftpuser" / "123456" (530 Login incorrect)**.
2271.238	FTP	192.168.40.129	192.168.40.135	Successful FTP login **(230 Login successful)** after authentication using the password **"anonymous"**.
2569.615	FTP	192.168.40.135	192.168.40.129	FTP **RETR** command used to download **INVOICE.pdf**.

Table 6.2.1 — Network Traffic Log

Analysis of this evidence is discussed in Chapter 8.

6.3 Encrypted File Findings

This section presents the findings obtained from the forensic examination of encrypted and password-protected files recovered during the investigation. The examination focused on identifying protected files, evaluating the encryption methods used, and documenting the outcome of password recovery and decryption attempts. All findings are presented objectively without interpretation.

6.3.1 Hidden Data and Password Recovery

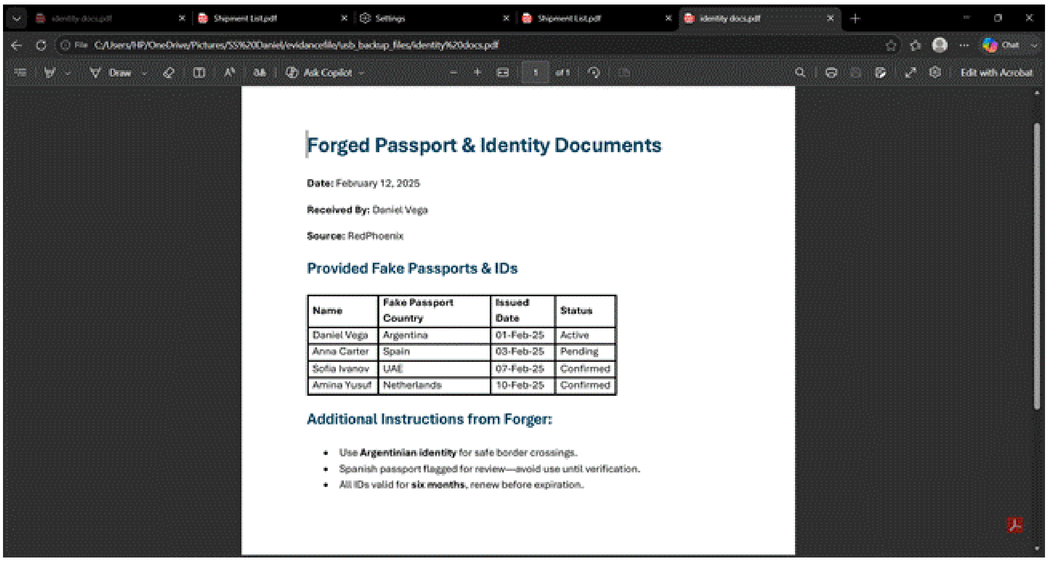
This subsection presents the findings obtained from the examination of encrypted and password-protected files recovered during the investigation. The examination focused on identifying protected files, verifying their file signatures, and attempting password recovery using forensic password-cracking techniques.

File Signature and Header Verification

File signature (magic number) verification was performed on the files recovered after password recovery from encrypted.zip and usb_backup_files.zip (cross-reference Section 6.3.2). Each file's actual content was visually and structurally confirmed to match its stated extension.

File	Stated Extension	Verified Content / Signature	Match
contacts.txt	.txt	ASCII plain text — dark-web contact list	Yes
safehouse_locations.txt	.txt	ASCII plain text — safehouse address list	Yes
identity docs.pdf	.pdf	PDF document (%PDF header) — forged passport record	Yes
transaction logs.xlsx	.xlsx	Microsoft Excel 2007+ (PK\x03\x04 / OOXML) — transaction ledger	Yes
messages.txt.txt	.txt	ASCII plain text — chat log transcript	Yes

Result: no file extension mismatches were identified among the files recovered from within encrypted.zip and usb_backup_files.zip. This complements the single mismatch identified separately at the container level (client_list, Section 6.3.2, Table 6.3.2) and is reported here as a negative finding for this specific set of files.



Forged Passport & Identity Documents

Date: February 12, 2025

Received By: Daniel Vega

Source: RedPhoenix

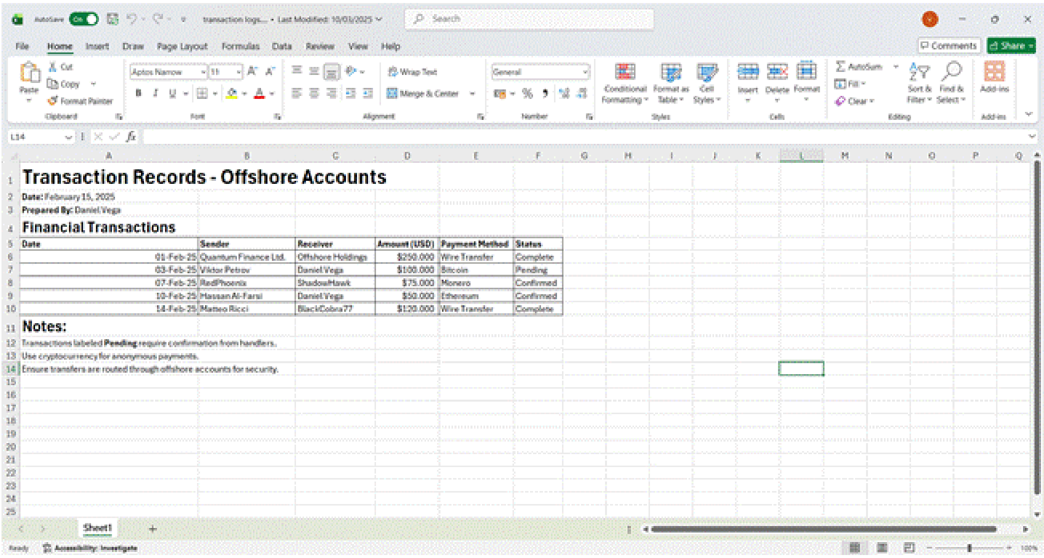
Provided Fake Passports & IDs

Name	Fake Passport Country	Issued Date	Status
Daniel Vega	Argentina	01-Feb-25	Active
Anna Carter	Spain	03-Feb-25	Pending
Sofia Ivanov	UAE	07-Feb-25	Confirmed
Amina Yusuf	Netherlands	10-Feb-25	Confirmed

Additional Instructions from Forger:

- Use **Argentinian identity** for safe border crossings.
- Spanish passport flagged for review—avoid use until verification.
- All IDs valid for **six months**, renew before expiration.

Figure 6.3.1a — Content of identity docs.pdf, confirming genuine PDF structure and content (Forged Passport & Identity Documents record).



Transaction Records - Offshore Accounts

Date: February 15, 2025

Prepared By: Daniel Vega

Financial Transactions

Date	Sender	Receiver	Amount (USD)	Payment Method	Status
01-Feb-25	Quantum Finance Ltd.	Offshore Holdings	$250,000	Wire Transfer	Complete
03-Feb-25	Viktor Petrov	Daniel Vega	$100,000	Bitcoin	Pending
07-Feb-25	RedPhoenix	ShadowHawk	$75,000	Monero	Confirmed
10-Feb-25	Hassan Al-Farsi	Daniel Vega	$50,000	Ethereum	Confirmed
14-Feb-25	Matteo Ricci	BlackCobra77	$120,000	Wire Transfer	Complete

Notes:

Transactions labeled **Pending** require confirmation from handlers.

Use cryptocurrency for anonymous payments.

Ensure transfers are routed through offshore accounts for security.

Figure 6.3.1b — Content of transaction logs.xlsx, confirming genuine Excel/OOXML structure (Transaction Records — Offshore Accounts).

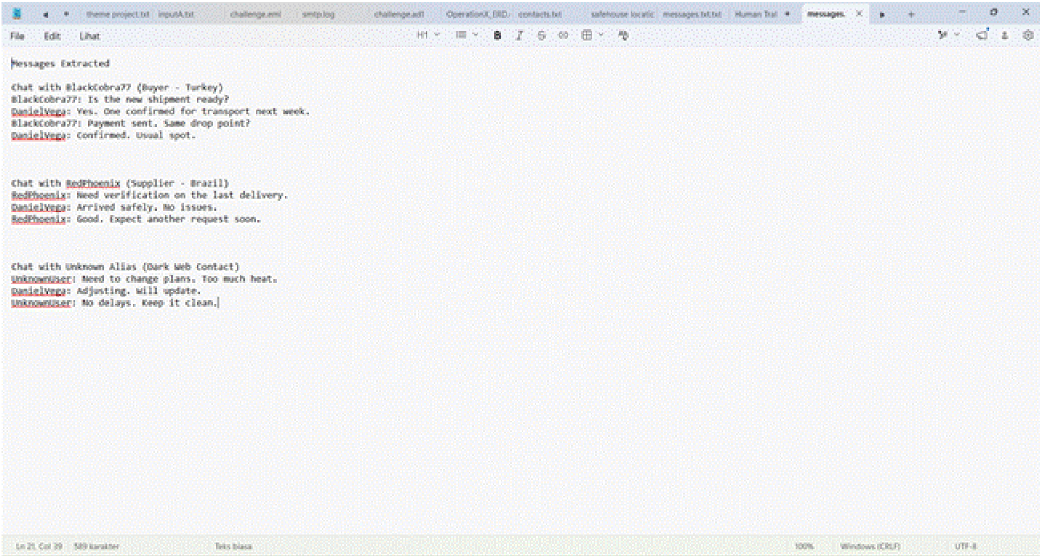


Figure 6.3.1c — Content of messages.txt.txt, confirming plain-text structure (extracted chat log).

Identify of Password-Protected Files

All files transferred and recovered under this scope were checked to identify which required a password before their contents could be accessed.

File	Password-Protected?	Verification Method
encrypted.zip	Yes — AES encrypted	Password prompt on manual open attempt; confirmed as WinZip AE (method 99) via zipinfo -v (Section 6.3.2)
usb_backup_files.zip	No	Opened directly without a password prompt; consistent with standard Deflate compression (Table 6.3.2)

Result: encrypted.zip was confirmed as the only password-protected archive within this analyst's scope. Both encrypted.zip and HiddenMessage.bmp were located within the OneDrive/Documents/Hidden folder of Daniel_Laptop.E01, alongside safehouse location.bmp/.jpg and usb_backup_files.zip.

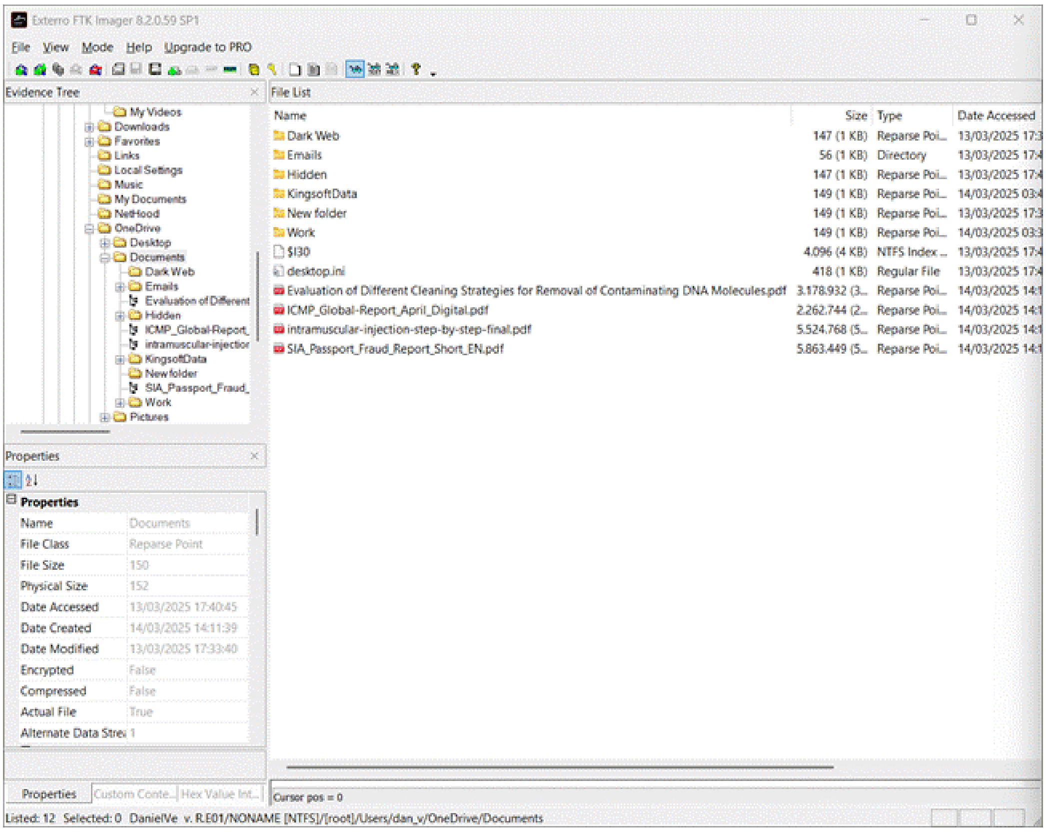


Figure 6.3.1d — FTK Imager Evidence Tree showing the Hidden folder contents, including encrypted.zip and usb_backup_files.zip.

Password Recovery

A password hash was extracted from encrypted.zip using the zip2john utility (John the Ripper suite), and a dictionary attack was performed using the rockyou.txt wordlist:

- zip2john encrypted.zip > zip_hash.txt
- john --wordlist=rockyou.txt zip_hash.txt
- john --show zip_hash.txt

Result: the password for encrypted.zip was recovered as 123456. The recovered password is consistent with the password obtained independently using an alternative tool chain described in Section 6.3.2. Once decrypted, encrypted.zip was confirmed to contain contacts.txt and safehouse_locations.txt, consistent with the extraction reported in Section 6.3.2.

Password Recovery Session

The following screenshots corroborate the SHA-256 hash values already recorded in Table 4.1 for the disk image segments (E01–E13).

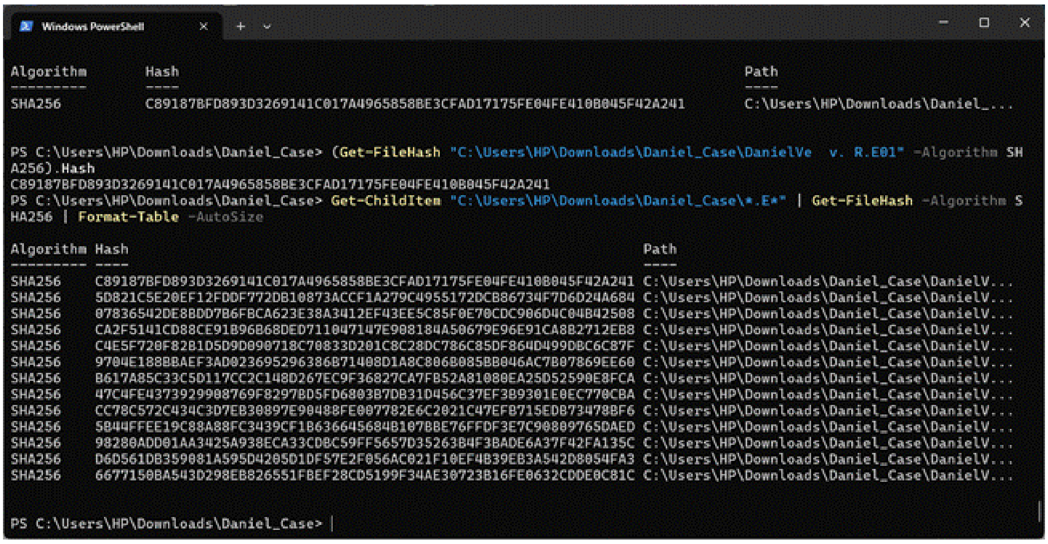


Figure 6.3.1e — Get-FileHash (SHA-256) output for DanielVe v. R.E01, matching Table 4.1.

Figure 6.3.1f — Get-FileHash (MD5 and SHA-256) verification session for DanielVe v. R.E01–R.E13.

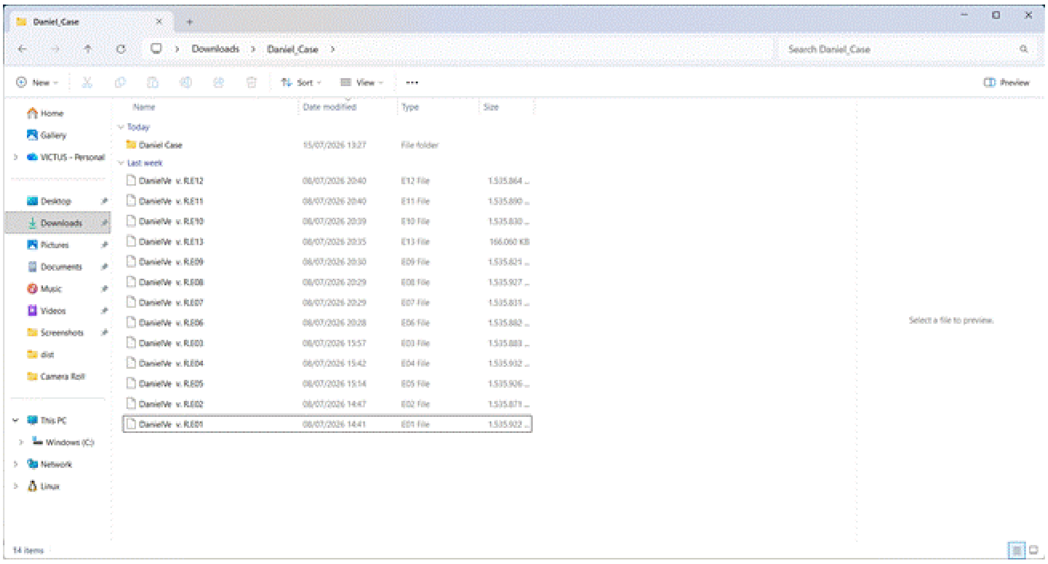


Figure 6.3.1g — File Explorer listing of DanielVe v. R.E01–R.E13 disk image segments.

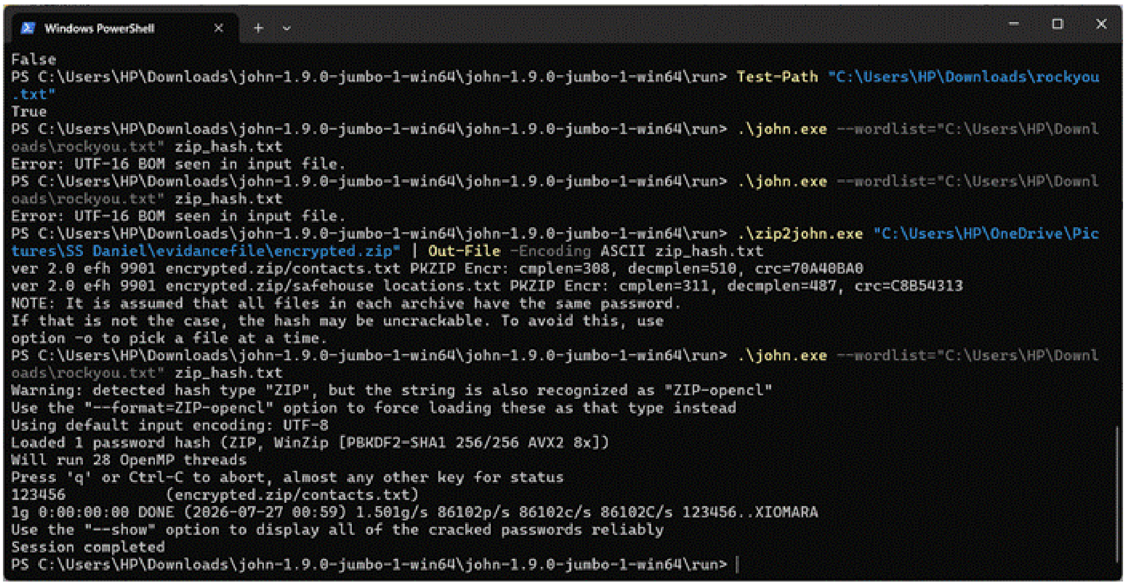


Figure 6.3.1h — zip2john hash extraction from encrypted.zip, producing zip_hash.txt (both $zip2$ and $pkzip2$ hash formats), followed by locating the rockyou.txt wordlist.

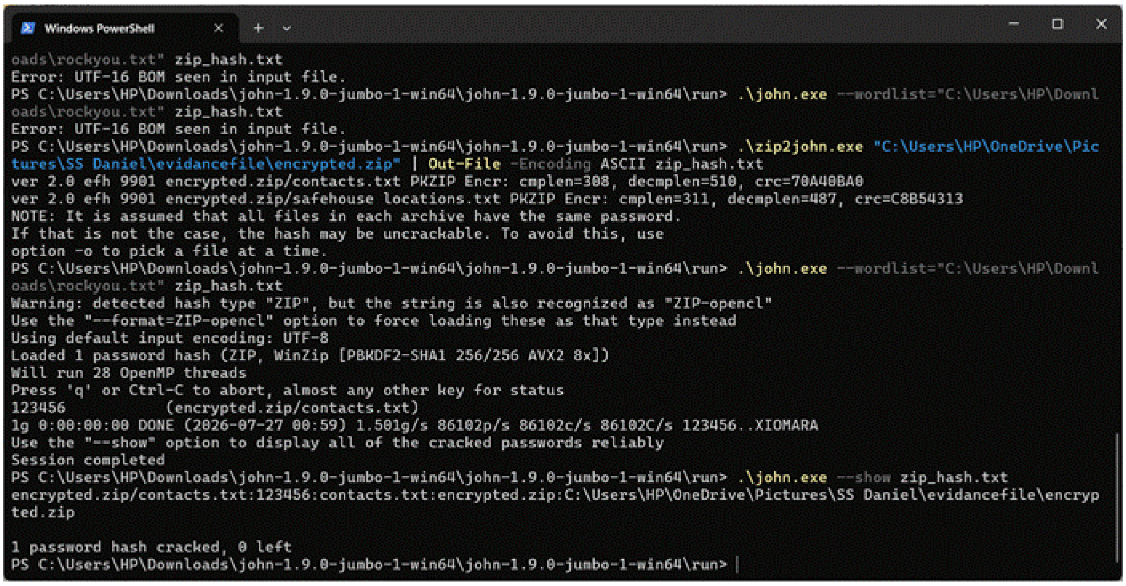


Figure 6.3.1i — John the Ripper dictionary attack (--wordlist=rockyou.txt) against zip_hash.txt, successfully recovering the password 123456 for encrypted.zip/contacts.txt in under 1 second.

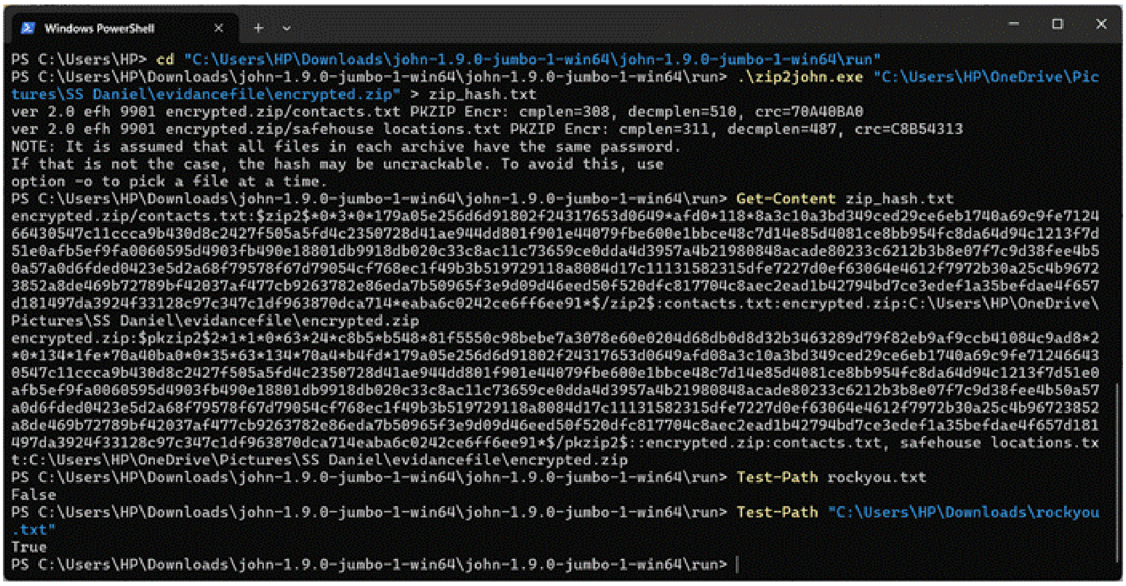


Figure 6.3.1j — Confirmation of the recovered password via john --show, displaying encrypted.zip/contacts.txt:123456 ("1 password hash cracked, 0 left").

These figures document the password recovery process, including hash extraction, dictionary attack, and verification of the recovered password. The recovered password (123456) is consistent with the password obtained independently using pyzipper, as described in Section 6.3.2, confirming the result through two independent recovery methods.

6.3.2 Encryption and File Cracking

This subsection presents the findings obtained from the examination of encrypted files and recovered artifacts. All findings are presented objectively without interpretation. Correlation with other evidence is discussed in Chapter 8.

File Extension Mismatch Detection

All recovered files were checked using the file command and verified with xxd when necessary. Only client_list showed a mismatch: it had no extension but its signature (50 4B 03 04 / “PK..”) identified it as a Microsoft Excel 2007+ (.xlsx) file, indicating the extension had been intentionally removed. All other files matched their stated extensions.

File	Stated Extension	file Output / Signature	Mismatch
client_list	(none)	Microsoft Excel 2007+ (hex 50 4B 03 04 / "PK..")	Yes — extension removed
Shipment List.pdf	.pdf	PDF document, version 1.7	No
HiddenMessage.bmp	.bmp	PC bitmap, Windows 3.x, 474x266x24	No
safehouse location.bmp	.bmp	PC bitmap, Windows 3.x, 474x316x24	No
safehouse location.jpg	.jpg	JPEG image data, JFIF 1.01	No
encrypted.zip	.zip	Zip archive, AES Encrypted	No

usb_backup_files.zip	.zip	Zip archive, deflate	No

Table 6.3.2 — file Extension Mismatch Results

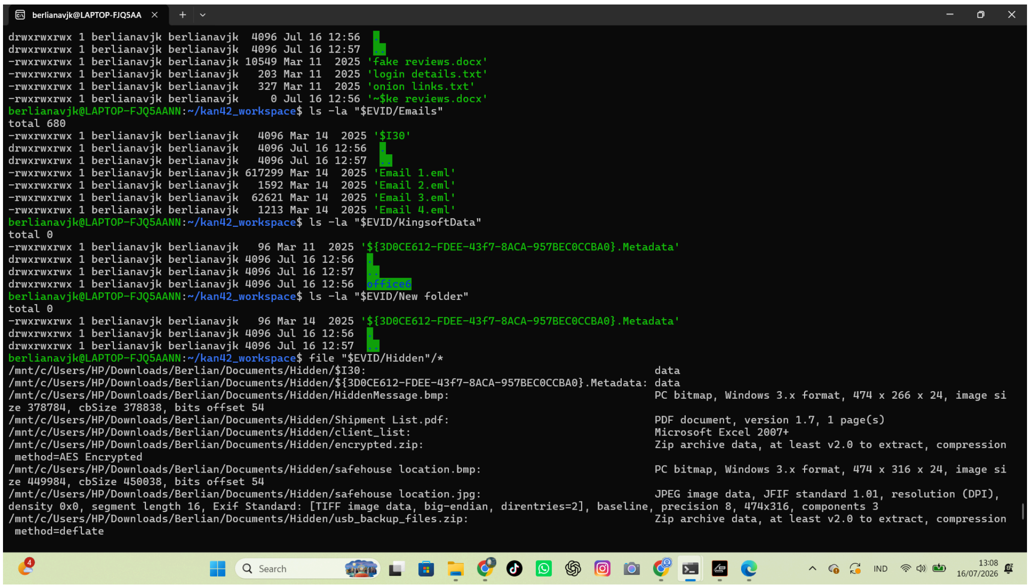


Figure 6.3.2a — file command output confirming client_list as a disguised Excel file

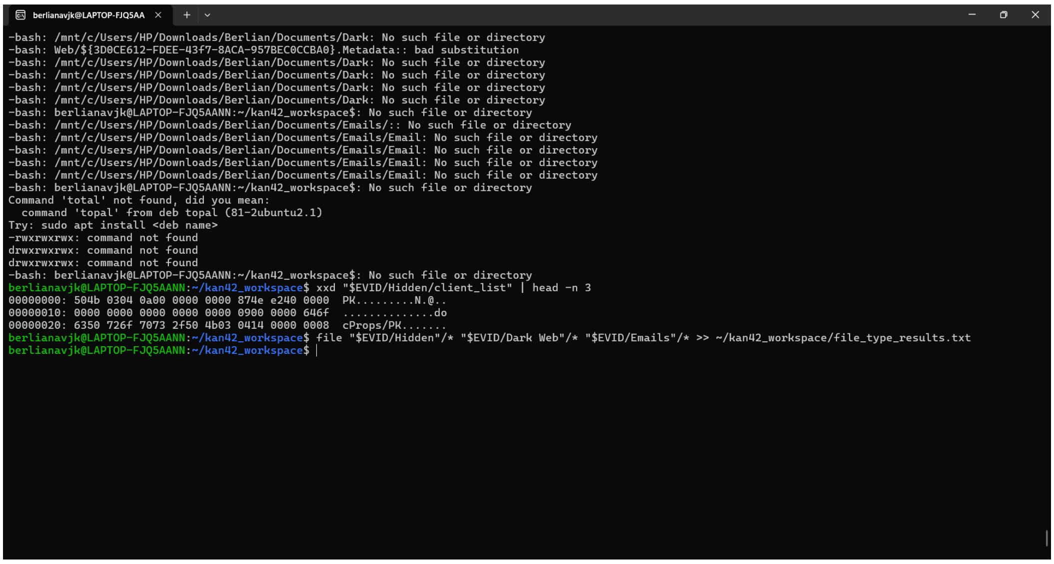


Figure 6.3.2b — xxd output confirming the Microsoft Excel OOXML signature (50 4B 03 04) of client_list.

Binwalk Analysis on Suspicious Files

HiddenMessage.bmp, safehouse location.bmp, and safehouse location.jpg showed only their native image headers, normal entropy values, and no embedded or appended data. No GPS metadata was found in the images. usb_backup_files.zip was successfully extracted and contained identity docs.pdf, messages.txt.txt, and transaction logs.xlsx, all of which were verified with the file command. After assigning the .xlsx extension, client_list was confirmed to contain buyer names and payment amounts.

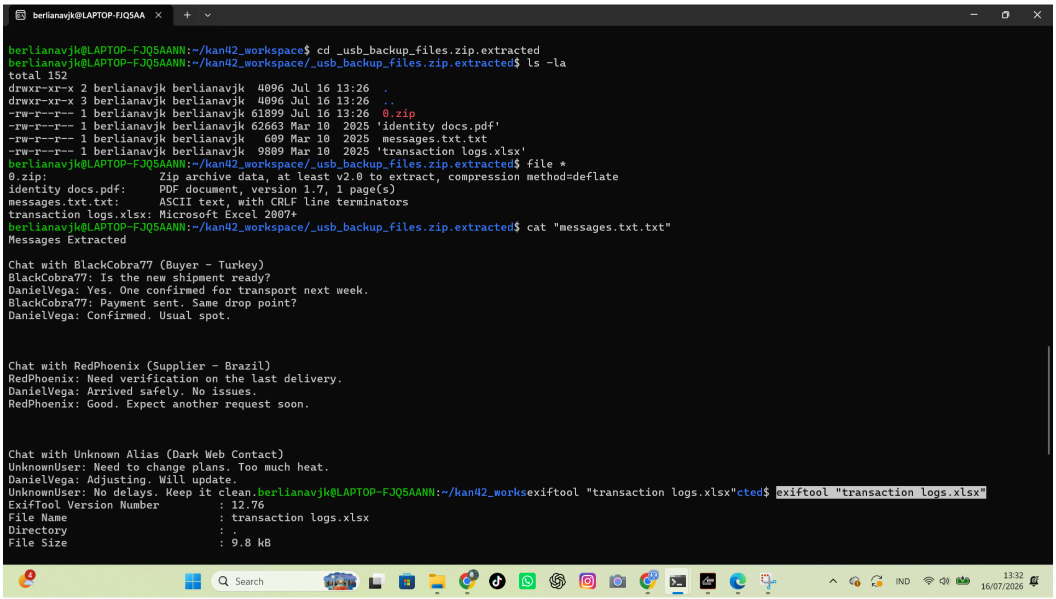


Figure 6.3.2c — Extracted contents of usb_backup_files.zip verified with the file command

Crack VeraCrypt/Encrypted Volumes :

No VeraCrypt container or encrypted volume was found in the provided evidence. The only encrypted artifact was encrypted.zip, identified as WinZip AES (method 99) using zipinfo. Initial password attempts with unzip failed due to AES incompatibility, while 7z also failed against a case-derived wordlist. A dictionary attack using pyzipper and the rockyou.txt wordlist successfully recovered the password 123456. The archive was extracted with 7z, revealing contacts.txt and safehouse_locations.txt. The original evidence file remained unchanged throughout the process.

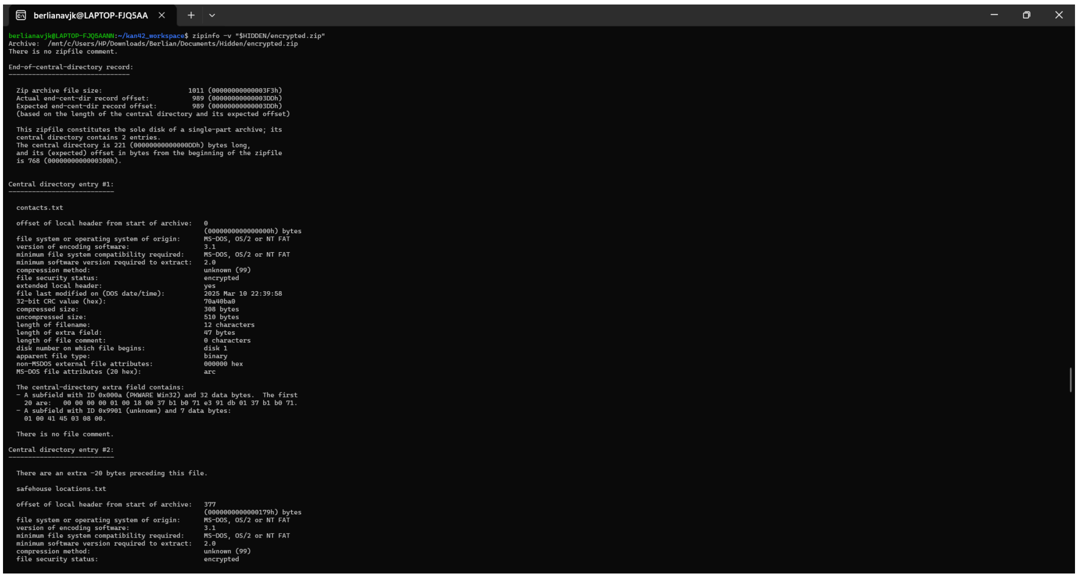


Figure 6.3.2d — zipinfo -v output identifying encrypted.zip as a WinZip AES encrypted archive.

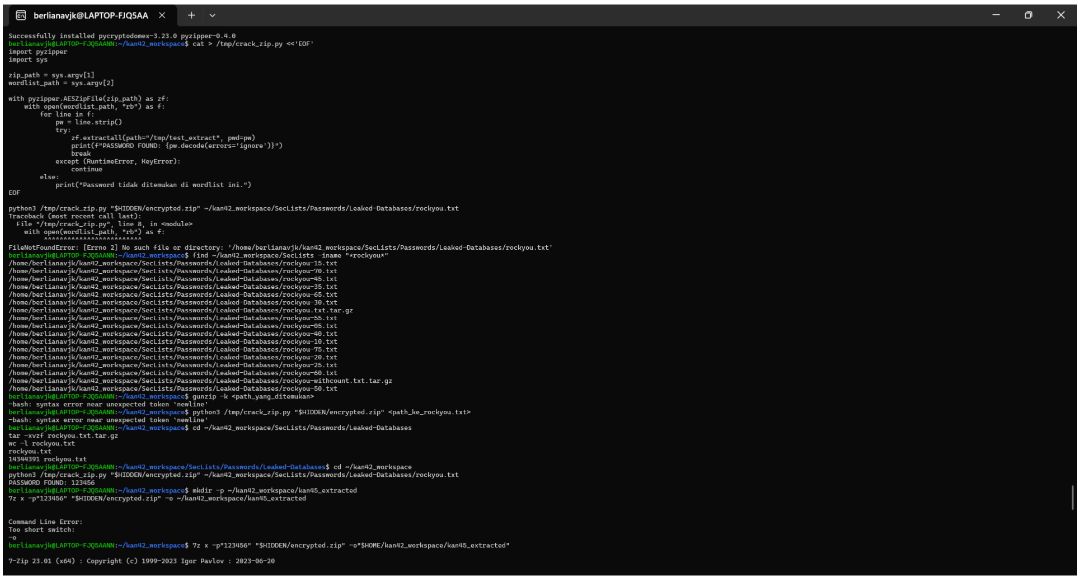


Figure 6.3.2e — pyzipper dictionary attack using the rockyou.txt wordlist, successfully recovering the password for encrypted.zip.

Analysis of this evidence is discussed in Chapter 8.

6.4 Steganography Findings

This section presents the findings obtained from the steganographic examination of image files recovered during the network analysis. The examination focused on identifying hidden data, embedded files, metadata, and other concealed information within the recovered image files. All findings are presented objectively without interpretation.

6.4.1 Carrier Files Examined

Both files were selected for steganographic examination because they were image files recovered during the investigation and were considered relevant to the forensic examination.

6.4.2 Metadata Analysis

ExifTool was used to examine the metadata of both image files. No unusual or evidential metadata was identified in either HiddenMessage.bmp or safehouse location.bmp.

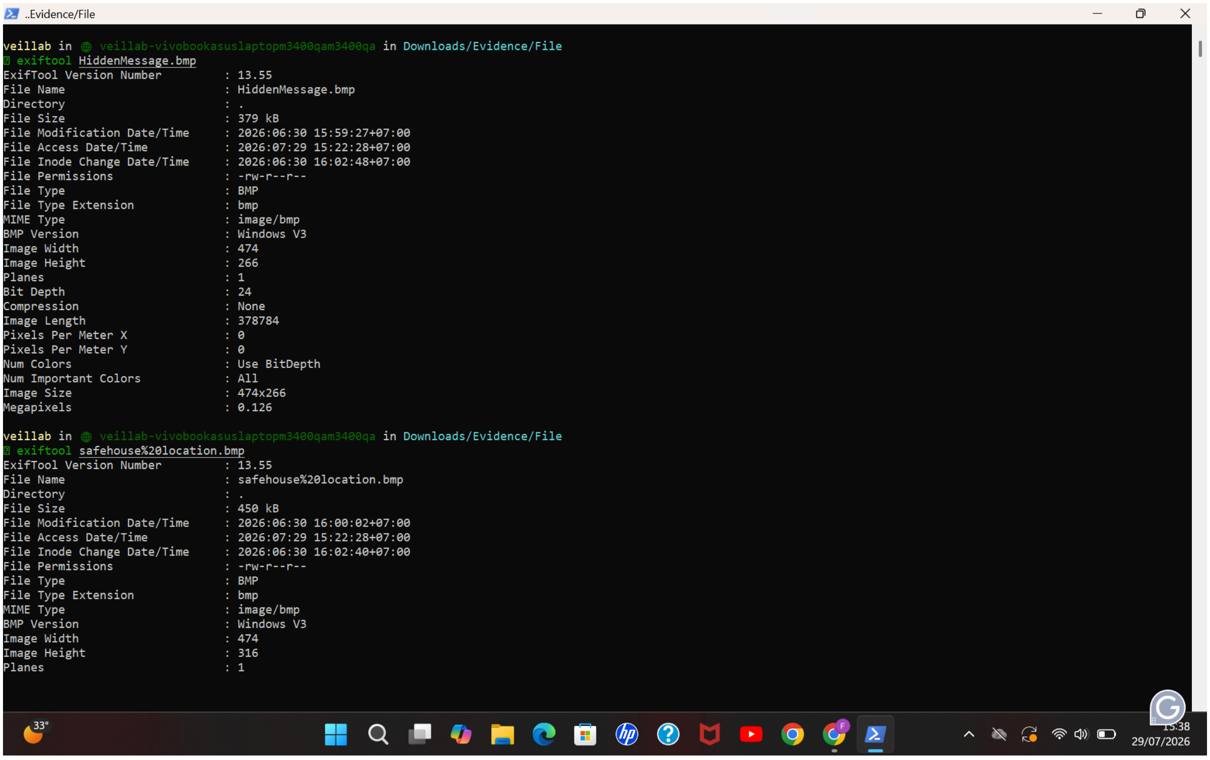


Figure 6.4.1 — ExifTool metadata output for HiddenMessage.bmp and safehouse location.bmp.

6.4.3 String and Hex Analysis

Both image files were examined using strings and xxd to identify readable text and inspect the raw hexadecimal structure of the files. No plaintext payloads or suspicious appended data were identified.

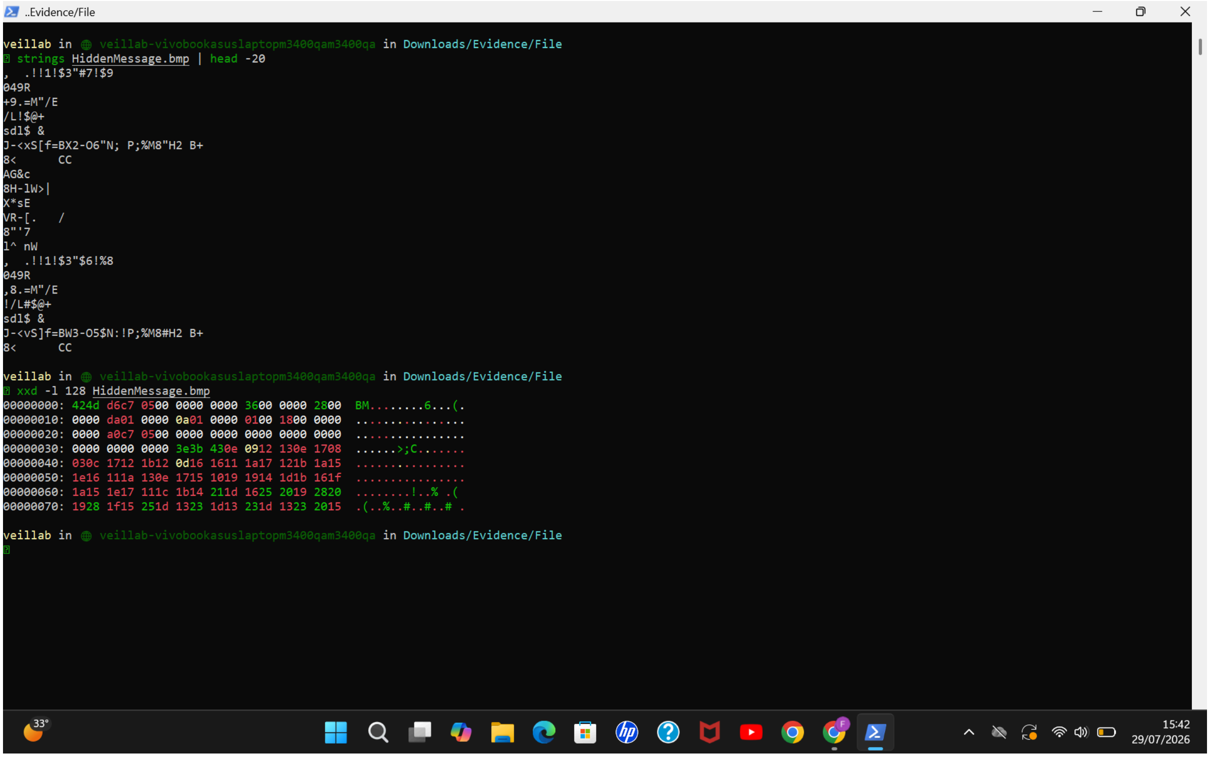


Figure 6.4.2 — GNU strings and xxd output showing printable strings and the initial hexadecimal structure of HiddenMessage.bmp.

6.4.4 Embedded File Detection

Binwalk was used to scan both BMP image files for embedded or appended file signatures. No additional signatures were identified in HiddenMessage.bmp. During the examination of safehouse location.bmp, Binwalk reported a JBOOT STAG header at offset 0x5DA3D. Manual hexadecimal inspection of the reported offset did not identify a valid embedded file or recoverable payload. Accordingly, the reported JBOOT STAG signature was assessed as a false positive, and no embedded files were confirmed.

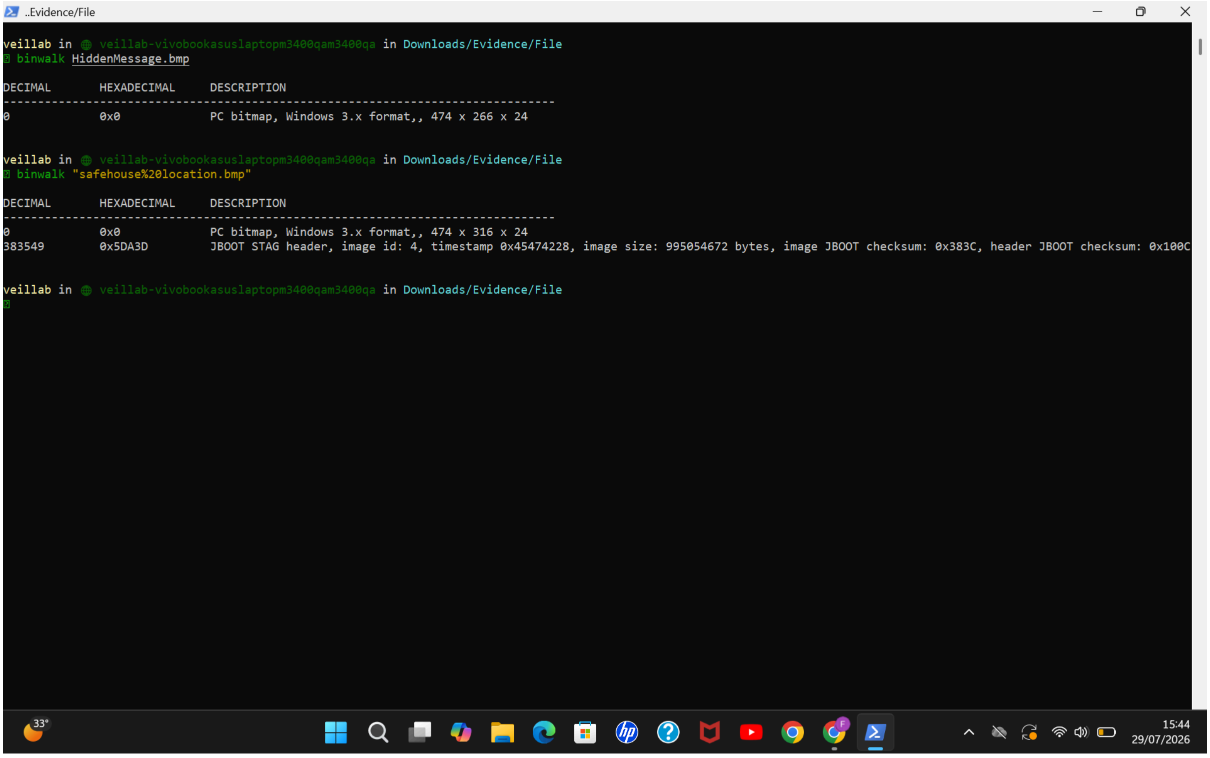


Figure 6.4.3 — Binwalk scan identifying a reported JBOOT STAG signature in safehouse location.bmp prior to manual verification.

6.4.5 Steganography Analysis

zsteg was used to examine both BMP image files for potential least significant bit (LSB) steganography and hidden data patterns. The tool reported multiple potential OpenPGP Public Key and OpenPGP Secret Key signatures within both images. Manual verification of the reported locations using hexadecimal inspection did not identify valid OpenPGP key material or recoverable hidden payloads. Based on the combined results of zsteg and manual verification, no confirmed steganographic payload was identified within either image file.

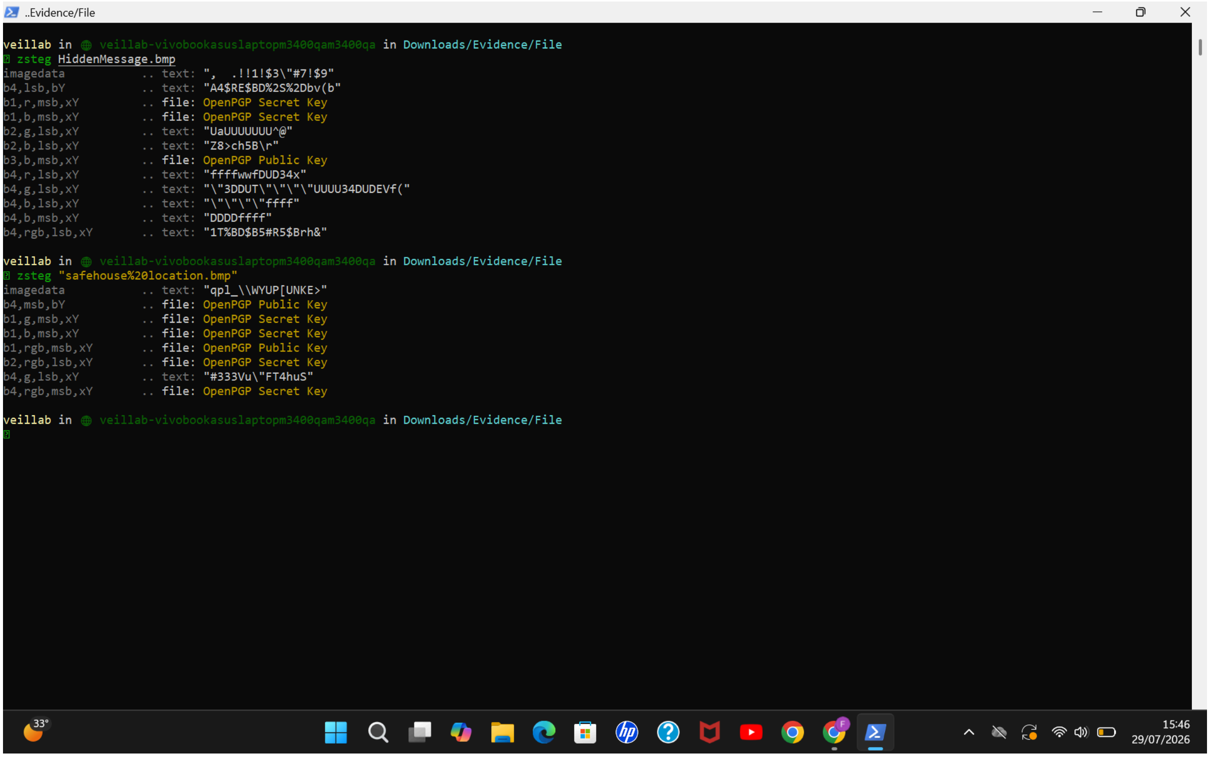


Figure 6.4.4 — zsteg output reporting potential OpenPGP Public Key and OpenPGP Secret Key signatures prior to manual verification.

Analysis of this evidence is discussed in Chapter 8.

6.5 Document Findings

This section presents the findings obtained from the forensic examination of documents recovered from the evidence. The examination focused on identifying document metadata, document contents, and other artifacts relevant to the investigation. All findings are presented objectively without interpretation.

6.5.1 Recovered Documents

The forensic examination recovered several documents and related artifacts from the disk image, including PDF documents, spreadsheet files, email messages, and Windows shortcut (.lnk) files. Examples of recovered documents include Shipment List.pdf, identity docs.pdf, INVOICE.pdf, and transaction logs.xlsx, together with email artifacts and recent document shortcuts. These recovered items may provide information regarding previous user activities and are preserved as digital evidence.

In addition to the documents already listed in Table 6.5.1, the following recovered items provide additional factual content:

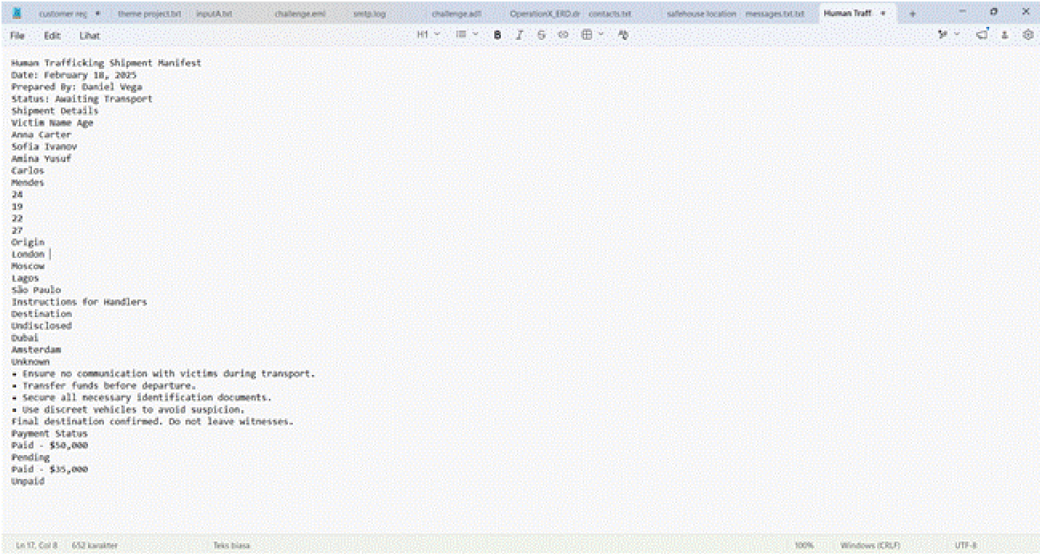

Human Trafficking Shipment Manifest
Date: February 18, 2025
Prepared By: Daniel Vega
Status: Awaiting Transport
Shipment Details
Victim Name Age
Anna Carter
Sofia Ivanov
Amina Yusuf
Carlos
Mendes
24
19
22
27
Origin
London
Moscow
Lagos
São Paulo
Instructions for Handlers
Destination
Undisclosed
Dubai
Amsterdam
Unknown
• Ensure no communication with victims during transport.
• Transfer funds before departure.
• Secure all necessary identification documents.
• Use discreet vehicles to avoid suspicion.
Final destination confirmed. Do not leave witnesses.
Payment Status
Paid - $50,000
Pending
Paid - $35,000
Unpaid

Figure 6.5.2 — Content of the recovered Human Trafficking Shipment Manifest, listing named individuals, origin/destination cities, handler instructions, and payment status.

Recovered File	**Descripted**
Shipment List.pdf	Shipment manifest recovered from the disk image.
identity docs.pdf	Identity-related document recovered during examination.
INVOICE.pdf	Financial document recovered from the evidence.
transaction logs.xlsx	Transaction records recovered from the evidence

Table 6.5.1 Summary of Recovered Documents

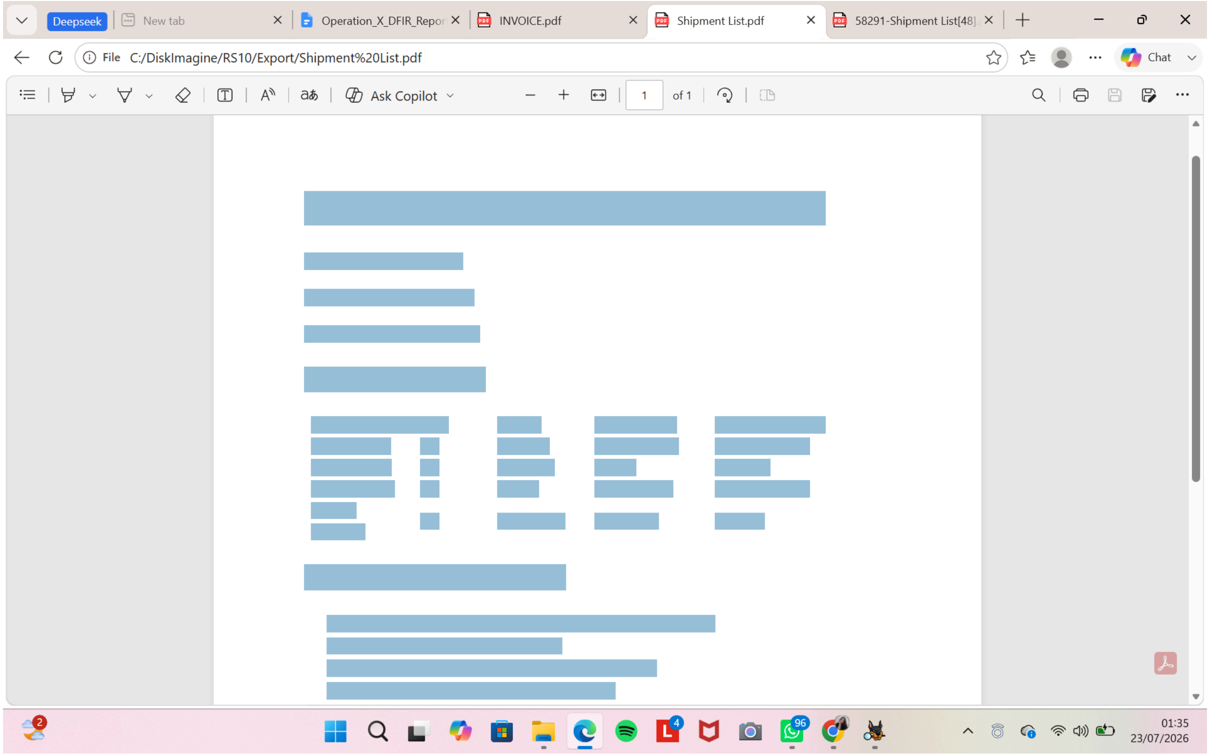


Figure 6.5.1 — Recovered documents and related artifacts identified during forensic examination (Shipment List.pdf).

Human Trafficking Shipment Manifest

Date: February 18, 2025

Prepared By: Daniel Vega **Status:**

Awaiting Transport

Shipment Details

Victim Name	Age	Origin	Destination	Payment Status
Anna Carter	24	London	Undisclosed	Paid - $50,000
Sofia Ivanov	19	Moscow	Dubai	Pending
Amina Yusuf	22	Lagos	Amsterdam	Paid - $35,000
Carlos Mendes	27	São Paulo	Unknown	Unpaid

Instructions for Handlers

- Ensure no communication with victims during transport.
- Transfer funds before departure.
- Secure all necessary identification documents.
- Use discreet vehicles to avoid suspicion.

Final destination confirmed. Do not leave witnesses.

Figure 6.5.1 — Recovered documents and related artifacts identified during forensic examination (Shipment List.pdf).

Analysis of this evidence is discussed in Chapter 8.

7 TIMELINE RECONSTRUCTION

This chapter presents a chronological reconstruction of events identified during the forensic examination. The timeline consolidates timestamps recovered from the forensic disk image, network traffic captures, recovered documents, email artifacts, and other digital evidence to reconstruct the sequence of activities relevant to the investigation. All timeline entries are derived from the examination results presented in Chapter 6.

Date and Time	Evidence Source	Description	Related Section
2025-03-07 17:33:06	Daniel_Laptop.E01	Tor Browser Portable installer identified from downloaded artifacts	6.1

Date and Time	Evidence Source	Description	Related Section
2025-03-11 00:22:19	Daniel_Laptop.E01	Nightclub.jpg recovered from the Downloads directory.	6.1
2025-03-11 00:22:19	encrypted.zip	Encrypted ZIP archive identified within Hidden folder.	6.3
2025-03-11 01:06:26	Daniel_Laptop.E01	transaction logs.xlsx modified within usb_backup_files.zip archive	6.1
2025-03-11 01:14:50	Daniel_Laptop.E01	identity docs.pdf modified within the same archive.	6.1
2025-03-11 01:16:32	encrypted.zip	Password-protected ZIP archive successfully accessed after password recovery, allowing forensic examination of its contents.	6.3
2025-03-11 09:23:40	Daniel_Laptop.E01	Email "Everything is in place" recovered; decoded message confirmed payment acknowledgement.	6.1
2025-03-13 00:55:01	Daniel_Laptop.E01	Browser history showed a search for OpenStego followed by the installer download.	6.1
2025-03-13 10:49:23	Daniel_Network.pcap	TLS Client Hello traffic observed during encrypted communication.	6.2
2025-03-13 11:08:45	Daniel_Network.pcap	HTTP browsing session to internal web server (192.168.40.135:8080) identified.	6.2
2025-03-13 11:14:32	Daniel_Network.pcap	encrypted.zip downloaded via an HTTP GET request.	6.2

Date and Time	Evidence Source	Description	Related Section
2025-03-13 11:32:13	Daniel_Network.pcap	FTP RETR command used to retrieve INVOICE.pdf.	6.2
2025-03-13 11:32:27	Daniel_Laptop.E01	Tor Browser and OpenStego execution recorded within a one-second interval.	6.1
2025-03-22 14:50:27	Daniel_Laptop.E01	Found a few Ciphertext in a each email that already decoded with Cryptii in Caesar Cipher	6.1.3
2025-03-12 10:10:14	Daniel_Laptop.E01	Subject: Urgent Cleanup Needed email from dan_vega_11041990@outlook.com to : fixer@darkops.xyz;	6.1.3
2004-11-27 10:41:44	Daniel_Laptop.E01	msg_43.txt Subject: Banned auto__mail.python.bat in mail from you	6.1.3
2001-04-07 00:23:06	Daniel_Laptop.E01	Msg25.txt Subject: Returned mail: Too mant hops 19 (17 max): from <linuxuser-admin@www.linux.org.uk> via [199.164.235.226], to <scoffman@wellpart…	6.1.3
2019-12-07 09:17:28	Daniel_Laptop.E01	Noted and followed up several installed program during investigation	6.1
2025-03-07 17:33:06	Daniel_Laptop.E0	Identified several downloaded jpg,png and application in Web Download	6.1

Date and Time	Evidence Source	Description	Related Section
2025-03-13 13:50:29	Daniel_Laptop.E01	Identified many document that suspicious inn Recent Document	6.1
2025-03-06 13:25:00	Daniel_Laptop.E01	Running Program during the Human trafficking and murder investigation	6.1
2025-03-08 21:25:09	Daniel_Laptop.E01	Remote monitoring/management software found installed, indicating potential remote access to the system.	6.1
2025-03-13 00:59:59	Daniel_Laptop.E01	File or folder suspected of being encrypted, identified during the examination of system artifacts.	6.1
2025-03-14 21:11:59	Daniel_Laptop.E01	Encryption application found installed on the system, indicating possible use of encryption to conceal data.	6.1
2023-03-19 09:14:13	Daniel_Laptop.E01	User-owned content suspected of being relevant to the investigation, found during examination of the user directory.	6.1
2025-03-13 19:18:35	Daniel_Laptop.E01	Evidence of an external USB device previously connected to the system, indicating potential data transfer to removable storage media.	6.1
2019-03-13 19:18:24	Daniel_Laptop.E01	Additional (child device) detail of the detected USB device, further confirming the identification of the external storage device used.	6.1

Table 7.1 — Master Investigation Timeline

8 ANALYSIS

This chapter presents the interpretation of the forensic findings obtained during the examination. Unlike Chapter 6, which reports factual findings, this chapter correlates evidence from multiple sources to explain the significance of the recovered artifacts in relation to the investigation.

8.1 Temporal Analysis

The reconstructed timeline presented in Chapter 7 shows that the recovered evidence is interconnected rather than representing isolated events. On 7 March 2025, Tor Browser Portable was identified as having been downloaded and installed, indicating the availability of an anonymised browsing tool before other relevant activities occurred. Four days later, on 11 March 2025, several files, including *Nightclub.jpg*, *transaction logs.xlsx*, and *identity docs.pdf*, were accessed or modified within a relatively short period. Later on the same day, the email titled *"Everything is in place"* was recovered, with its decoded content confirming that a payment had been received. The close timing of these activities suggests that they were part of the same operational session rather than unrelated user actions.

One of the most notable findings in the timeline is the near-simultaneous execution of **Tor Browser Portable** and **OpenStego** on 13 March 2025 at 11:32:27–11:32:28 WIB. The minimal time difference between the execution of both applications suggests that they were likely used during the same activity rather than independently. On the same day, the **Shipment List.pdf** attachment was also recovered from the email evidence. Although the steganography examination did not identify any hidden payload within the analysed BMP files, the combined presence of Tor Browser, OpenStego, recovered documents, and network activity indicates an attempt to utilise privacy and concealment-related tools during the investigation period.

8.2 Relational Analysis

This section analyzes the relationships between evidence obtained from different sources to identify connections between digital artifacts, recovered documents, network communications, and other entities relevant to the investigation.

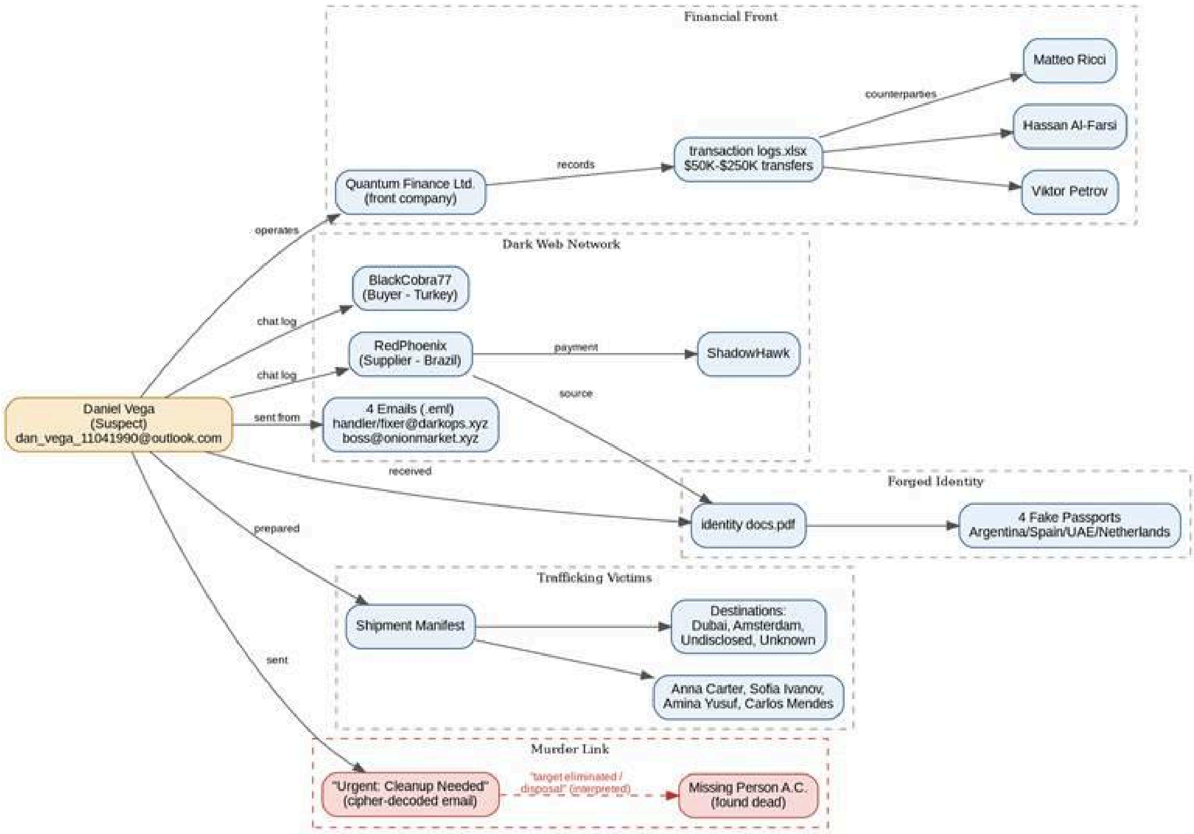


Figure 8.2.1 — Evidence relationship diagram connecting the suspect, financial front, dark-web network, forged identities, trafficking victims, and the murder-linked email.

Figure 8.2.1 illustrates how the recovered evidence is interconnected rather than existing as isolated artefacts. The relationships identified during the forensic examination can be grouped into five major evidence clusters: financial activities, communication network, forged identity documents, shipment records, and recovered encoded communications. Each cluster contributes contextual information that strengthens the overall reconstruction of the investigated activities.

The first cluster represents the financial records recovered from *transaction logs.xlsx*. The spreadsheet references **Quantum Finance Ltd.** together with several counterparties, including **Matteo Ricci**, **Hassan Al-Farsi**, and **Viktor Petrov**, indicating documented financial transactions associated with the investigation.

The second cluster focuses on recovered email communications involving aliases such as **BlackCobra77**, **RedPhoenix**, and **ShadowHawk**. These communications demonstrate interactions between multiple parties and provide supporting context for document exchange and operational coordination.

The third cluster consists of the recovered *identity docs.pdf* and associated counterfeit passport records. These artefacts indicate the preparation or exchange of forged identity documents involving multiple nationalities.

The fourth cluster relates to the recovered *Shipment List*, which records multiple individuals together with their intended destinations. When analysed alongside the recovered financial records and communications, this document provides additional contextual links between different evidence sources.

The final cluster concerns the decoded email entitled *"Urgent: Cleanup Needed"*. This communication establishes a relationship between the recovered email evidence and the reported murder case involving Missing Person A.C. While the email alone cannot establish criminal responsibility, it provides contextual evidence when interpreted together with the other recovered artefacts.

8.3 Functional Analysis

The forensic findings from the disk image and network capture indicate a sequence of activities that can be interpreted as four functional stages. These stages describe how the recovered digital evidence is related throughout the investigation.

1. Preparation

The forensic examination identified several files related to the investigation, including the shipment list, identity documents, financial records, and other supporting files. Some of these files were stored within hidden directories and protected using an encrypted ZIP archive, indicating that the data was organised before further activities were performed.

2. Concealment

The examination identified the presence of Tor Browser and OpenStego on the examined system. Timeline analysis also showed that both applications were accessed within a very short time interval. In addition, an encrypted ZIP archive was recovered and successfully decrypted during the investigation. These findings indicate that encryption and privacy-related tools were available and actively used during the period covered by the investigation.

3. Transmission

Network traffic analysis identified HTTP and FTP communication involving several recovered files, including **HiddenMessage.bmp**, **safehouse location.bmp**, **encrypted.zip**, **Shipment List.pdf**, and **INVOICE.pdf**. The network capture also recorded encrypted TLS sessions during the same investigation period. These findings indicate that multiple files were transmitted across the network while encrypted communication was also observed.

4. Coordination and confirmation

Recovered email messages and decoded Caesar cipher text provide additional context for the activities identified during the forensic examination. Messages such as **"Everything is in place," "Safehouse Ready,"** and **"Urgent: Cleanup Needed"** indicate communication between involved parties during the investigated period. The decoded email stating **"Everything is in place"** also confirms that payment had been received, providing supporting evidence that complements the recovered documents and network artifacts.

Overall, the forensic findings indicate a consistent sequence of activities involving document preparation, data protection, network transmission, and communication between related parties. The combination of disk artifacts, recovered documents, network traffic, encrypted archives, and decoded messages provides a coherent reconstruction of the events identified during the investigation.

9 CONCLUSIONS

This chapter summarizes the results of the digital forensic examination by addressing the investigation objectives and describing the limitations encountered during the examination process.

9.1 Answers to Investigation Objectives

Objective 1: Identify digital evidence related to the suspect's activities

Status : Confirmed

Digital evidence associated with the suspect's account (dan_vega_11041990@outlook.com) was successfully recovered from the forensic image. The recovered evidence includes trafficking-related documents, forged identity records, financial records, browser artifacts, and installation artifacts associated with Tor Browser and OpenStego.

Objective 2: Analyze communications and transferred files associated with the investigation

Status : Confirmed

Four recovered email messages and multiple HTTP and FTP file transfers were successfully analyzed and correlated with system artifacts recovered from the forensic image. The decoded email messages and network activity collectively support the transfer of investigation-related files between communicating systems.

Objective 3: Reconstruct the sequence of events based on recovered digital evidence

Status: Highly Likely

A chronological sequence of events was successfully reconstructed in Chapter 7 by correlating timestamps recovered from the forensic disk image with supporting network evidence. The recovered timestamps are internally consistent and describe a logical sequence of user activities throughout the investigation period. The confidence level is assessed as **Highly Likely** rather than **Confirmed** because the network capture timestamps could not be fully verified against an independent system clock.

Objective 4: Produce a forensic report supporting the investigation

Status: Confirmed

This report documents the forensic examination methodology, findings, timeline reconstruction, and analysis in accordance with NIST SP 800-86. It serves as the consolidated forensic report produced by the examination team to support the investigation.

9.2 Limitations of Examination

9.2.1 Disk Image Acquisition Limitations

- No reference hash values were provided at the time the evidence was received. Therefore, evidence integrity could only be verified by generating a baseline SHA-256 hash rather than comparing it with a previously known reference hash.

- The forensic image was examined as a static, read-only source. Live system memory (RAM) was not available during the examination; therefore, artifacts that existed only in volatile memory could not be recovered.

9.2.2 Deleted File and Artifact Recovery Limitations

- Deleted file recovery was limited to artifacts that remained recoverable through file system metadata or were identified within Autopsy's **Deleted Files** view. Files that had been overwritten at the storage level could not be recovered.
- Some shortcut (.lnk) artifacts referenced file paths whose original target files were no longer available. As a result, the existence of the shortcut could be confirmed, but the contents of the referenced files could not always be independently verified.

9.2.3 Network and Steganography Limitations

- No hidden content was recovered from **HiddenMessage.bmp** or **safehouse location.bmp** using ExifTool, Strings, XXD, Binwalk, or Steghide. The examination was limited to the steganography tools available in the analysis environment and does not exclude the possibility that other specialized techniques could reveal additional hidden content.
- The presence of Tor Browser is supported by installation artifacts recovered from the forensic image. However, the associated network traffic could not be conclusively identified as Tor traffic because the captured communications remained encrypted.
- No cryptocurrency transaction data was observed in cleartext within **Daniel_Network.pcap**. If cryptocurrency transactions occurred, they were not directly observable within the available network capture.

9.2.4 Encryption & File Cracking Limitations

- The standard **unzip** utility available in the analysis environment does not support WinZip AE (AES) encryption. As a result, password recovery using unzip -P was unsuccessful despite the correct password being available. This limitation was resolved by using 7-Zip and Pyzipper, both of which support WinZip AE encryption.
- No standalone VeraCrypt volume or other full-disk encrypted container was identified within the evidence provided for examination. The only encrypted file encountered during this investigation was **encrypted.zip**.

- Password recovery for **encrypted.zip** relied on dictionary-based attacks using a custom wordlist followed by **rockyou.txt**. The effectiveness of this approach depends on the completeness of the available wordlists.
- The recovered **client_list** file required a working copy with the **.xlsx** extension to be created for examination. The original evidence file remained preserved in its original state without modification.

10 EXAMINATION NOTES / INVESTIGATION LOG

This chapter records the activities performed during the forensic examination in chronological order. The log provides an audit trail of the examination process, including the analyst responsible, the tools used, and the outcome of each examination activity.

10.1 Disk Image and Deleted File Recovery Examination Log

Date & Time	Analyst	Action / Tool Used	Outcome
2026-07-06 13:56	Disk Image	Reviewed Operating System Information – Autopsy	Confirmed OS version, computer name, and installation date on Daniel_Laptop.E01
2026-07-12 23:59	Disk Image	Reviewed Installed Programs artifacts — Autopsy	Identified installed encryption and remote monitoring software on the system
2026-07-15 13:01	Disk Image	Reviewed Run Programs / execution artifacts — Autopsy	Confirmed execution history of encryption tool
2026-07-16 11:47	Disk Image	Reviewed Recent Documents artifact — Autopsy	Confirmed document access timeline consistent with recovered PDF/XLSX files

2026-07-16 11:50 11:52	Disk Image	Reviewed USB Device attachment and child device artifacts (USBDeviceAttached.png, USBChild.png) — Autopsy	artifacts (USBDeviceAttached.png, USBChild.png) — AutopsyIdentified external USB device connection and associated device details, consistent with potential data transfer
2026-07-16 11:55	Disk Image	Reviewed encryption-related artifacts (EncryptionProgram.png, EncryptionSuspected.png) — Autopsy	Identified installed encryption software and files/folders suspected to be encrypted
2026-07-16 04:38	Disk Image	Reviewed Remote Monitoring/Management software artifact — Autopsy	Identified installed remote access/monitoring tool on the system
2026-07-16 04:37	Disk Image	Reviewed User Content directory artifact (UserContentSuspected.png) — Autopsy	Flagged user-generated files suspected to be relevant to the investigation
2026-07-08 18:00 2026-07-12 16:16	Disk Image	Reviewed Browser Search and Download history — Autopsy	Confirmed search query for OpenStego followed by installer download

Table 10 — Examination Log Examination Log (Disk Image Analyst)

10.2 Network and Steganography Examination Log

Date & Time	Analyst	Action / Tool Used	Outcome
2026-06-30 15:07	Network & Steganography	capinfos Daniel_Network.pcap	Verified capture file information (packet count, capture duration, interfaces).
2026-06-30 15:07	Network & Steganography	tshark -q -z io,phs	Identified major protocols present in the capture (HTTP, FTP, DNS, TLS).
2026-06-30 15:08	Network & Steganography	tshark -q -z endpoints,ip	Enumerated communicating IP addresses.
2026-06-30 15:08	Network & Steganography	tshark -q -z conv,tcp	Reviewed TCP conversations between hosts.
2026-07-06 13:05	Network & Steganography	tshark HTTP request filtering	Analysed HTTP request activity and reconstructed communications.
2026-07-06 13:11–13:17	Network & Steganography	Multiple tshark filtering commands	Reviewed DNS queries, FTP sessions, and TLS handshakes for suspicious traffic.
2026-07-07 10:40–10:53	Network & Steganography	Additional tshark analysis	Verified protocol findings and cross-checked previous observations.

Date & Time	Analyst	Action / Tool Used	Outcome
2026-07-19 18:59	Network & Steganography	HTTP traffic analysis (tcp.port==8080)	Identified web communication over TCP port 8080.
2026-07-19 19:01	Network & Steganography	DNS and HTTP correlation	Correlated DNS requests with HTTP activity.
2026-07-19 19:06	Network & Steganography	FTP session analysis	Confirmed FTP file transfer activity.
2026-07-28 12:44	Network & Steganography	HTTP inspection	Reviewed complete HTTP communications.
2026-07-28 12:45	Network & Steganography	FTP inspection	Verified FTP control and data sessions.
2026-07-28 12:46	Network & Steganography	DNS inspection	Examined DNS query records.
2026-07-28 12:47	Network & Steganography	TLS handshake inspection	Reviewed encrypted TLS connections.
2026-07-29 15:06	Network & Steganography	Final HTTP verification (tshark)	Final verification before report preparation.
2026-06-30 16:26	Network & Steganography	exiftool HiddenMessage.bmp	Metadata examined; no suspicious metadata identified.
2026-06-30 16:26	Network & Steganography	exiftool safehouse location.bmp	Metadata examined; no anomalies detected.

Date & Time	Analyst	Action / Tool Used	Outcome
2026-06-30 16:26	Network & Steganography	strings	Extracted printable strings; no hidden plaintext recovered.
2026-07-06 12:56	Network & Steganography	strings	Repeated string extraction for verification
2026-07-06 12:58	Network & Steganography	xxd	Examined file header and hexadecimal structure.
2026-07-07 11:12	Network & Steganography	file	Verified actual file format of BMP evidence.
2026-07-07 11:14	Network & Steganography	ExifTool	Revalidated metadata consistency.
2026-07-07 11:49	Network & Steganography	strings	Reviewed printable contents of BMP images.
2026-07-07 11:57	Network & Steganography	zsteg	Attempted extraction of hidden payload; no recoverable payload found.
2026-07-07 12:03	Network & Steganography	binwalk	Scanned for embedded files and signatures; none detected.
2026-07-08 16:51	Network & Steganography	ExifTool	Final metadata validation.

Date & Time	Analyst	Action / Tool Used	Outcome
2026-07-08 16:52	Network & Steganography	strings	Searched for hidden keywords and credentials; negative result.
2026-07-08 16:53	Network & Steganography	xxd	Verified hexadecimal contents.
2026-07-29 15:22	Network & Steganography	binwalk	Final embedded-file scan; no hidden objects identified.
2026-07-29 15:46	Network & Steganography	zsteg	Final steganography verification; negative result confirmed.

Table 10.1 — Examination Log (Network & Steganography Analyst)

10.3 Encryption and File Cracking Examination Log

Date	Analyst	Action / Tool Used	Outcome
[Day 1]	Encryption & File Cracking	File type verification — file, xxd (Documents root + Hidden folder)	One extension mismatch identified: client_list (Excel, no extension)
[Day 1]	Encryption & File Cracking	Binwalk scan + extraction — HiddenMessage.bmp, safehouse location.bmp/.jpg, usb_backup_files.zip	No embedded signatures in BMP/JPG files; usb_backup_files.zip yielded 3 archived files
[Day 2]	Encryption & File Cracking	Encryption identification — zipinfo -v, unzip -v on encrypted.zip	Identified as WinZip AE (compression method 99)

[Day 3]	Encryption & File Cracking	Dictionary attack — pyzipper against rockyou.txt	Password recovered: 123456
[Day 3]	Encryption & File Cracking	Extraction — 7z x	contacts.txt and safehouse_locations.txt recovered and reviewed

10.4 Email and Document Recovery Examination Log

Date	Analyst	Action / Tool Used	Outcome
[Day 1]	Email & Document Recovery	Reviewed E-Mail Messages module for account dan_vega_11041990@outlook.com Autopsy	Recovered 4 email artifacts (Everything is in Place, Safehouse Ready, Urgent: Cleanup Needed, File Transfer – Urgent)
[Day 1]	Email & Document Recovery	Cross-checked email attachment-save events against Recent Documents Autopsy	Confirmed Shipment List[48].pdf saved as email attachment at 2025-03-13 01:11:24 WIB
[Day 2]	Email & Document Recovery	Reviewed recovered PDF and spreadsheet documents (Shipment List.pdf, identity docs.pdf, INVOICE.pdf, transaction logs.xlsx) Autopsy Data Artifacts	Confirmed document content consistent with trafficking manifest, forged identity records, and financial ledger
[Day 2]	Email & Document Recovery	Decoded Caesar-cipher (shift 3) body text in "Everything is in place[47].eml" — manual decoding / Cryptii	Recovered plaintext message confirming payment and operational instructions

[Day 3]	Email & Document Recovery	Metadata cross-check of recovered documents against .lnk shortcut artifacts (Emails.lnk, Dark Web.lnk) Autopsy	Confirmed consistent creation/access/modification timestamps; no evidence of metadata manipulation
[Day 3]	Email & Document Recovery	Compiled recovered document summary for Section 6.5.1 Autopsy Report export	Finalized Table 6.5.1 (Summary of Recovered Documents)
[Day 3]	Email & Document Recovery	Decoded additional recovered ciphertext messages — manual decoding / Cryptii	Recovered plaintext content confirming further operational/payment details
[Day 3]	Email & Document Recovery	Reviewed Remote Monitoring/Management software artifact — Autopsy	Identified installed remote access/monitoring tool on the system

Table 10.4 — Examination Log (Email & Document Recovery)

11 REFERENCES

This chapter lists the standards, forensic tools, and external references cited throughout this report. All references are provided to support the methodology and findings presented during the digital forensic examination.

11.1 Standards

- National Institute of Standards and Technology. (2006). NIST Special Publication 800-86: Guide to Integrating Forensic Techniques into Incident Response. https://csrc.nist.gov/publications/detail/sp/800-86/final
- Association of Chief Police Officers. (2012). Good Practice Guide for Digital Evidence.
- International Organization for Standardization. (2012). ISO/IEC 27037:2012: Guidelines for Identification, Collection, Acquisition and Preservation of Digital Evidence.

11.2 Tool References

- Basis Technology. (n.d.). Autopsy. https://www.autopsy.com/
- ExifTool by Phil Harvey. (n.d.). ExifTool. https://exiftool.org/
- Exterro. (n.d.). FTK Imager. https://www.exterro.com/ftk-imager
- GNU Project. (n.d.). GNU Binutils. https://www.gnu.org/software/binutils/
- Hashcat Team. (n.d.). Hashcat. https://hashcat.net/hashcat/
- Netresec. (n.d.). NetworkMiner. https://www.netresec.com/
- Openwall. (n.d.). John the Ripper. https://www.openwall.com/john/
- ReFirm Labs. (n.d.). Binwalk. https://github.com/ReFirmLabs/binwalk
- Steghide Project. (n.d.). Steghide. https://steghide.sourceforge.net/
- Wireshark Foundation. (n.d.). TShark User's Guide. https://www.wireshark.org/docs/man-pages/tshark.html
- Wireshark Foundation. (n.d.). Wireshark. https://www.wireshark.org/

11.3 Additional Tool References

- Darwin, I. (n.d.). file(1): Determine File Type Using Magic Numbers. https://www.darwinsys.com/file/
- Miessler, D. (n.d.). SecLists. https://github.com/danielmiessler/SecLists
- Pavlov, I. (n.d.). 7-Zip. https://www.7-zip.org/
- pyzipper Contributors. (n.d.). pyzipper. https://github.com/danifus/pyzipper

12 APPENDIX

This appendix contains supplementary materials supporting the findings presented throughout this report. The appendices include command histories, raw tool outputs, supplementary screenshots, and technical references that were not included in the main chapters.

Appendix A — Network Traffic Analysis Commands

The following commands were executed during the network traffic investigation using **TShark** to analyse HTTP, FTP, DNS, TLS, and Dark Web (.onion) activities extracted from the packet capture.

```
tshark -r Daniel_Network.pcap -q -z conv,tcp

tshark -r Daniel_Network.pcap -q -z io,phs

tshark -r Daniel_Network.pcap -Y http

tshark -r Daniel_Network.pcap -Y ftp

tshark -r Daniel_Network.pcap -Y ftp-data

tshark -r Daniel_Network.pcap -Y dns

tshark -r Daniel_Network.pcap -Y tls.handshake



tshark -r Daniel_Network.pcap \

-Y "http.request" \

-T fields \

-e frame.time \

-e http.request.method \

-e http.request.uri \

-e http.host



tshark -r Daniel_Network.pcap \

-Y 'ftp.request.command=="RETR"' \

-T fields \
```

```
-e frame.time \
-e ftp.request.command \
-e ftp.request.arg

tshark -r Daniel_Network.pcap \
-Y "dns.qry.name contains onion" \
-T fields \
-e frame.time_relative \
-e ip.src \
-e ip.dst \
-e dns.qry.name

tshark -r Daniel_Network.pcap \
-Y "tls.handshake.type==1" \
-T fields \
-e frame.time \
-e ip.src \
-e ip.dst \
-e tls.handshake.extensions_server_name

tshark -r Daniel_Network.pcap \
--export-objects http,http_objects_full
```

Appendix B — File, Steganography, and Archive Analysis Commands

The following commands were used to inspect file metadata, verify file integrity, analyse bitmap files for hidden content, and examine encrypted archives.

```
File Type Identification
```

```
file *.pdf

Metadata Analysis

exiftool HiddenMessage.bmp
exiftool "safehouse location.bmp"

String Extraction

strings HiddenMessage.bmp

Hex Inspection

xxd -l 128 HiddenMessage.bmp

Embedded Data Analysis

binwalk HiddenMessage.bmp
binwalk "safehouse location.bmp"

Archive Inspection

zipinfo encrypted.zip
zipinfo -v encrypted.zip
unzip -l encrypted.zip

Integrity Verification

sha256sum Daniel_Network.pcap
```

Appendix C — Supplementary Screenshots

This appendix contains supplementary screenshots referenced throughout the report but not included in the main chapters.

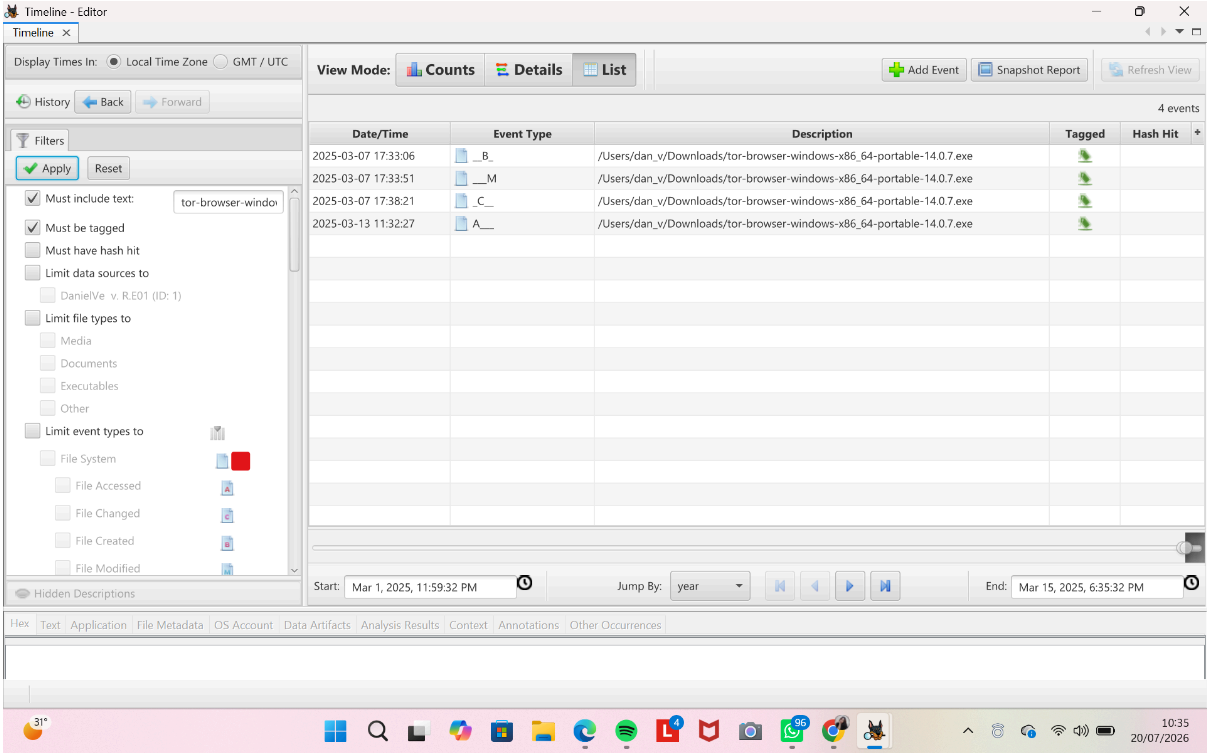


Figure C.1 - Tor Browser access timeline

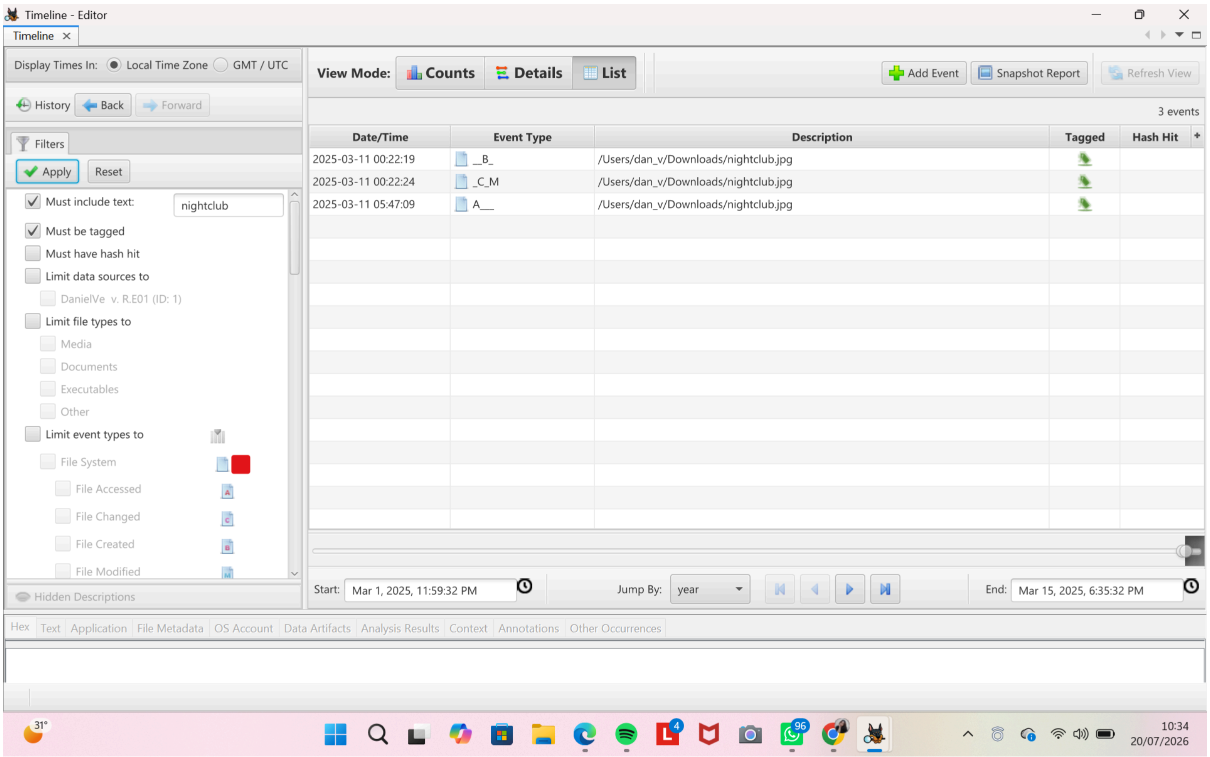


Figure C.2 - nightclub.jpg file timeline

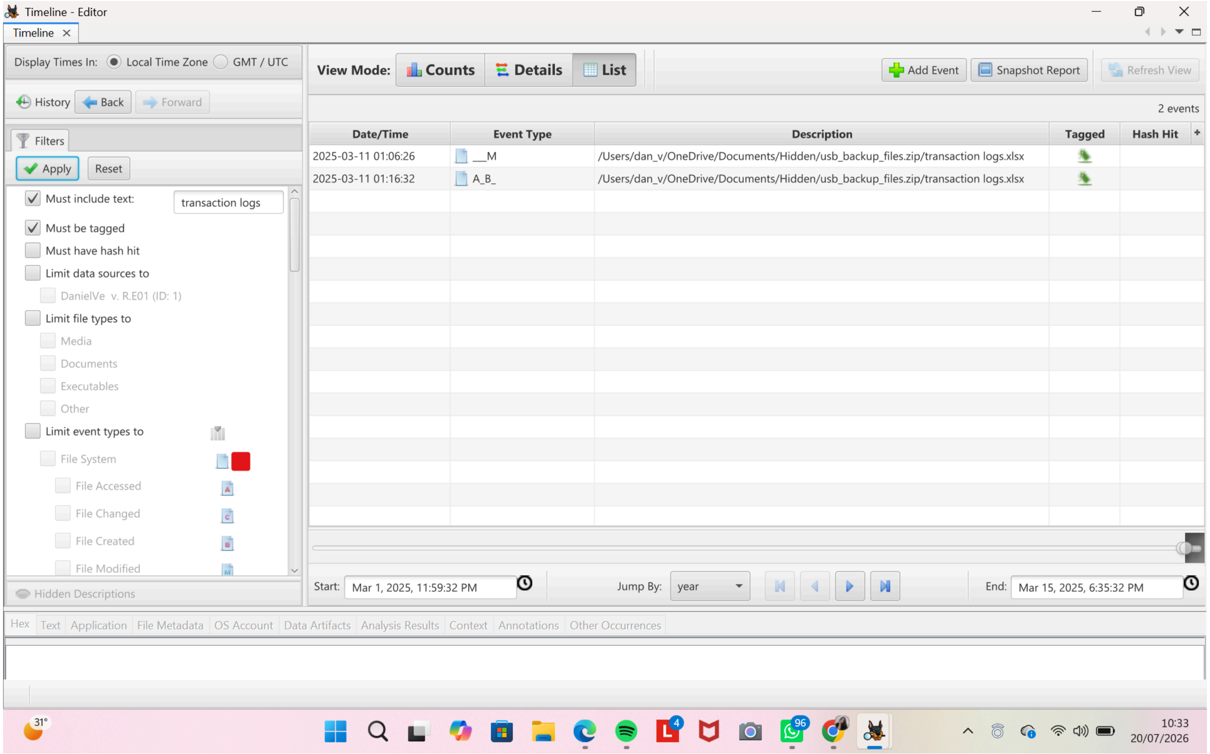


Figure C.3 - transaction logs.xlsx timeline

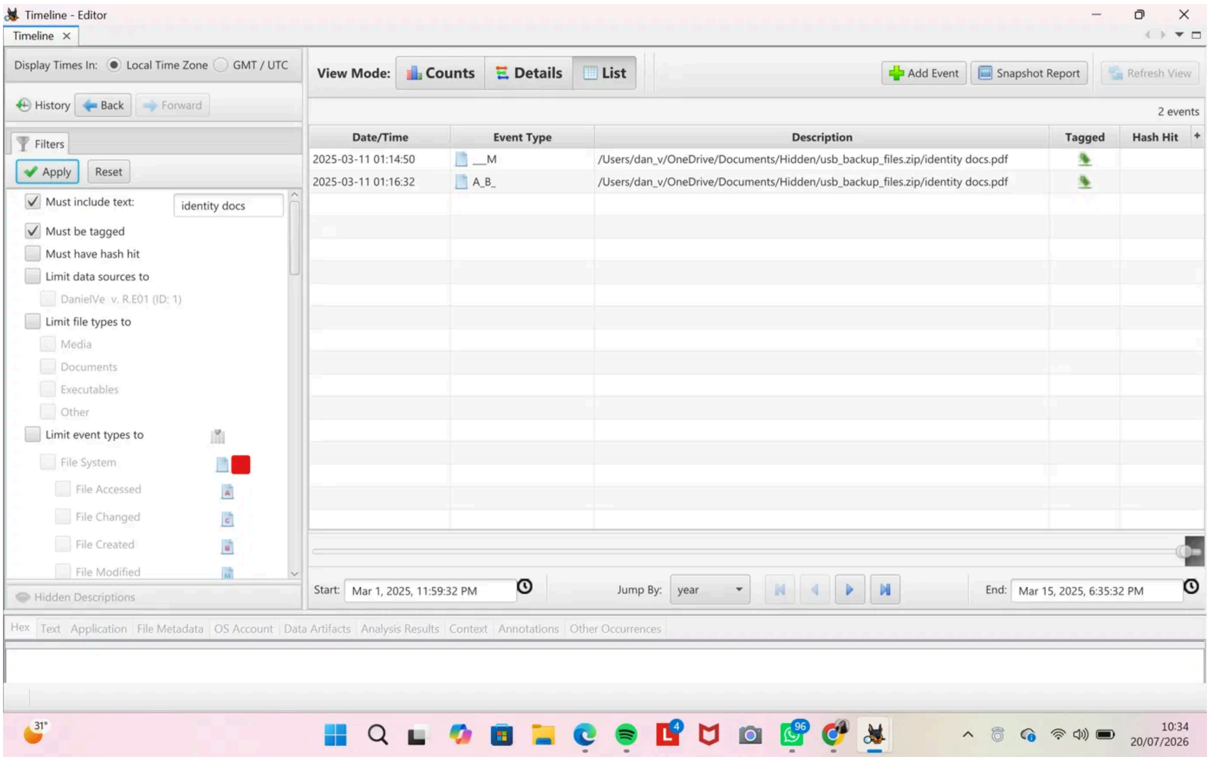


Figure C.4 - identity docs.pdf timeline

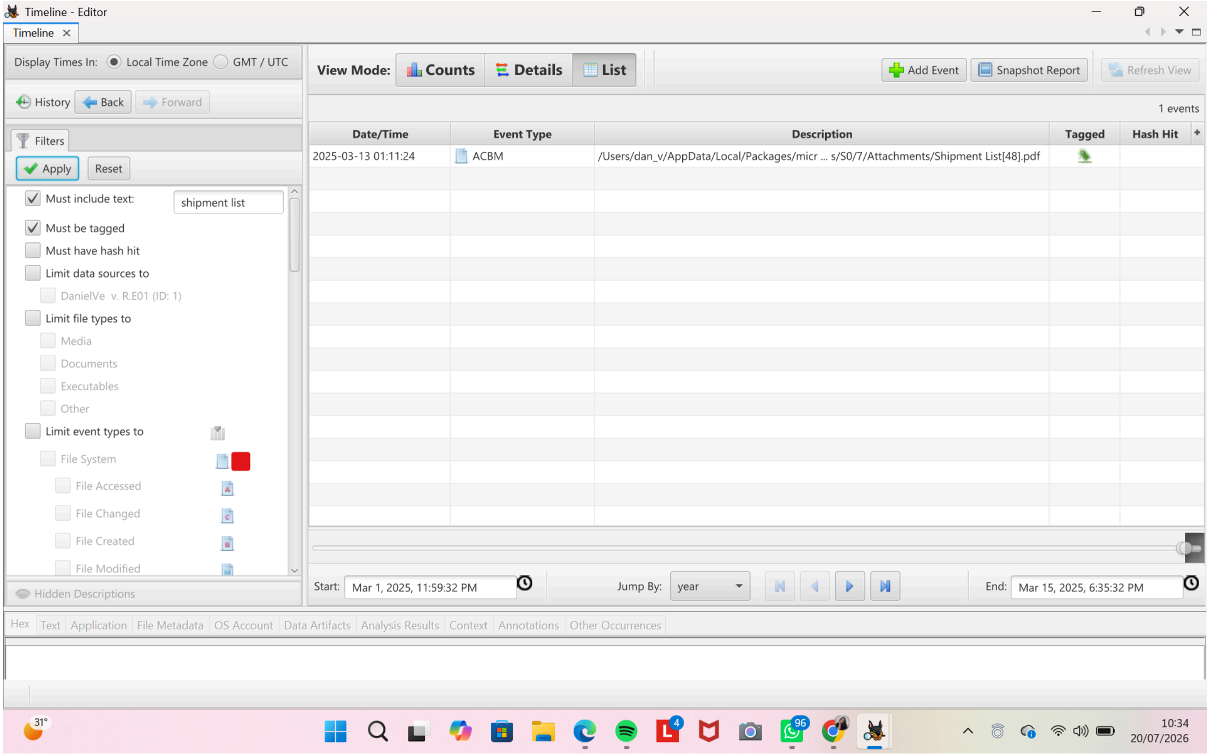


Figure C.5 - Shipment List attachment event

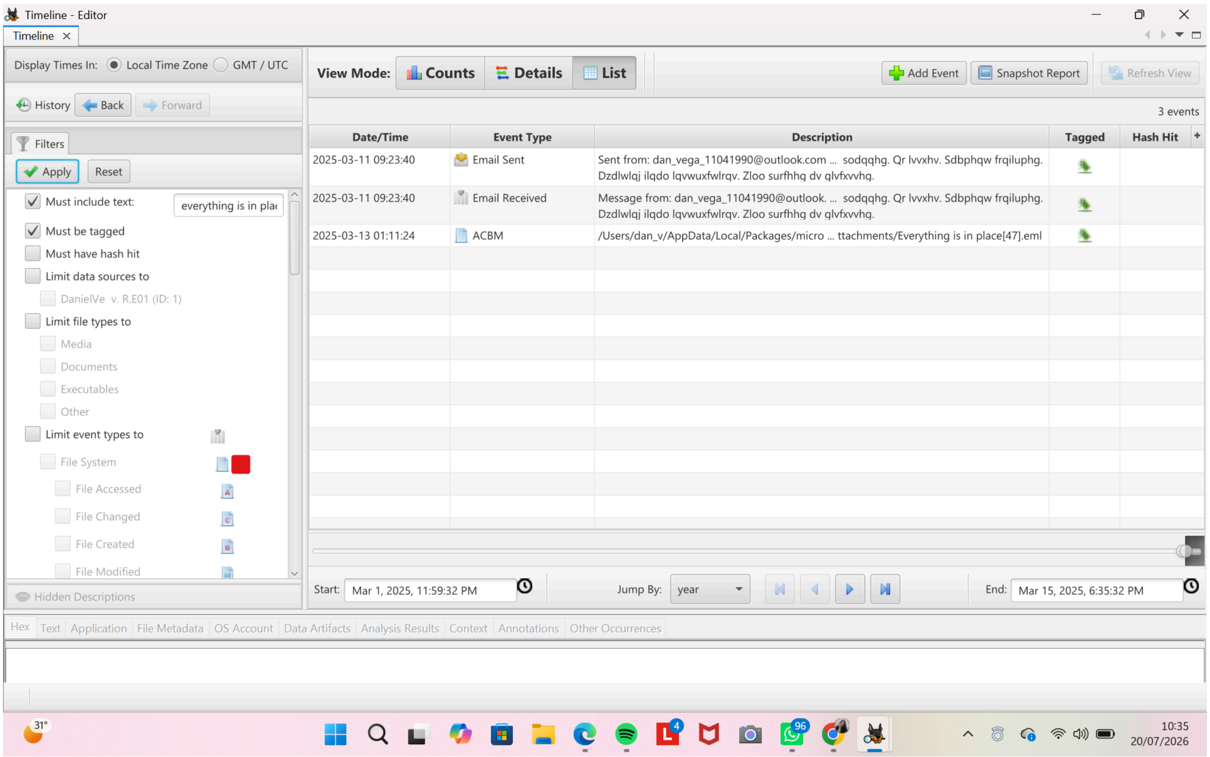


Figure C.6 - Email Everything is in place timeline

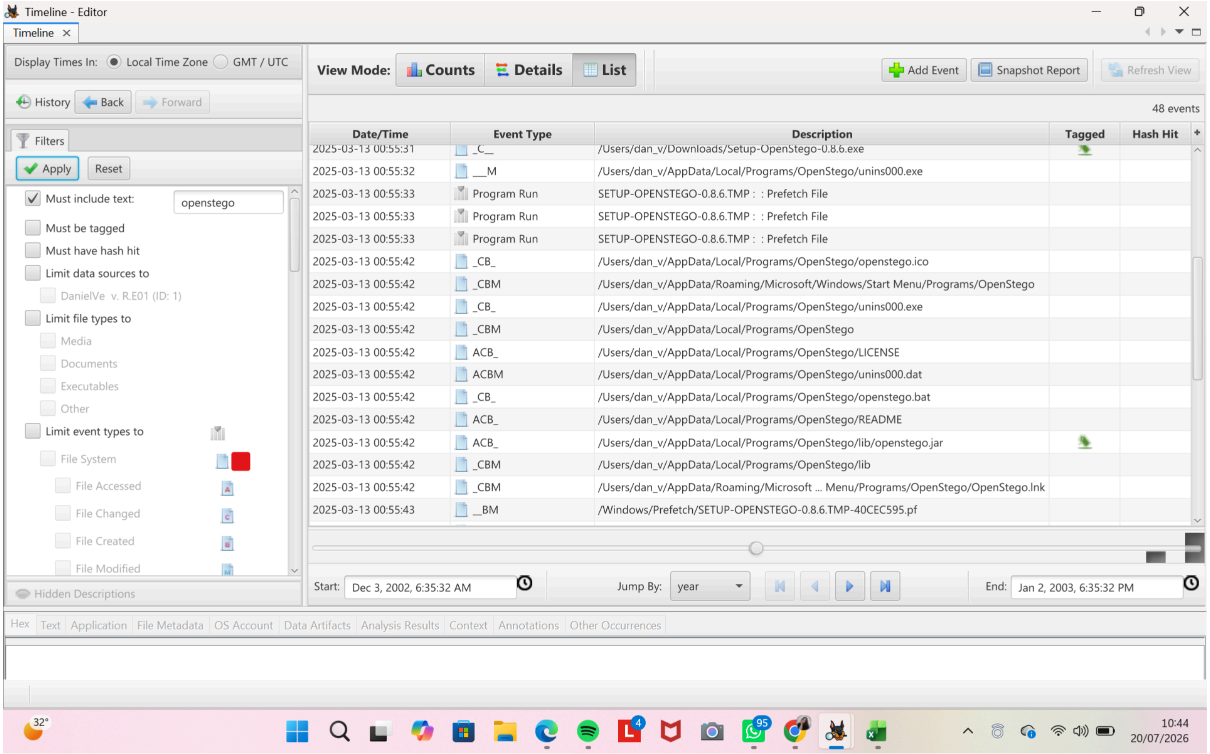


Figure C.7 - OpenStego installation detail

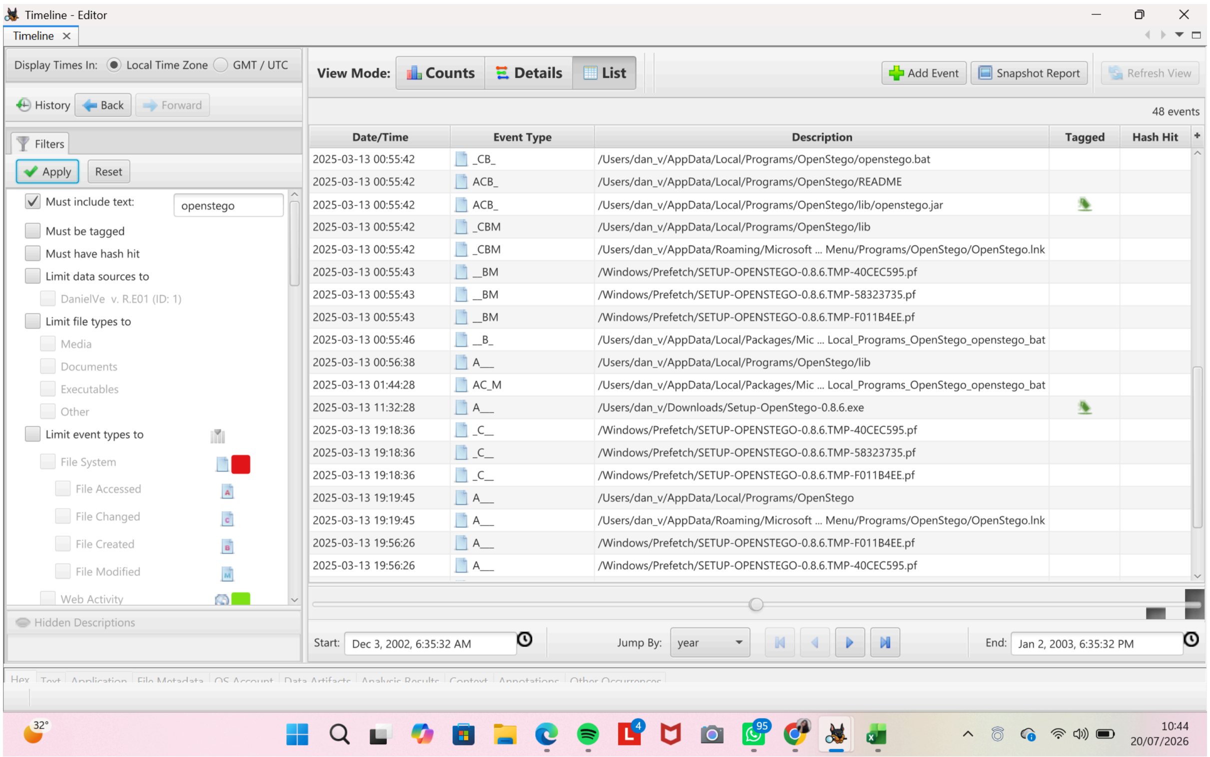


Figure C.8 - OpenStego installation detail

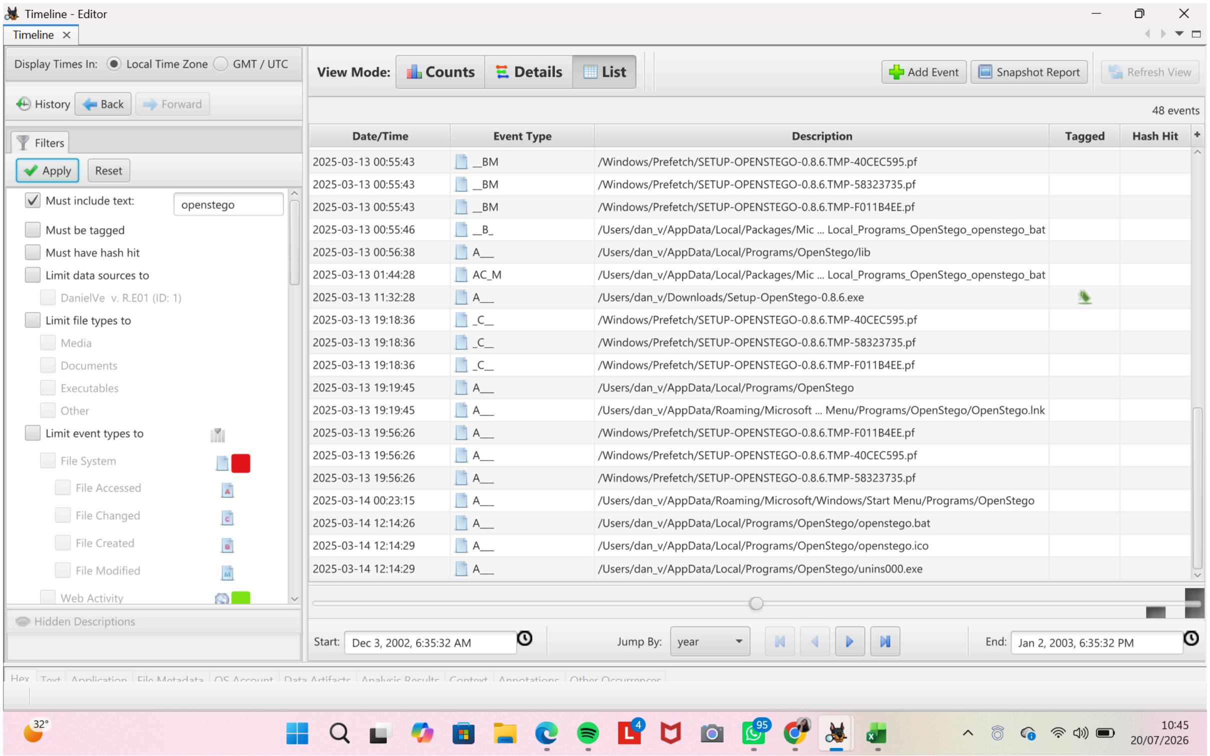


Figure C.9 - OpenStego installation detail

Appendix D — Glossary

The following glossary defines technical terms used throughout this report

- PCAP a saved recording of network traffic, viewable in tools like Wireshark.
- TCP Stream the full back-and-forth of a single network conversation, reassembled from individual packets.
- Tor a network that routes traffic through multiple relays to hide who is talking to whom.
- Steganography hiding data inside another file, such as an image, so its presence isn't obvious.
- Hash a fixed-length fingerprint of a file or password, used to verify integrity or attempt password recovery without altering the original file.
- Brute-force trying many possible passwords systematically until the correct one is found.

- E01 the EnCase disk image format, a common forensic format for full disk/drive images.